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Table of Contents

1	Introduction	1
1.1	Introduction to Topic	1
1.2	Problem Scenario	2
1.3	Project as a Solution	3
1.4	Aims and Objectives	4
1.4.1	Aims.....	4
1.4.2	Objectives	4
2	Background.....	5
2.1	Trend in the Field	5
2.2	Benefits of Hisab Khata	6
2.3	Challenges	7
2.4	Future Possibilities	8
2.5	Similar Real-World Projects	9
2.5.1	Karobar App.....	9
2.5.2	Khatabook, OkCredit, TallyPrime.....	9
2.6	Literature review (feature related)	11
2.6.1	OCR Receipt Scanning & Automated Financial Data Capture.....	11
2.6.2	AI & Chatbot Integration in Business and Financial Apps	13
2.7	Review of the Technical Aspects	14
2.7.1	Tech Stack.....	14
2.7.2	Packages and Libraries	14
2.8	Methodology	14
2.9	Considered Methodology	14
2.9.1	Waterfall Model.....	14
2.9.2	Agile/Scrum	15
2.9.3	Prototype-Based / Iterative-Prototyping Methodology.....	15

2.10	Selected Methodology – Rational Unified Process (RUP)	16
3	Development till Date	17
3.1	Software Requirements Specification (SRS).....	17
3.2	Product Backlog.....	17
3.3	Pre Survey Description	17
3.4	System Design.....	18
3.4.1	System Architecture/Context Diagram	18
3.4.2	ERD	19
3.4.3	Collaboration Diagrams	20
3.4.4	Sequence Diagrams	22
3.4.5	Use case Diagram	30
3.4.6	Class Diagram	34
3.4.7	Activity Diagrams	35
3.4.8	Data Flow Diagram (DFD)	40
3.5	Style guide	41
3.6	UI Figma Design	42
3.6.1	Welcome Screen	42
3.6.2	Authentication Screens.....	43
3.6.3	Customer Dashboard Screens.....	47
3.6.4	Business Dashboard Screen	57
3.7	Velocity Chart (Completed Story Points).....	65
3.8	Burndown Chart.....	66
4	Analysis of Progress	66
4.1	Progress Table.....	67
4.2	Gantt Chart	68
5	Future Works.....	70
6	Conclusion	71

7	References.....	72
8	Appendix	75
8.1	SRS	75
8.2	Pre Survey	75
8.2.1	Report on Customer.....	75
8.2.2	Report on Business	81
8.3	Physical Pre Survey	87
8.3.1	Customers	87
8.3.2	Business	93
8.4	Product Backlogs	99
8.4.1	Sprint 1 (Foundation: Auth, Profiles, Basic Dashboards)	99
8.4.2	Sprint 2 (Customer Management: Connections + Lists)	100
8.4.3	Sprint 3 (Transaction Management: Normal + Credit + History)	101
8.4.4	Sprint 4 (Payments + Notifications + Reminders)	102
8.4.5	Sprint 5 (Chat + Analytics + PDF Reports + Loyalty)	103
8.4.6	Sprint 6 (Admin Panel + Support Tickets + AI Enhancements).....	104
8.5	Acronyms and Abbreviations	105
8.6	Market context of digital	105
8.7	Milestone Diagram	106
8.8	WBS.....	106
8.9	Reason for not selecting waterfall model	106
8.10	Reasons for not selecting Agile/Scrum methodology	107
8.11	Reasons for not selecting Prototype based methodology	107
8.12	Reason for selecting RUP	107

Table of Figures

Figure 1: Internet Penetration in Nepal.....	1
Figure 2: Karobar App Logo	9
Figure 3: Khatabook APP	9
Figure 4: OkCredit App.....	9
Figure 5: TallyPrime	9
Figure 6: (Garcia & Claour, 2021) - Mobile Receipt Scanner for Financial Management	11
Figure 7: (Septiawan et al., 2025) - Receipt Scanning with EasyOCR and LLM for Mobile Finance Apps.....	12
Figure 8: (Miklosik et al., 2021) - Chatbots in Digital Business Transformation.....	13
Figure 9: Waterfall model	14
Figure 10: Agile Methodology.....	15
Figure 11: Prototype based methodology.....	15
Figure 12: Process of RUP Methodology	16
Figure 13: System Architecture/Context Diagram	18
Figure 14: Entity Relationship Diagram	19
Figure 15: Collaboration Diagram – Make Payment.....	20
Figure 16: Collaboration Diagram – Manage Transaction	20
Figure 17: Collaboration Diagram – Register User.....	21
Figure 18: Collaboration Diagram – User Management	21
Figure 19: Collaboration Diagram – Realtime Chat.....	22
Figure 20: Sequence Diagram – Make Payment.....	22
Figure 21: Sequence Diagram – Manage Transaction	23
Figure 22: Sequence Diagram – Register User.....	24
Figure 23: Sequence Diagram – Dashboard and Analytics.....	25
Figure 24: Sequence Diagram – User Management	26
Figure 25: Sequence Diagram – AI Features	27
Figure 26: Sequence Diagram – Notification and Reminders.....	28
Figure 27: Sequence Diagram – Realtime Chat.....	29
Figure 28: Use case Diagram.....	30
Figure 29: Class Diagram.....	34
Figure 30: Activity Diagram – Transaction Management.....	35
Figure 31: Activity Diagram – Customer Management	36

Figure 32: Activity Diagram – Dashboard and Analytics.....	37
Figure 33: Activity Diagram – AI Features.....	38
Figure 34: Activity Diagram – Notification and Reminder	39
Figure 35: Dataflow Diagram.....	40
Figure 36: Color Palette	41
Figure 37: Hisab Khata Logo.....	41
Figure 38: Welcome Scree.....	42
Figure 39: Login Screen	43
Figure 40: Signup Screen for business	44
Figure 41: Signup Screen as Customer	45
Figure 42: OTP Verification Screen.....	46
Figure 43: Chat Screen	47
Figure 44: Business Favorite Active Screen.....	48
Figure 45: Business Favorite Passive Screen.....	49
Figure 46: All Added Business List Screen	50
Figure 47: Add Business Screen	51
Figure 48: Customer Profile Screen	52
Figure 49: Payment Gateway Screen.....	53
Figure 50: Notification Screen	54
Figure 51: Customer Dashboard Screen.....	55
Figure 52: Customer Profile Edit Screen	56
Figure 53: Add Transaction Screen.....	57
Figure 54: Add Transaction Options Screen	58
Figure 55: Add Customer Screen	59
Figure 56: Analytics Screen	60
Figure 57: Business Dashboard Screen.....	61
Figure 58: Customer Profile View.....	62
Figure 59: Business Profile Screen	63
Figure 60: All Connected Customers.....	64
Figure 61: Velocity Chart.....	65
Figure 62: Burndown Chart	66
Figure 63: Pre Survey Question 1	75
Figure 64: Pre Survey Question 2	75
Figure 65: Pre Survey Question 3	76

Figure 66: Pre Survey Question 4	76
Figure 67: Pre Survey Question 5	76
Figure 68: Pre Survey Question 6	77
Figure 69: Pre Survey Question 7	77
Figure 70: Pre Survey Question 8	77
Figure 71: Pre Survey Question 9	78
Figure 72: Pre Survey Question 10	78
Figure 73: Pre Survey Question 11	78
Figure 74: Pre Survey Question 12	79
Figure 75: Pre Survey Question 13	79
Figure 76: Pre Survey Question 14	79
Figure 77: Pre Survey Question 15	80
Figure 78: Pre Survey Question 16	80
Figure 79: Pre Survey Question 17	80
Figure 80: Pre Survey Question 18	81
Figure 81: Pre Survey Question 19	81
Figure 82: Pre Survey Question 20	81
Figure 83: Pre Survey Question 21	82
Figure 84: Pre Survey Question 22	82
Figure 85: Pre Survey Question 23	82
Figure 86: Pre Survey Question 24	83
Figure 87: Pre Survey Question 25	83
Figure 88: Pre Survey Question 26	83
Figure 89: Pre Survey Question 26	84
Figure 90: Pre Survey Question 27	84
Figure 91: Pre Survey Question 28	84
Figure 92: Pre Survey Question 29	85
Figure 93: Pre Survey Question 30	85
Figure 94: Pre Survey Question 31	85
Figure 95: Pre Survey Question 32	86
Figure 96: Milestone Diagram	106
Figure 97: WBS Diagram	106

Table of Tables

Table 1: Comparison of Similar Systems	10
Table 2: Progress Table	67
Table 3: Sprint 1 (Foundation: Auth, Profiles, Basic Dashboards)	99
Table 4: Sprint 2 (Customer Management: Connections + Lists)	100
Table 5: Sprint 3 (Transaction Management: Normal + Credit + History)	101
Table 6: Sprint 4 (Payments + Notifications + Reminders)	102
Table 7: Sprint 5 (Chat + Analytics + PDF Reports + Loyalty)	103
Table 8: Sprint 6 (Admin Panel + Support Tickets + AI Enhancements)	104
Table 9: Acronyms and Abbreviations	105

1 Introduction

1.1 Introduction to Topic

Small businesses shops, retail stalls make a very important branch of the economy of most developing states, including Nepal. But great numbers of these businesses are still dependent upon hand record-keeping, handwritten ledger, or unofficial notebook, to record credit sales, dues, payments and any other financial transactions. Human errors, the absence of standardization and poor availability of data plague such practices and hence the inability to control customer credit and evaluate the performance of business accurately. As more smartphones and mobile payment systems are penetrated, it is a better time to create a customized system to use by Nepalese small businesses (Khanal, 2025). The contribution of MSME or SME to the Nepalese economy is high. It contributes 80% of the employment, 70% of the gross value of exports and 90% of industrial GDP (Madan, 2020).

Hisab Khata is a project that is meant to create an elaborate digital credit and transaction management solution, which incorporates mobile applications and an administration web interface that will allow tracking credit and digital payments, analytics, and other innovative capabilities to bridge the gap between the old business model and the new fintech-based model of operation.

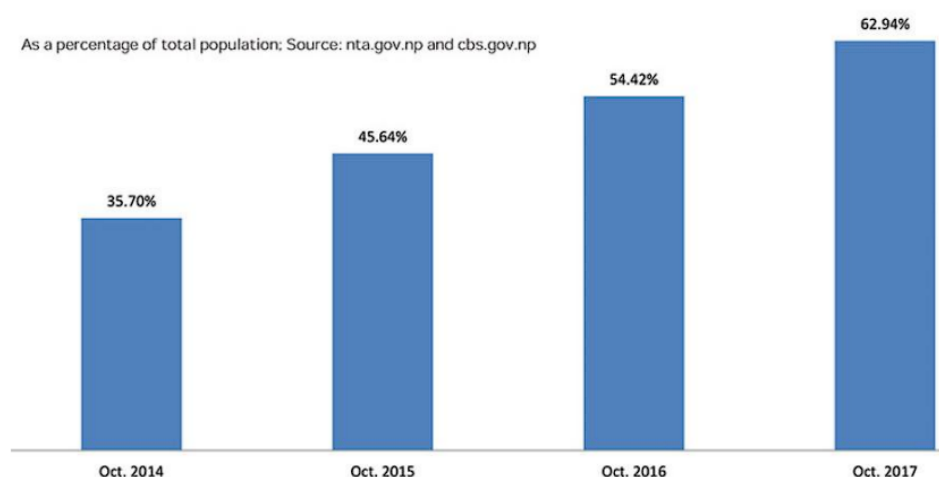


Figure 1: Internet Penetration in Nepal

About Market context of digital transformation [read more...](#)

1.2 Problem Scenario

Small businesses in the Nepal face significant challenges with their credit and Transaction management practices.

- **Manual Record-Keeping leads to errors:**
 - Paper-based records are used by many small firms. These are handwritten ledgers or spread sheets, which can be lost or misplaced, or contain errors or inconsistencies. Empirical research on SMEs around the world indicates that many of them do not keep proper accounting records (Qubbaja & Talahmeh, 2020).
- **Lack of transparency in credit which creates distrust:**
 - Without an official or shared system both customer and business shopkeepers may disagree about the current balance, transactions history, payment history and due amount and vice versa.
- **Manual payment collection is time consuming:**
 - The process of collecting payments either in cash or tracking partial payment manually is time consuming and inefficient.
- **Business insights are not-existent:**
 - Traditional record-keeping rarely provides proper information: e.g. total outstanding dues, who owes most, income vs expense breakdowns, or trends over time, and business owners can hardly make data-driven decisions.
- **Digital payment integration gap:**
 - Although digital payments are becoming popular in Nepal. A large number of small businesses in Nepal are not integrating it appropriately into their credit systems. The trend in Nepal with the increasing use of fintech among SMEs is a sign that the adoption is increasing, however, there are gaps in the adoption because of trust, literacy and infrastructure (Manoj Kumar Chaudhary et al., 2024).

1.3 Project as a Solution

These are some solutions to overcome the problems:

- **Digital transaction management:**
 - All credit and payment transactions are digitally recorded in real time, and this removes the errors of manual recording.
- **Transparent customer access:**
 - Customers have access to their own dues, payment history and transaction statements directly via their own mobile application and this means that misunderstandings are minimized and trust is established.
- **Digital payment integration:**
 - Integration with common Nepal payment gateways (e.g. Khalti and eSewa) to make integration of digital payments straightforward and avoid the use of cash.
- **AI powered efficiency:**
 - An automatic receipt scanning (photo) or voice-based data entry to accelerate the recording of transactions and reduce manual efforts.
- **Business intelligence dashboard:**
 - Analytics designed to allow shopkeepers to have a simple view of total outstanding dues, customers with highest/lowest dues, income and expenses, and business health overtime.
- **Loyalty point reward system:**
 - Encourage timely paying and frequently paying customers by Gazab Customer Points to build customer trust.
- **Localization for Nepal:**
 - The entire interface including currency, words, UI design, data and even dates are localized for Nepali users.

1.4 Aims and Objectives

1.4.1 Aims

The primary aim of the Hisab Khata is to develop a mobile-first digital credit and transaction management platform that enables small businesses in Nepal to efficiently track, manage, and collect credit transactions while providing complete transparency to customers.

1.4.2 Objectives

- To enable businesses and customers to record transactions and due.
- To allow customers to view their payment, transactions, and pay through the system.
- To develop Real-Time communication features between business and customer.
- To enable generation and download statements as PDF
- To provide admin function for user management, support and system monitoring.
- To provide a user-friendly interface that is easy to use and understand.
- To build AI powered mobile application for both shopkeepers and customers.

2 Background

2.1 Trend in the Field

In the developing economies, micro, small and medium enterprises (MSMEs) are rapidly adopting digital technologies in order to update their financial management systems. The emergence of fintech solutions around the globe has changed the approach of small companies in terms of accounting, credit tracking, and payments. This has been catalyzed by the high penetration of smartphones and expansion of mobile payment systems (eSewa and Khalti) in Nepal. (Bhattarai et al., 2023) discovered that digital payments have a strong positive impact on MSME performance through better performance of transactions and the decrease of the necessity to use cash (Kim, 2022).

On the same note, Kunwar and Chhetri also highlighted that fintech services have a strong enabling role in financial inclusion, especially in semi-urban and rural areas of Nepal (Kunwar & Chhetri, 2025). Their evidence indicates that mobile-based fintech solutions increase access to credit and assist business people to formalize their transactions. The trend is not limited to payments, digital bookkeeping, cloud-based ledges, and mobile accounting systems are now at the center of modern MSME activities in South Asia. Similar proofs in Indonesia demonstrate that the usage of digital bookkeeping in small companies enhances financial transparency and the quality of decisions (Rustendy et al., 2024).

The fintech ecosystem in Nepal is still immature with banks, telecoms, and startups partnering to provide hybrid services, i.e. a combination of payment, analytics, and bookkeeping. Such convergence puts tools such as Hisab Khata as indispensable digital enablers of MSMEs that require ready and available localized financial solutions.

2.2 Benefits of Hisab Khata

Hisab Khata appeals to the pain points experienced by Nepali MSMEs directly: their inability to have a structured accounting, their reliance on manual ledgers, and their very low level of financial visibility. It reduces human error since there is a tracking of income and expense which is automated and gives real-time updates that enhance trust between the owners of a business and customers. Research demonstrates that the use of fintech leads to the enhancement of operational efficiency and perceived professionalism of small enterprises (Bhattarai et al., 2023). The bilingual interface and the user-friendly nature of this system enables the non-technical users to create reports, access dues, and manage inventories without much training.

Additionally, the inclusion is enhanced by fintech platforms which link informal businesses to digital ecosystems. (Kunwar & Chhetri, 2025) showed that mobile based finance applications broaden the involvement in the formal economy particularly when localized in terms of language and payment integration. In following this wisdom, Hisab Khata fosters transparency and access of entrepreneurs of urban and rural Nepal.

2.3 Challenges

Though the benefits are evident, there are still issues with implementing digital financial systems. The main obstacle is the digital literacy. Nepali entrepreneurs do not have access to organized digital finance in many areas, particularly in the rural areas. Gangaram Biswakarma and team observed that in the banking industry of Nepal, even the user innovativeness has a lot of influence on adoption of fintech; people who have not been exposed to new interfaces are generally reluctant to switch to digital bookkeeping (Biswakarma et al., 2024).

The other issue is the privacy and confidence of the data. Small companies might be afraid of placing financial information on cloud repositories. Dwi Saraswati and team emphasized that infrastructure safety and apparent governmental assistance were the key to developing trust in digital transformation. Weak internet connectivity, maintenance, and access to technical assistance are also a constraint on the sustainability of fintech system in low-income environments. These barriers need to be overcome by the concerted efforts of the developers, regulators, and financial institutions in order to achieve reliability and data security (Saraswati et al., 2025).

2.4 Future Possibilities

The future of digital accounting tools such as Hisab Khata looks bright since Nepal is improving under the national Digital Nepal Framework. The increasing smartphone sales and the government digital-payment programs are broadening the prospects which allow MSMEs to use automated bookkeeping. The literature, including (Saraswati et al., 2025) notes that the incorporation of fintech adoption into the supportive policy frameworks improves the general quality of financial decisions.

New technologies will also increase the potential of Hisab Khata: artificial intelligence can provide predictive cost analytics, blockchain can provide security to transaction integrity, and data visualization can allow companies to comprehend patterns without their formal accounting training. The combination of fintech startups and established banks may establish financial ecosystems that are both informal and formal vendors and formal credit systems (Kunwar & Chhetri, 2025). With the growing digital literacy and the increased internet infrastructure, digital tools such as Hisab Khata will become normalized in the small-business financing sector.

2.5 Similar Real-World Projects

2.5.1 Karobar App

The Karobar App is one of the most downloaded bookkeeping applications in Nepal among the owners of small businesses. It enables users to take note of the sales made on daily basis, customer dues and generate reports in English and Nepali. The application is also integrated with the local payment systems, such as eSewa and Khalti, which allows Nepali shopkeepers to operate their accounts online (BizSphere , 2025).



Figure 2: Karobar App Logo

2.5.2 Khatabook, OkCredit, TallyPrime

In India, other companies that offer the same service are Khatabook and OkCredit. Both applications assist store owners to store credit data and remind customers to send them their money. Khatabook is an online based company whereas ok credit is offline based. On a bigger scale, TallyPrime is a professional accounting software applied in medium-sized firms, which also provides such functions as inventory and tax management but is more technologically skilled.



Figure 3: Khatabook APP



Figure 4: OkCredit App



Figure 5: TallyPrime

Hisab Khata is made tailored to Nepali MSMEs, compared to these systems. It takes the Karobar App and Khatabook features and introduces more intelligent features like expense analysis and AI responses. It is also local language based and the one which is linked to payment gateways of Nepal. This renders Hisab Khata more convenient and suitable to the local business requirements.

Table 1: Comparison of Similar Systems

Systems	Country	Target Users	Payment Integration	Language Support	Offline Functionality	Analytics Features
Karobar App	Nepal	Micro-retailers, MSMEs	eSewa, Khalti	English/Nepali	Yes	Limited reports
Khatabook	India	Small retailers	UPI-linked	English/Hindi	Partial	Basic reports
OkCredit	India	Micro-merchants	-	English/Hindi	Full	Minimal
TallyPrime	Global	SMEs / enterprises	Bank integration	English	No	Advanced
Hisab Khata	Nepal	Local MSMEs	eSewa, Khalti (APIs)	English/Nepali	Planned	AI & cash-flow analytics

2.6 Literature review (feature related)

2.6.1 OCR Receipt Scanning & Automated Financial Data Capture

Relevance: Supports the receipt scanning feature in your app where transaction details are auto-filled from photographed receipts (Garcia & Claour, 2021) (Septiawan et al., 2025).

Mobile Bookkeeper: Personal Financial Management Application with Receipt Scanner Using Optical Character Recognition

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Abstract:

Personal financial management is undeniably a worthwhile practice to establish financial security during a struggling economy and make intelligent monetary decisions regardless of the plethora of spending temptations. Monitoring personal cash flow is part of achieving financial independence, and it is now undemanding to perform because of the available personal budget apps and finance tools. Nevertheless, a missing feature of these technology-driven innovations is the recording, tracking, and monitoring of receipts as well as the generation of personal expenses reports based on these collected pieces of paper. With this application, "*Mobile Bookkeeper*", financial enthusiasts can just scan the receipt using the inbuilt camera of any smartphone and details will be automatically transcribed using Optical Character Recognition (OCR). To measure the satisfaction and test the usability of the mobile app, subjective and objective measures via ISO 25062 and ISO 9241 standards were collected, and QUIS 7.0 questionnaire, respectively. The testing results established *Mobile Bookkeeper* particularly on its receipt scanner feature as a needed mobile finance app. Together with this acceptance is the report highlighting issues and challenges in developing such a mobile application, especially with OCR integration and its accuracy in text recognition.

Keywords:

Bookkeeping, Mobile Application Development, Optical Character Recognition, Personal Financial Management

Reference to this paper should be made as follows:

Garcia, M. B., & Claour, J. P. (2021). Mobile Bookkeeper: Personal Financial Management Application with Receipt Scanner Using Optical Character Recognition. *2021 1st Conference on Online Teaching for Mobile Education (OT4ME!)*. DOI: 10.1109/OT4ME53559.2021.9638794.

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Figure 6: (Garcia & Claour, 2021) - Mobile Receipt Scanner for Financial Management

Receipt Scanning with EasyOCR and ChatGPT-4o in a Mobile Finance App: an Agile Kanban Approach**M. Fiqry Septiawan^{1✉}, Siska Anraeni², Ramdaniah³**^{1, 2, 3} Informatics Engineering, Computer Science, Universitas Muslim Indonesia, Indonesia[✉] Corresponding Author : M.Fiqry Septiawan (e-mail: fikrisep130902@gmail.com)

Article Information	ABSTRACT
Article History Received : August 07, 2025 Revised : August 26, 2025 Published : October 04, 2025	Technological advancements have provided convenience for Generation Z in managing finances; however, many are still not accustomed to recording their financial activities regularly. Shopping receipts, which should serve as proof of transactions, are often ignored or poorly managed, despite their important role in tracking expenses. Therefore, this research aims to develop an Android-based financial recording application capable of handling both manual input and automated recording through receipt scanning using Optical Character Recognition (OCR) technology. The findings indicate that ChatGPT-4o significantly outperforms EasyOCR by providing more consistent accuracy and faster, stable processing, making it a more reliable solution for receipt-based financial recording. Developed using the Agile Kanban method, the application was validated through alpha testing and proven to function properly across all features. Beyond practical benefits for users, this research also contributes to the financial technology literature by demonstrating the integration of large language models (LLM) to enhance OCR performance in mobile finance applications.
Keywords: Agile Kanban; Finance Tracking; Generation Z; OCR; Receipt Scanning.	

INTRODUCTION

Low financial literacy remains a major challenge in Indonesia, including among Generation Z, who tend to have consumptive behavior and adopt the You Only Live Once (YOLO) principle, which is to live for the present without thinking too much about the future (Asrun & Gunawan, 2024; Lohita et al., 2022). Financial literacy is the ability to understand financial concepts and apply that knowledge effectively to manage income, expenses, and plan personal budget (Echchakoui & Heppell, 2023). Good literacy can help individuals maintain a balance between income and expenses, plan budgets wisely, and support more accurate financial decision-making (S. Bhuvanapriya et al., 2024; Ulifah, 2024).

However, financial record-keeping is often considered impractical because it requires time and attention to detail, especially when done manually through simple note-taking apps or digital spreadsheets (Felia Putri & Nurlaila, 2022). To support better financial literacy, accurate transaction recording through physical evidence such as shopping receipts plays a crucial role in monitoring an individual's financial condition (Kosadi et al., 2021). In this context, receipt scanning with OCR technology offers a more efficient alternative to manual input, providing automated extraction of transaction details that can be integrated into personal finance applications.

Receipts serve not only as proof of transactions but also as an essential means of monitoring expenses and making price comparisons (Matsui et al., 2023; Sari et al., 2023). However, many individuals neglect to store or manage receipts properly, which leads to inaccurate

Figure 7: (Septiawan et al., 2025) - Receipt Scanning with EasyOCR and LLM for Mobile Finance Apps

2.6.2 AI & Chatbot Integration in Business and Financial Apps

Relevance: Validates the *Byapar d-AI chatbot* feature by situating it within broader research showing the business value and adoption of chat-oriented AI systems (Miklosik et al., 2021).

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The Use of Chatbots in Digital Business Transformation: A Systematic Literature Review

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This paper is an output of RTIS seed funding grant from UniSA for a project on Digital Transformation of the Superannuation industry in Australia.

ABSTRACT The research on chatbots has gained momentum over the past few years. Academics and practitioners investigate how these tools for communication with customers or internal team can be improved in terms of their performance, acceptance, and deployment. Although there is a plethora of recent studies available, not all of them deal with the digital business transformation implications of chatbots. The main aim of the research presented in this paper was to conduct a systematic literature review of high-quality journal research papers in order to summarise the current state of research on chatbots, identify their role in digital business transformation and suggest the areas warranting further attention. 74 papers were included in the research. Topical (focus and applications), methodological (methods used, sample size, sample type, and countries studied) and bibliometric (publication outlet, citations, and Altmetric Attention Score) aspects are evaluated and described. Scholars and practitioners can use the results to identify topics, areas, and applications that are intensely discussed in the literature and require further attention, select a methodology for their research that is well established in the field or is emerging, identify the most influential publications not to be missed in their research or identify publication outlets for publishing their research on chatbots.

INDEX TERMS AI, chatbots, conversational agents, digital business transformation, digital disruption, ML

I. INTRODUCTION

We are currently witnessing an immense technological development, resulting in the need for organisations to adopt new technologies and information systems (IS) [1]. The technologies disrupt organisations' business processes, and trigger the need for digital transformation (DT) [2]–[4]. The need for DT has been reflected across all industries, including manufacturing (industry 4.0), retail, logistics, and services [5]–[8].

One category of IS with increasing capabilities are artificial intelligence (AI)-powered systems. Such systems have many potential applications in decision making support, production automation, learning, communication, etc [9]. The communication between online users and organisations is shifting towards interactions with AI-driven systems [10]. A chatbot is one example of technology that is used in computer-mediated communication where AI agents increasingly occupy roles once served by humans [11]. Not all chatbots are developed as Machine Learning (ML) or AI driven chatbots [12], but the advantage of deploying AI driven chatbots is that they give the impression of being intelligent as they get smarter with increased data and user interactions [13].

Chatbots can be defined as a 'software that accepts natural language as input and generates natural language as output, engaging in a conversation' [14]. Another definition accentuates their attempted human-like character: 'Chatbots are interactive virtual characters whose mission is to assist people in high-profile environments.' [15]. Apart from engaging in written conversations (text-based

chatbots), chatbots also have the ability to mimic human speech (voice-based chatbots) to improve user experience and cultivate customer loyalty [15], [16].

Chatbots can be found on websites, social media or instant messaging apps [17], [18]. They can be deployed within an organisation to assist with various services and processes such as internal support systems, IT Service Management (ITSM), learning or human resources management (HRM) [15], [19]–[24].

For external communication, standalone chatbots can also represent an alternative to branded websites [25]. They have been deployed to provide services in many areas such as customer relationships management (CRM), customer service or sales and marketing [26]–[30]. Chatbots are used to make product or service recommendations regarding shopping, financial or health related decisions [25], [31]–[34]. Researchers are, amongst others, focusing on investigating how to build better social bots for interaction in business or commercial environments, how to improve services with chatbots, which factors affect user perceptions of chatbots or how to encourage repetitive use of chatbots [35]–[42].

II. OBJECTIVES OF THE STUDY

Researchers have been examining the various uses of chatbots, the factors affecting their acceptance by users, and the creation of new algorithms and frameworks for chatbot deployment to increase their efficiency. The number of studies on chatbots has increased significantly over the past

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Figure 8: (Miklosik et al., 2021) - Chatbots in Digital Business Transformation

2.7 Review of the Technical Aspects

2.7.1 Tech Stack

- **Frontend:** Flutter
- **Backend:** Django
- **Database:** MySQL
- **Other Tools:** Git/GitHub

2.7.2 Packages and Libraries

- **State Management & Architecture:** flutter_bloc, equatable, injectable
- **Functional Programming & Data Handling:** dartz, intl
- **Networking & Connectivity:** http, internet_connection_checker
- **Local Storage & Media:** shared_preferences, image_picker
- **Design & UI:** cupertino_icons
- **Artificial Intelligence & Computer Vision:** google_ml_kit
- **Voice Recognition:** speech_to_text

2.8 Methodology

2.9 Considered Methodology

2.9.1 Waterfall Model

The Waterfall model is a traditional software development model, based on the plan. It is a series of clearly defined stages, and one stage should be successfully finished before another is started (Adobe for Business Team, 2025).

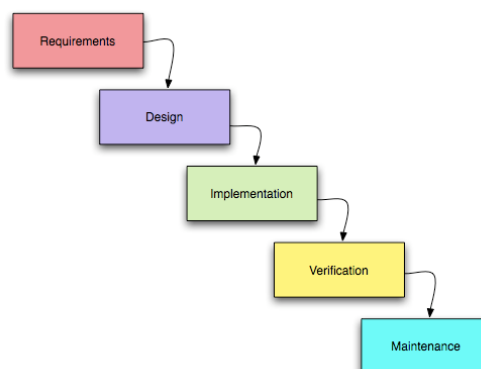


Figure 9: Waterfall model

Reasons for not selecting waterfall model...

2.9.2 Agile/Scrum

Agile is a mixture of software development methods that are found on iterative and incremental development and focus on collaboration, flexibility, customer feedback and adaptive planning (Laoyan, 2025).



Figure 10: Agile Methodology

Reasons for not selecting Agile/Scrum methodology:

2.9.3 Prototype-Based / Iterative-Prototyping Methodology

A prototype-based or iterative-prototyping methodology places developers in a situation where they create rough prototypes or mockups at the initial stages, collects user feedback, refines requirements and then creates successive versions in an iterative way (Indeed Editorial Team, 2025).

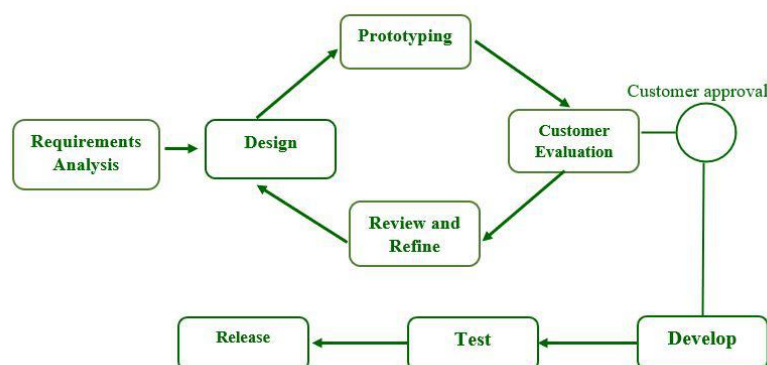


Figure 11: Prototype based methodology

Reasons for not selecting Prototype based methodology:

2.10 Selected Methodology – Rational Unified Process (RUP)

RUP is a phase-based process framework of software development, which is iterative. It breaks down the development into four key stages: Inception, Elaboration, Construction, and Transition, and promotes the iterative development of each of the stages, which enables refinement, risk reduction, and incremental delivery (Biscontinini, 2024).

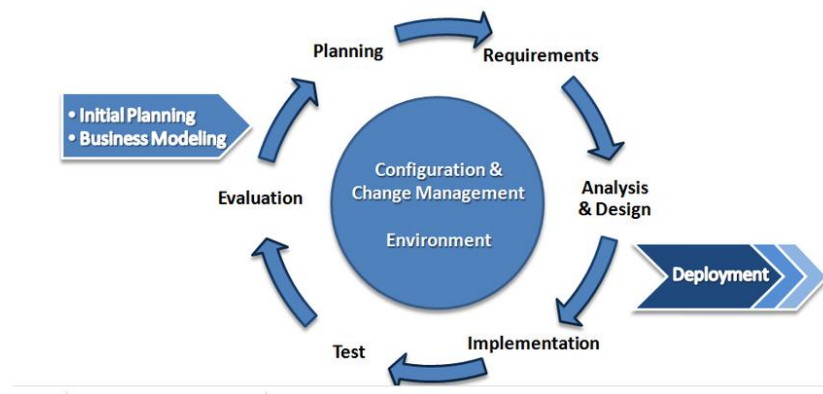


Figure 12: Process of RUP Methodology

Reasons for selecting RUP Methodology:

Considering the nature and scope of Hisab Khata, which includes several modules (mobile apps with various roles, an administrative panel), complex features and the necessity of reliability, maintainability, and scalability, a pure Waterfall would be overly rigid, a pure Agile too unpredictable and light in documentation, and prototyping would be too shallow and risky in terms of architectural integrity. [More..](#)

3 Development till Date

3.1 Software Requirements Specification (SRS)

Hisab Khata is mobile and web-based credit and transaction management system available to Nepalese small and medium-sized businesses. It transforms the old Baki Khata platform and allows companies to enter the sales, credit customer accounts, and track payments and real-time analytics. The platform allows making digital payments on eSewa and Khalti, provides an instant messaging system, automatic alerts, and AI-driven functionality, including support in scanning receipt codes and recording transactions via voice recognition. Hisab Khata, constructed on the security, reliability, and localization (Nepali and English), improves the transparency, lessens the manual errors, and improves the financial decision-making by the shopkeepers and customers.

[Read More...](#)

3.2 Product Backlog

Since I Have use RUP as a methodology, it does not have backlog as a sprint but still I have done it representing RUP phage as a sprint. [See complete sprint...](#)

3.3 Pre Survey Description

According to the answers provided to the questionnaire, it is evident that a significant number of customers often purchase products on credit in various shops, and the majority of them rely on paper khatas but demand digital solutions. The majority of them want an application to keep track of payments, get notifications, and pay directly. The customers would like to have a Nepali and English interface. The owners of shops are also interested in having an app to control credit and online payments. Both of the groups welcome AI functionality to enhance following and notifications. For more detailed view, [Click Here...](#)

3.4 System Design

3.4.1 System Architecture/Context Diagram

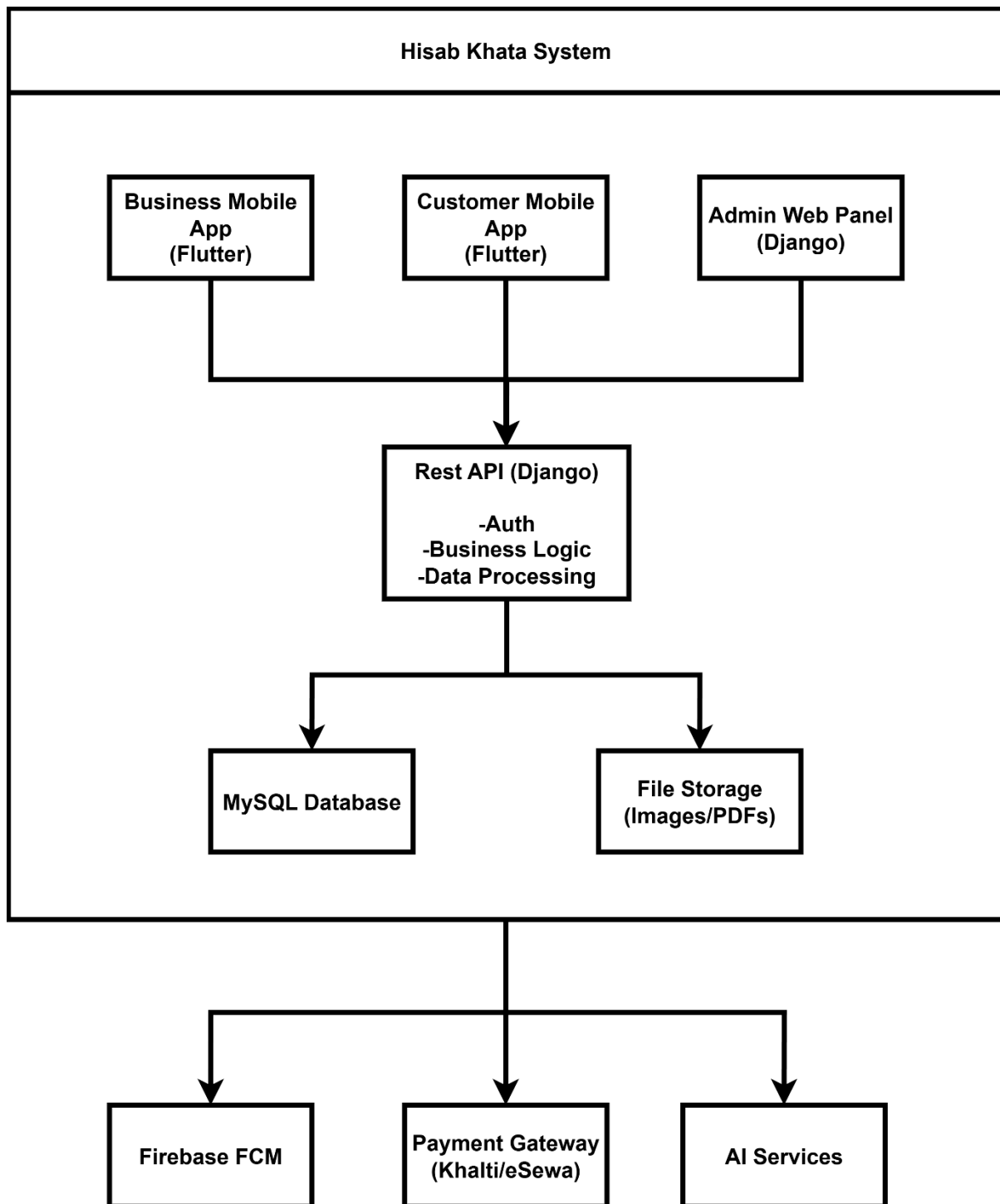


Figure 13: System Architecture/Context Diagram

3.4.2 ERD

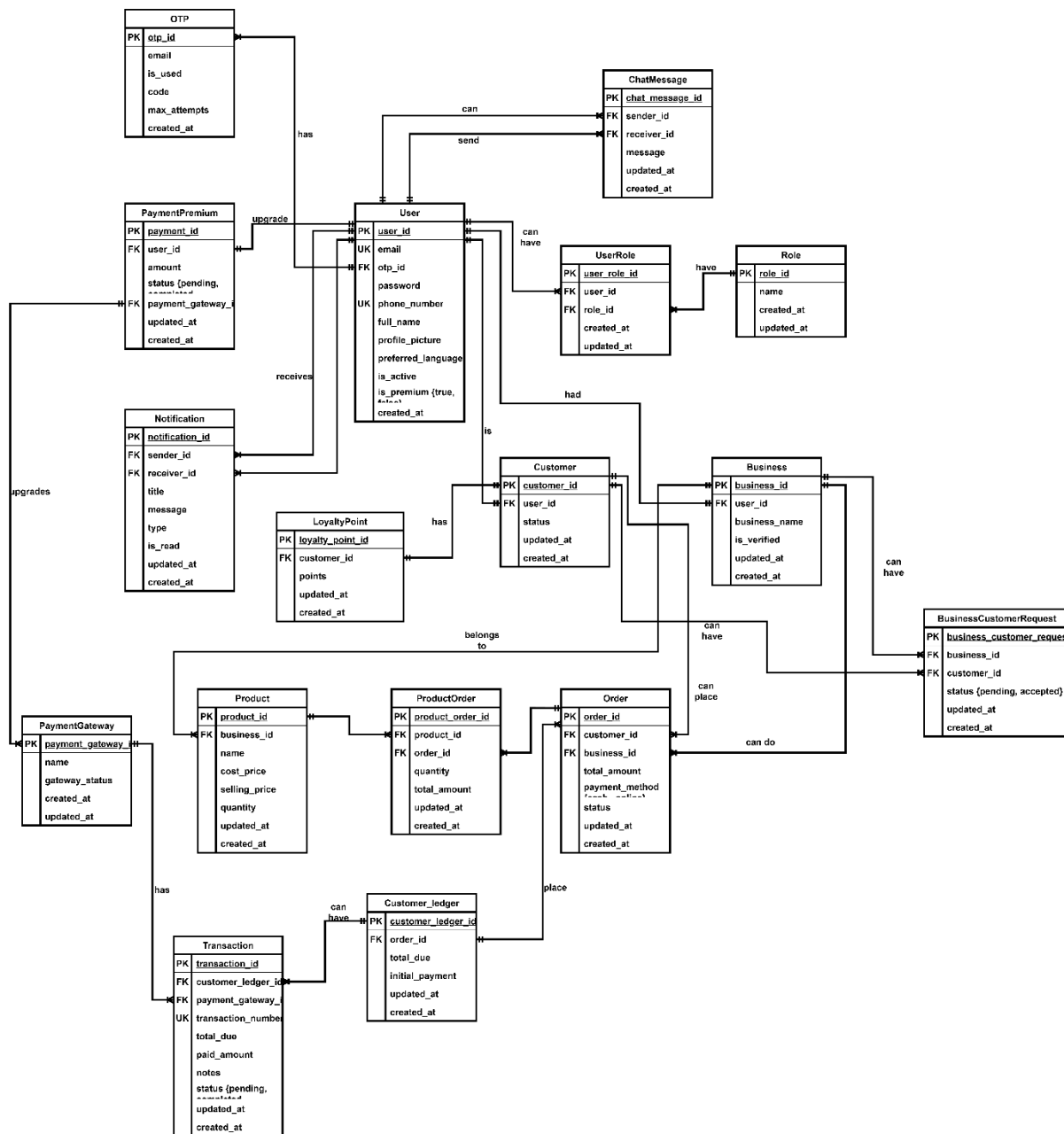


Figure 14: Entity Relationship Diagram

3.4.3 Collaboration Diagrams

3.4.3.1 Collaboration Diagram – Make Payment

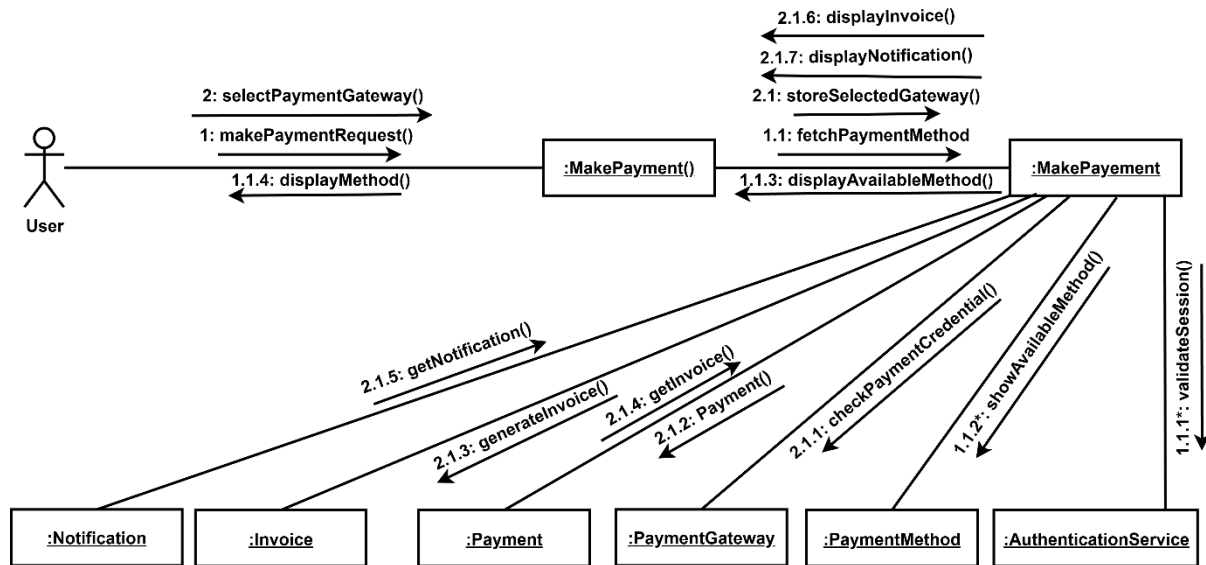


Figure 15: Collaboration Diagram – Make Payment

3.4.3.2 Collaboration Diagram – Manage Transaction

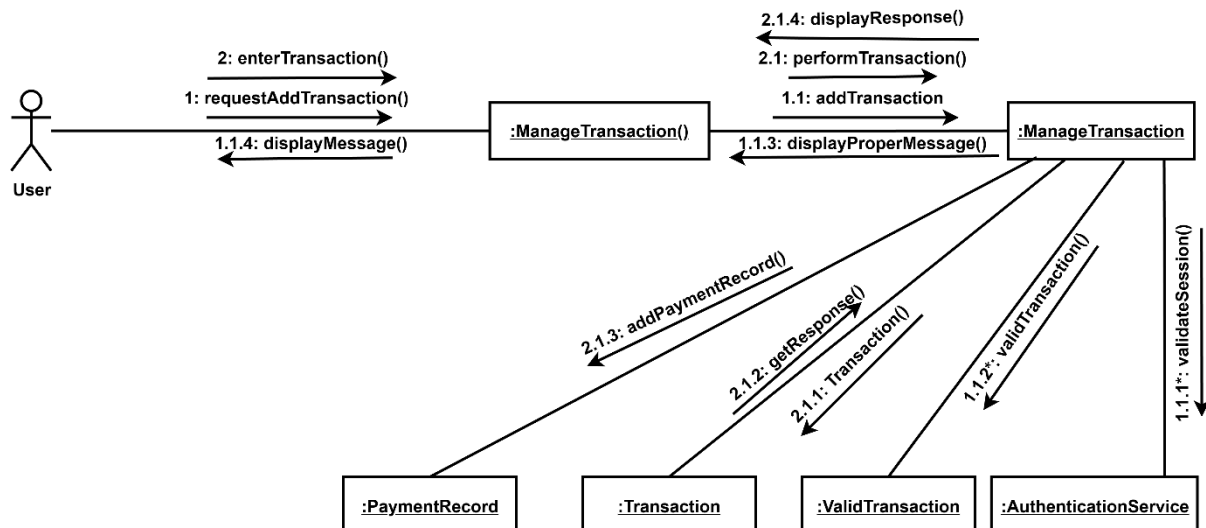


Figure 16: Collaboration Diagram – Manage Transaction

3.4.3.3 Collaboration Diagram – Register User

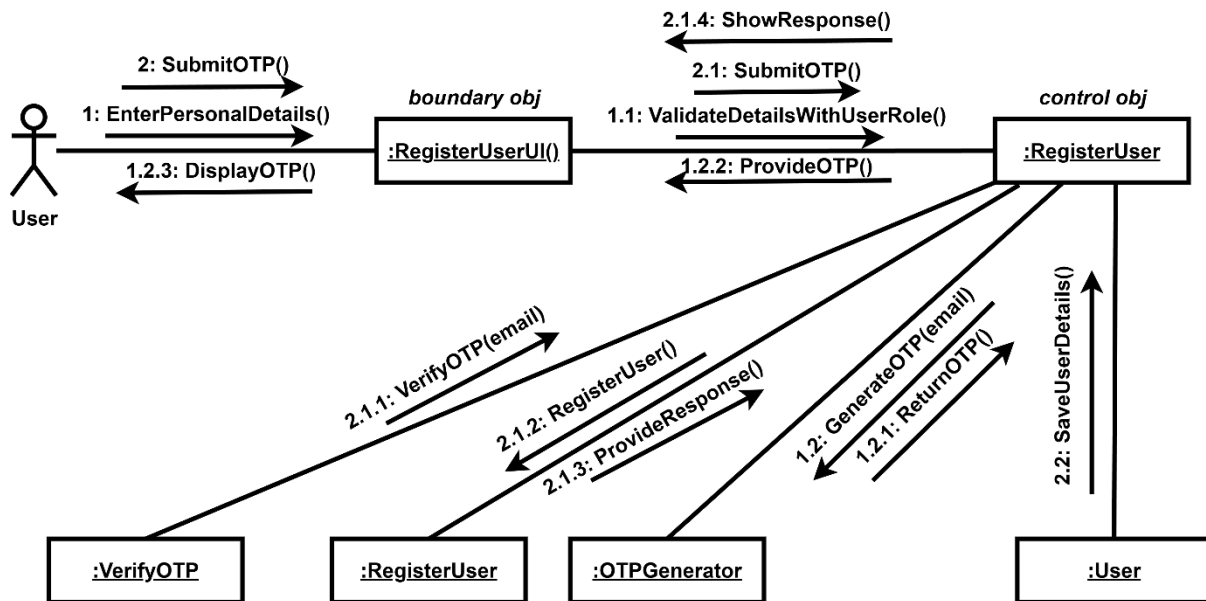


Figure 17: Collaboration Diagram – Register User

3.4.3.4 Collaboration Diagram – User Management

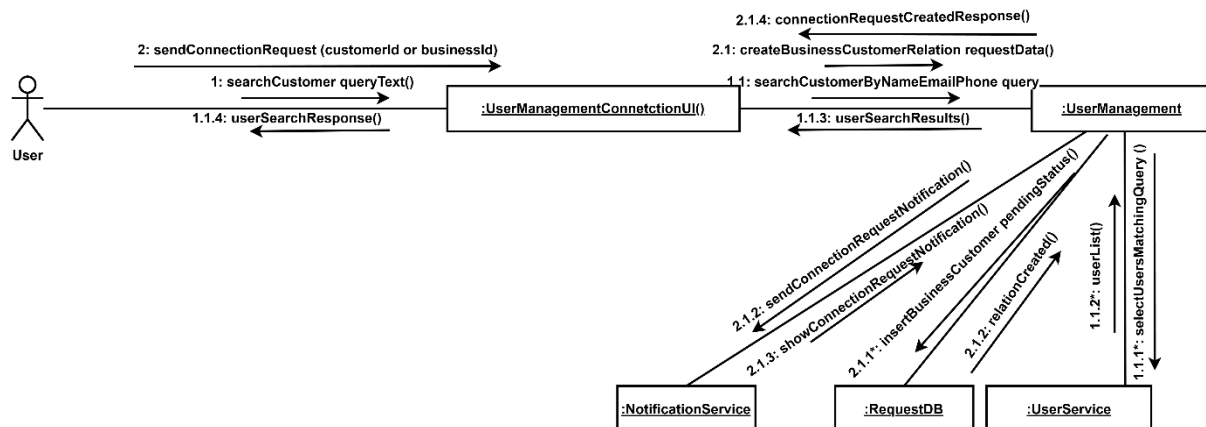


Figure 18: Collaboration Diagram – User Management

3.4.3.5 Collaboration Diagram – Realtime Chat

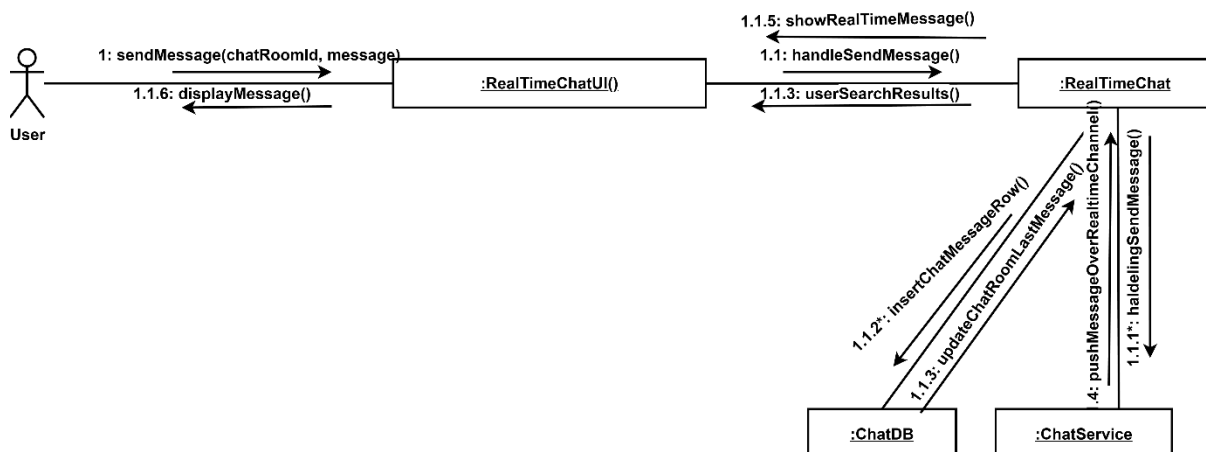


Figure 19: Collaboration Diagram – Realtime Chat

3.4.4 Sequence Diagrams

3.4.4.1 Sequence Diagram – Make Payment

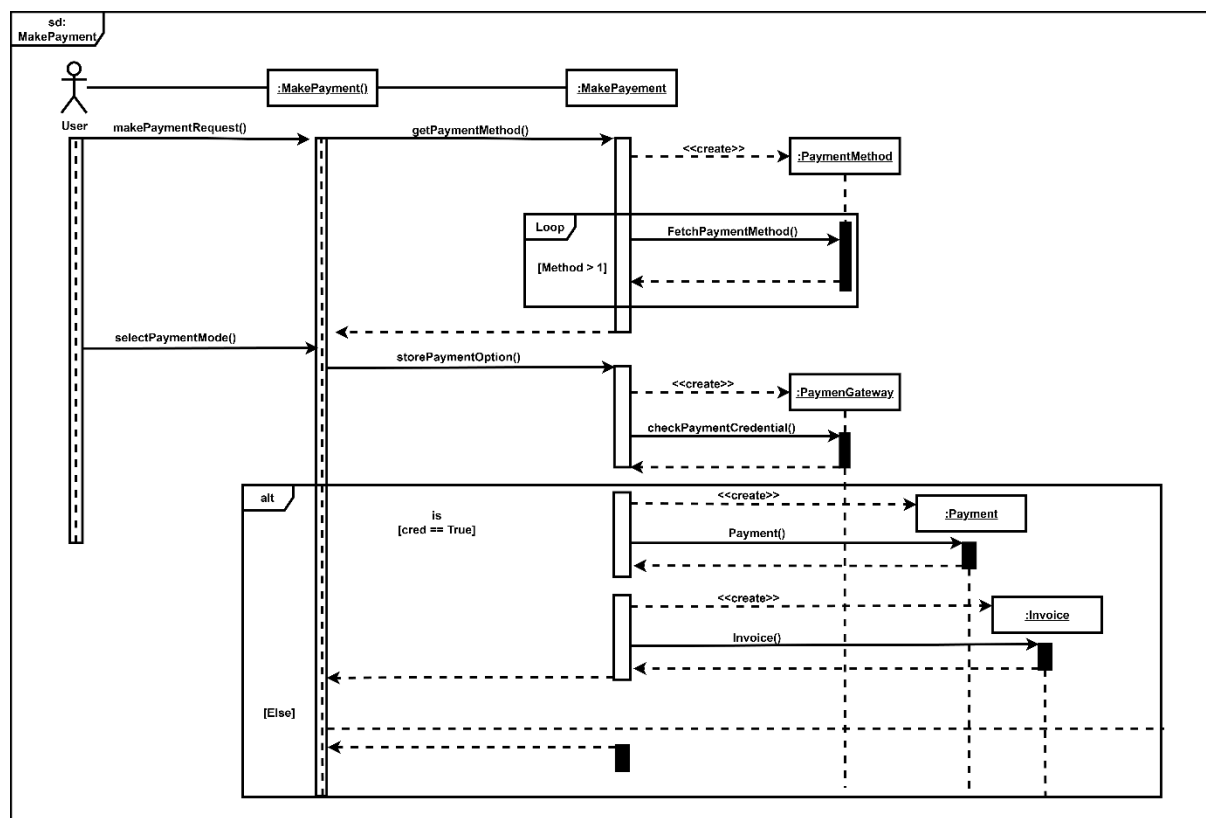


Figure 20: Sequence Diagram – Make Payment

3.4.4.2 Sequence Diagram – Manage Transaction

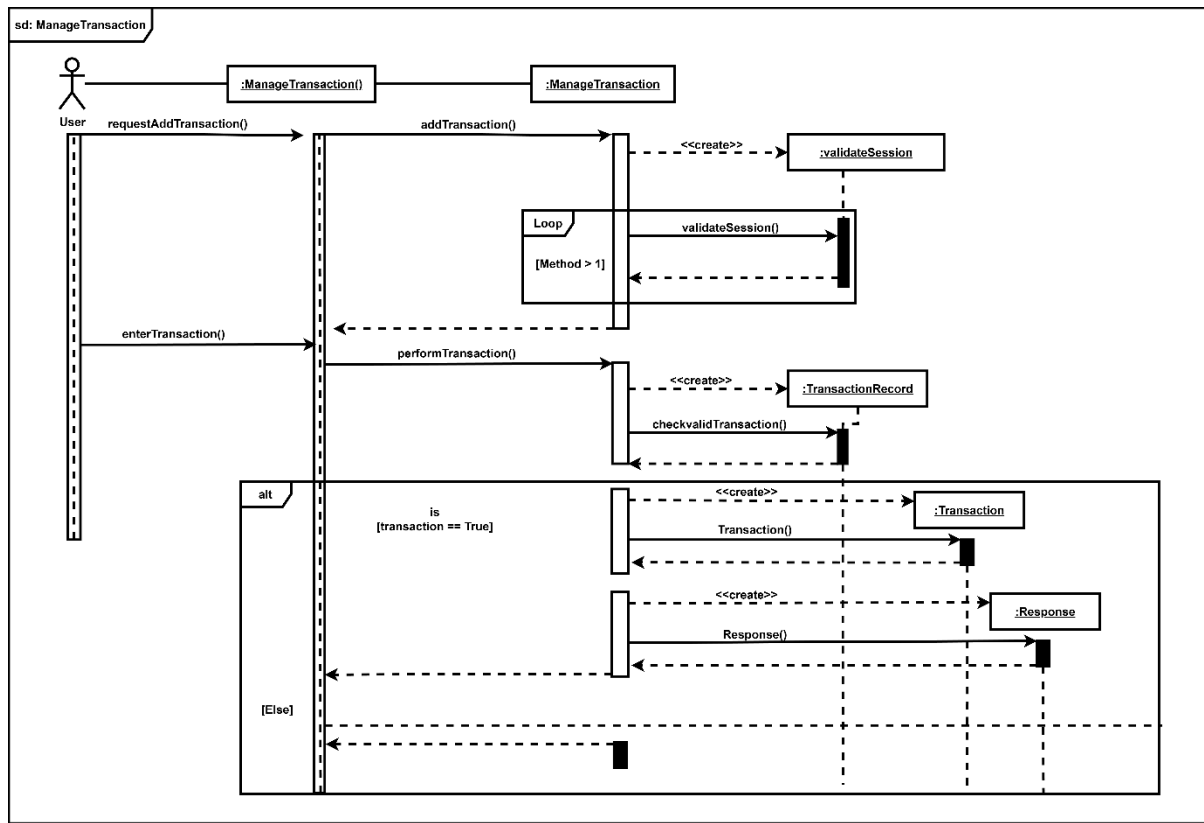


Figure 21: Sequence Diagram – Manage Transaction

3.4.4.3 Sequence Diagram – Register User

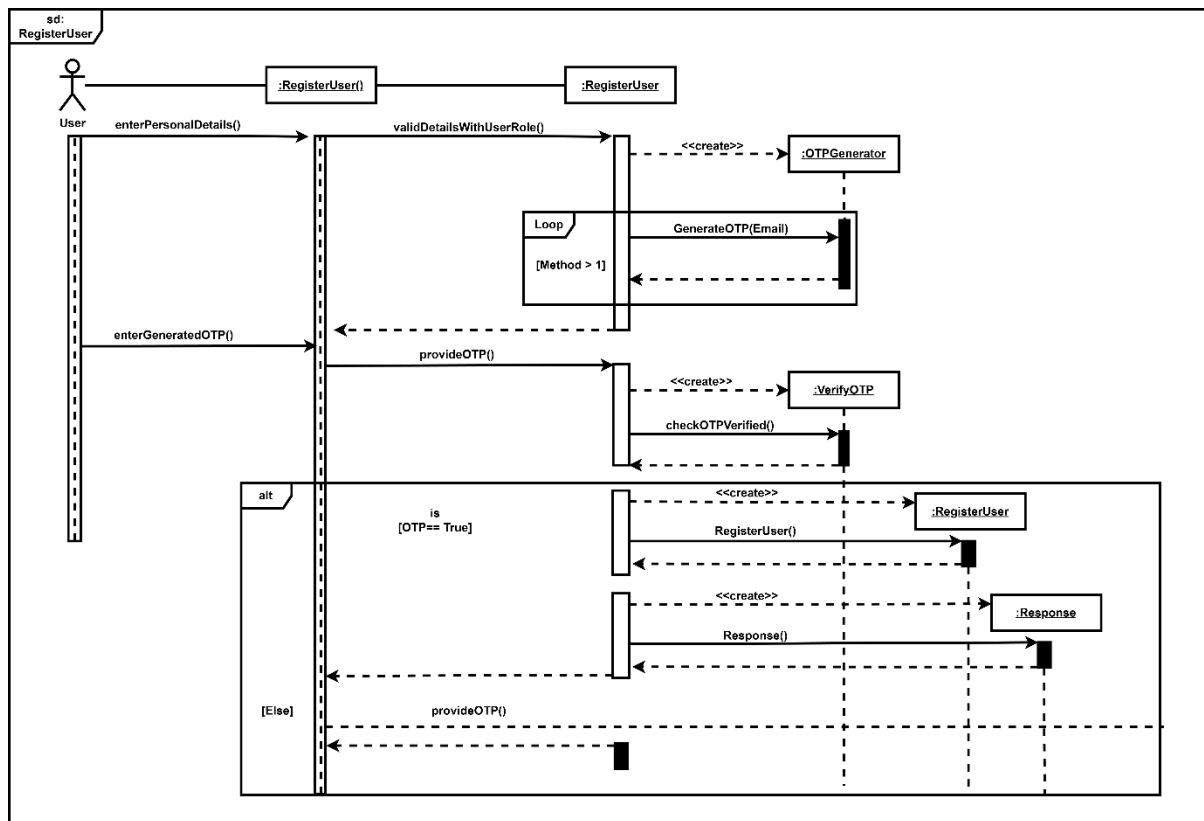


Figure 22: Sequence Diagram – Register User

3.4.4.4 Sequence Diagram – Dashboard and Analytics

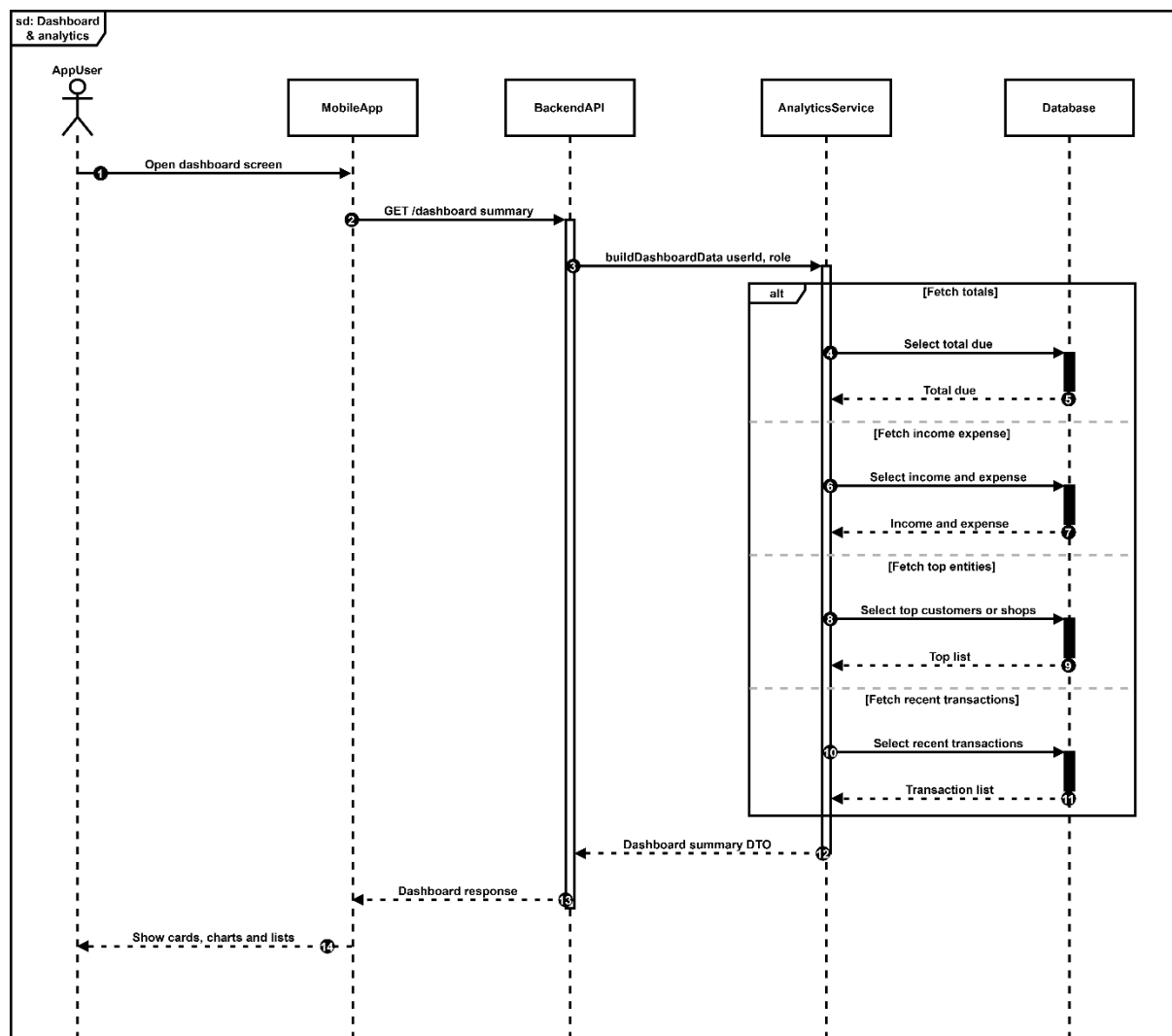


Figure 23: Sequence Diagram – Dashboard and Analytics

3.4.4.5 Sequence Diagram – User Management

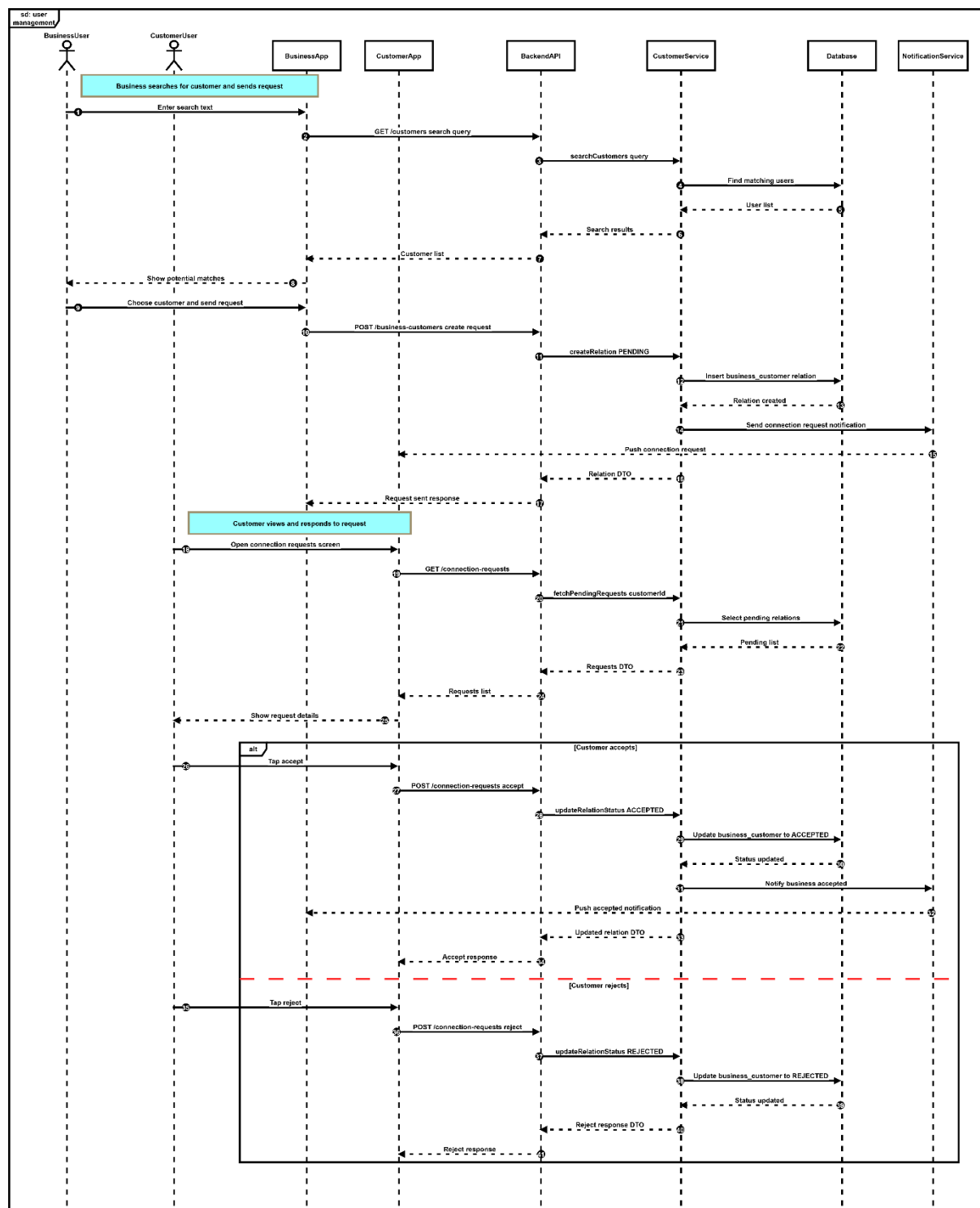


Figure 24: Sequence Diagram – User Management

3.4.4.6 Sequence Diagram – AI Features

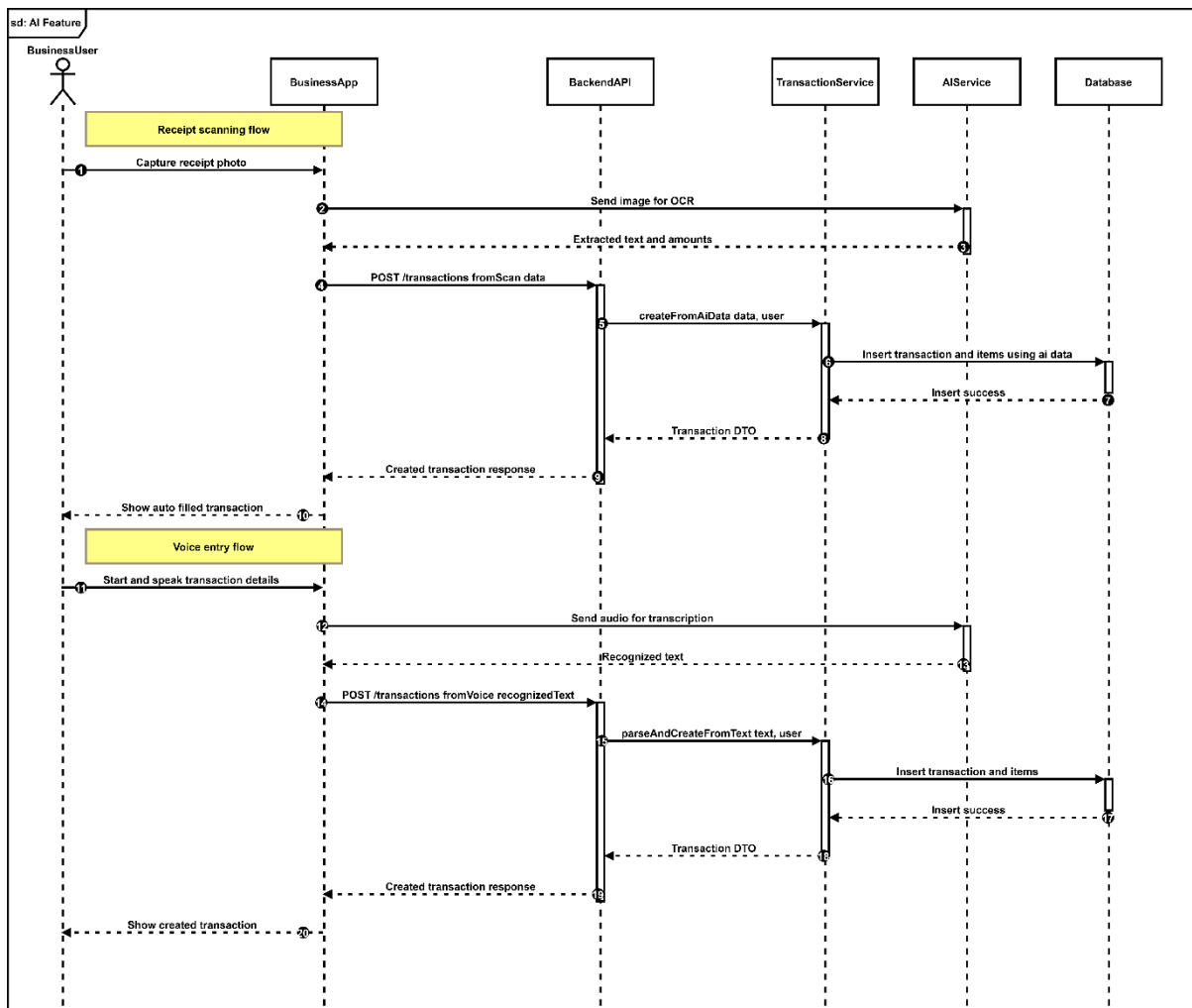


Figure 25: Sequence Diagram – AI Features

3.4.4.7 Sequence Diagram – Notification and Reminders

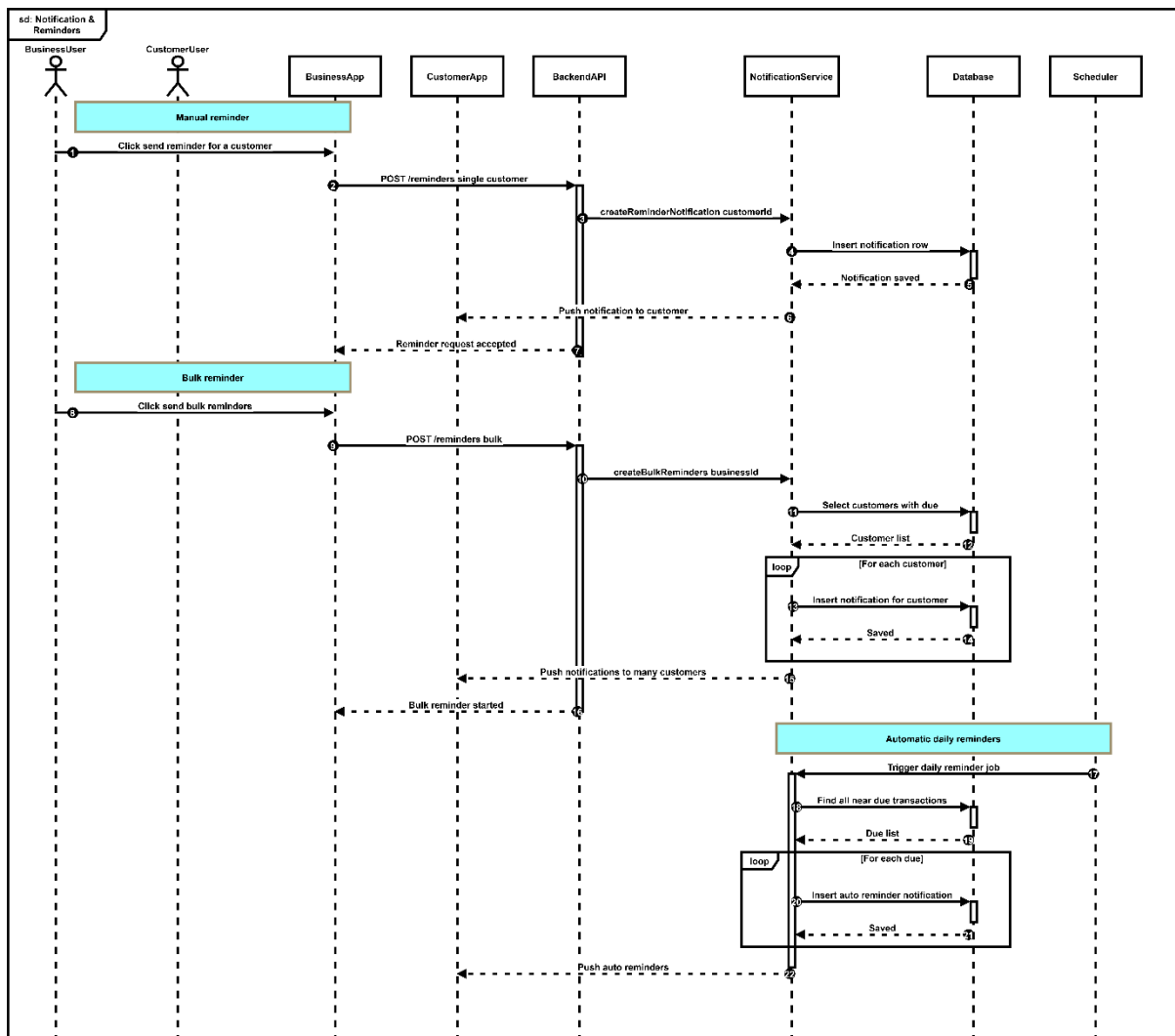


Figure 26: Sequence Diagram – Notification and Reminders

3.4.4.8 : Sequence Diagram – Real time Chat

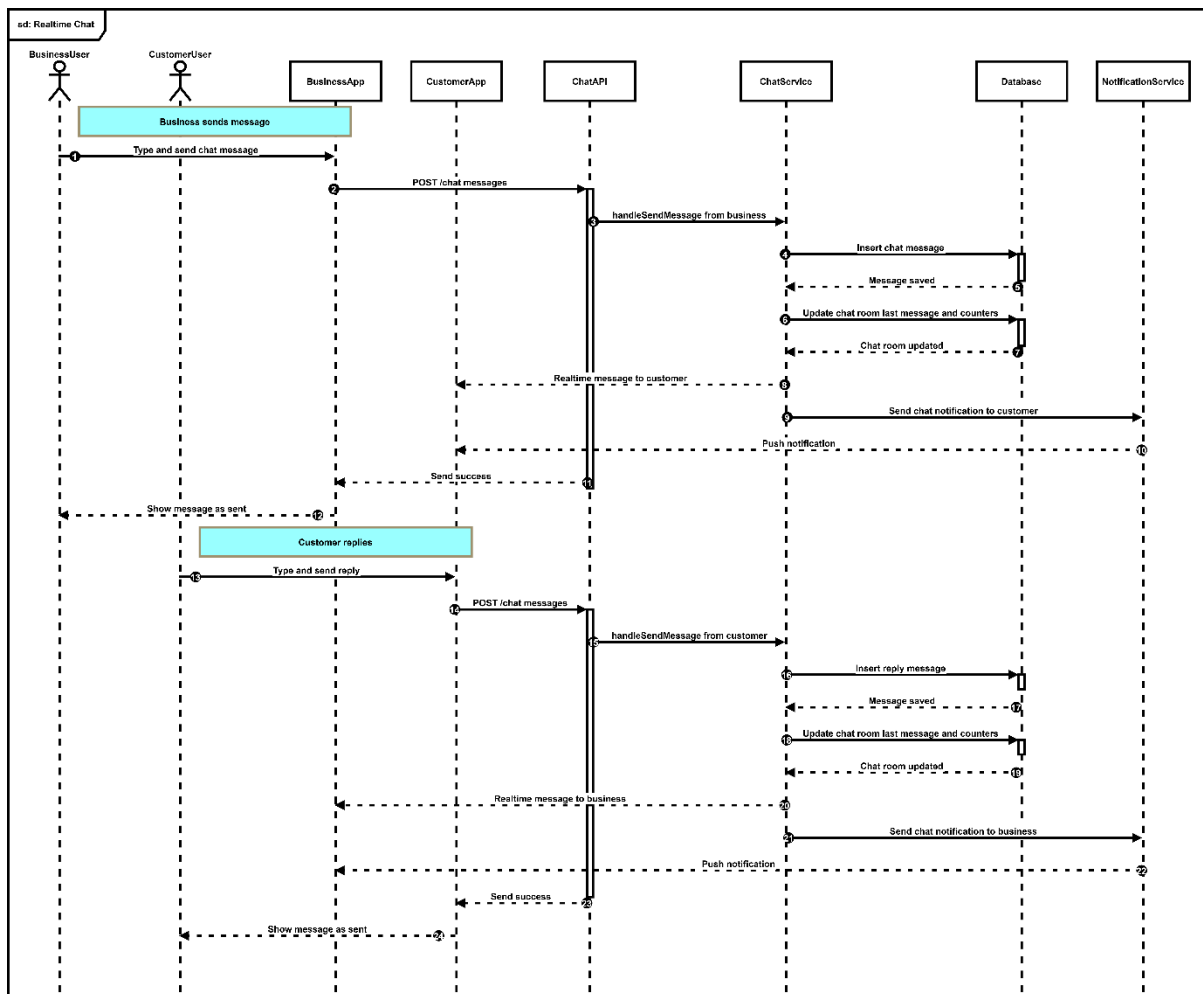


Figure 27: Sequence Diagram – Realtime Chat

3.4.5 Use case Diagram

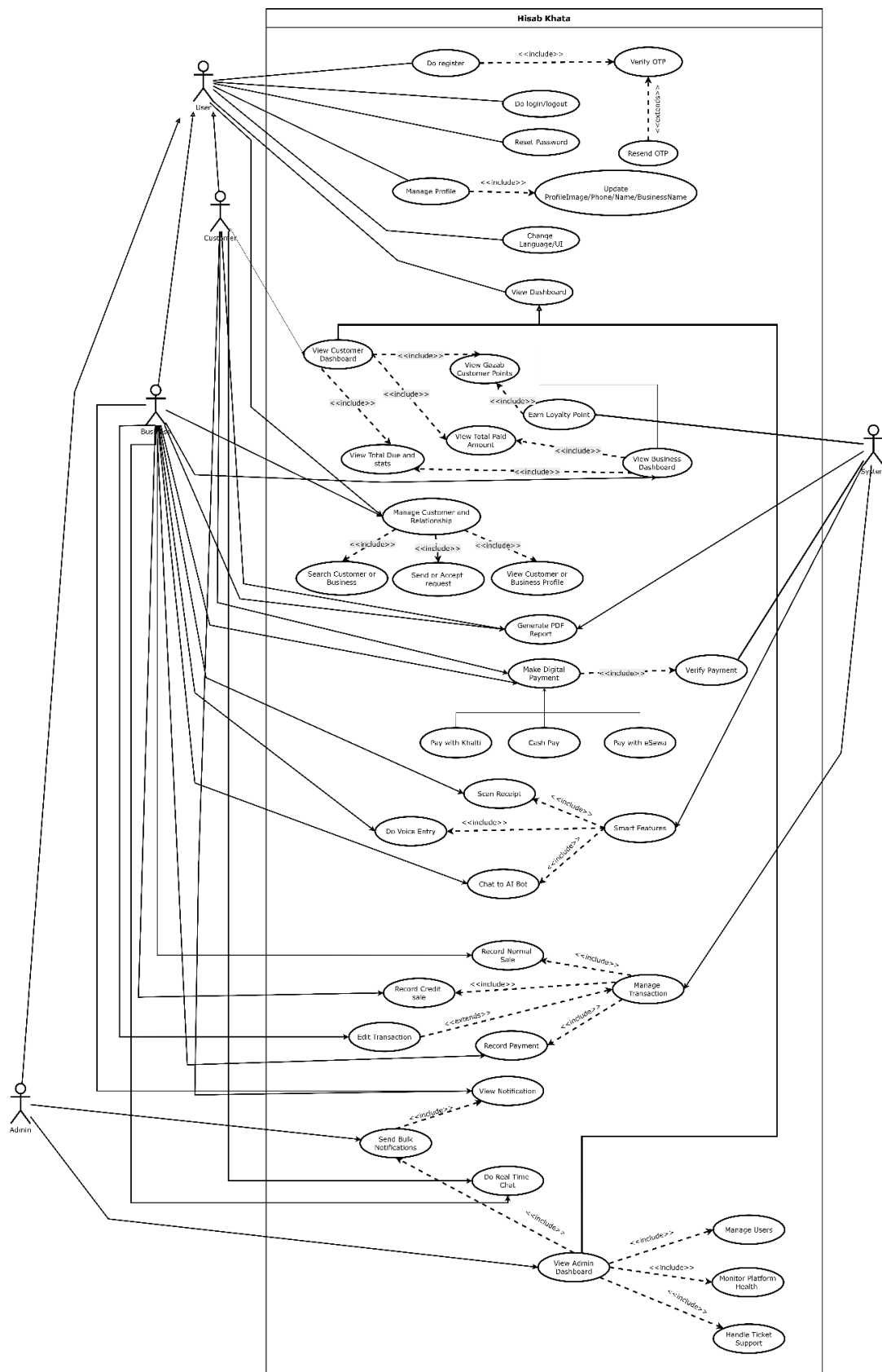


Figure 28: Use case Diagram

Higher Quality Image: [link](#)

High level Description (Hisab Khata)

Actors:

- **User:** This is a standard user of the system who can register, log in/out, create profile, change language, and be able to view the main dashboard depending on the role.
- **Customer:** A user that is connected to businesses, who sees dues and payment analytics, who makes payments, who sees notifications, who creates reports, and uses smart features.
- **Business:** A customer and relationship manager, sales and payment records, due and paid analytics, notification, and communication with customers.
- **Admin:** An administrative user that oversees the platform, controls users, sends bulk messages, and provides support of tickets.

Use Cases (with descriptions)

- **Do Register:** This is what enables a user to make a new account in the system.
- **Verify OTP:** authenticates the registration of the user based on OTP to verify account creation.
- **Resend OTP:** Re-sends OTP where the expiry or non-receipt of the earlier OTP has taken place.
- **DoLogin/Logout:** This gives a user an option of signing in and out of the system.
- **Reset Password:** allows the user to re-enter the account in case of forgetting the password.
- **Manage Profile:** Gives the user the opportunity to view and update his profile information.
- **Profile (Image/Name/ Number/ Business Name):** Updates profile information including profile picture, name, phone number and business name.
- **Change Language:** This is to enable the user to alter the language used in the app to enable easier access.

Dashboard & Analytics

- **View Dashboard:** This displays the user-level overview screen depending on the role of the user.
- **View Customer Dashboard:** Shows customer-specific dashboard, such as dues, payments and associated businesses.
- **View Graphs (Customer Paid Graph):** Displays the graphics of the customer payments and dues.
- **Visit Business Dashboard:** Customers due and payment insights - Business dashboard.
- **View Total Due and Stats:** Shows total due amount and the corresponding summary statistics.
- **View Total Paid Amount:** Displays the total paid amount to track and be transparent.
- **View Loyalty Point:** Shows the number of points the customer has earned in terms of loyalty.

Customer-Business Relationship

- **Manage Customer and relationship:** Stores customer-business relationship and customer-business relationship information.
- **Search Customer or Business:** Assists users with the connection and transactions with a customer or business.
- **Send or Accept Request:** Before the connections, the entire request (send, receive, accept/reject) is supported.
- **View Customer or Business Profile:** This feature allows users to see the profile of the connected customers or businesses.

Transactions & Payments

- **Manage Transaction:** Core use case to manage all the operations of the transactions such as sales and payments.
- **Record Normal Sale:** Enters a closed sale transaction which has been made.
- **Record Credit Sale:** This is a credit (due) sale transaction recorded.
- **Edit Transaction:** This gives an option of updating transaction details when necessary.
- **Record Payment:** This is a record of payment by the customer to decrease the outstanding due.

- **Make Digital Payments:** Customers can make payments online using the available payment methods.
- **Pay with Khalti:** Allows the digital payment with Khalti.
- **Pay using eSewa:** Allows the use of eSewa to make a digital payment.
- **Cash Pay:** Recording of payment through cash is done manually.
- **Verify Payment:** Enhances the payment status of the payment gateway and modifies the due based on that.
- **Create PDF Report:** Creates and downloads PDF documents in transactions and dues.

Notifications/ Communication.

- **View Notification:** This enables user to see notifications of the system like request, reminders and updates.
- **Send Bulk Notifications:** This option enables business/admin to send a notification to more than 10 users.
- **Real Time Chat:** Facilitates real time chat between the business and the customer.

Smart / AI Features

- **Smart Features:** The presence of a combined functions group that allows entering transactions to be quicker with the help of AI.
- **Scan Receipt:** Scan receipt is an OCR program to read transaction details on the image of receipts.
- **Do Voice Entry:** Voice-to-text faster entry of transactions.
- **Chat to AI Bot:** This is an AI chatbot that provides chatbot guidance and questions.

Admin Panel

- **Visit Admin Dashboard:** Gives insight into system usage and controls of an administrator.
- **Manage Users:** Allows the administration to maintain users and user statuses.
- **Monitor Platform Health:** This is used to track the performance and behavior of a system.

3.4.6 Class Diagram

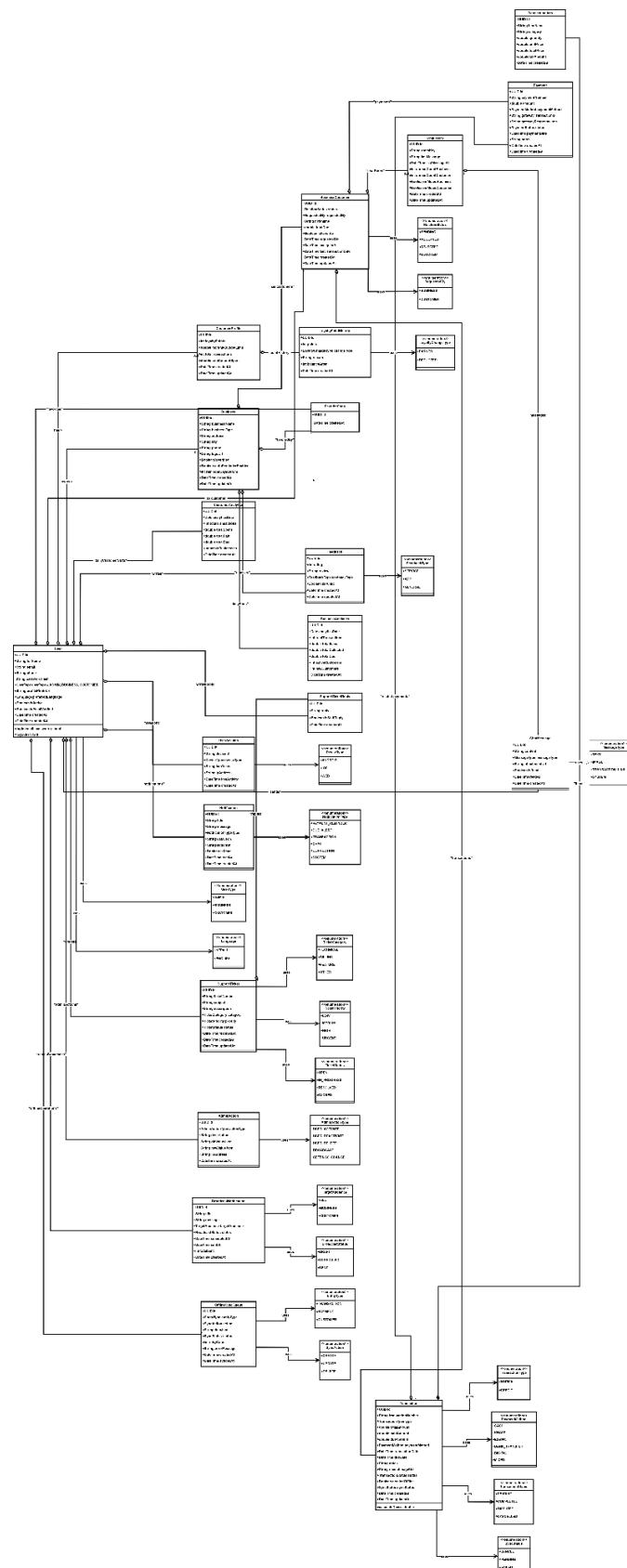


Figure 29: Class Diagram

Higher Quality Image: [link](#)

3.4.7 Activity Diagrams

3.4.7.1 Activity Diagram – Transaction Management

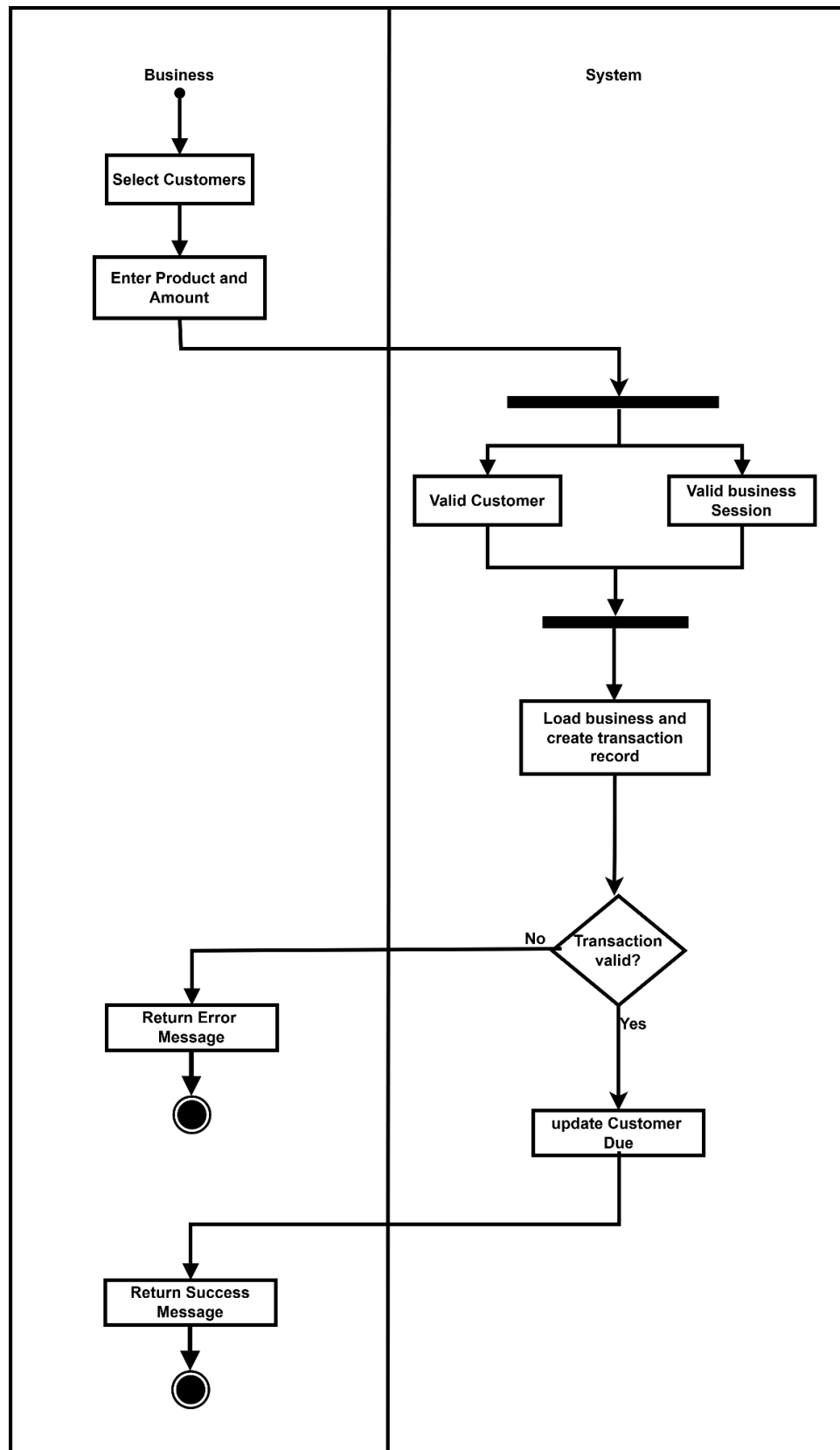


Figure 30: Activity Diagram – Transaction Management

3.4.7.2 Activity Diagram – Customer Management

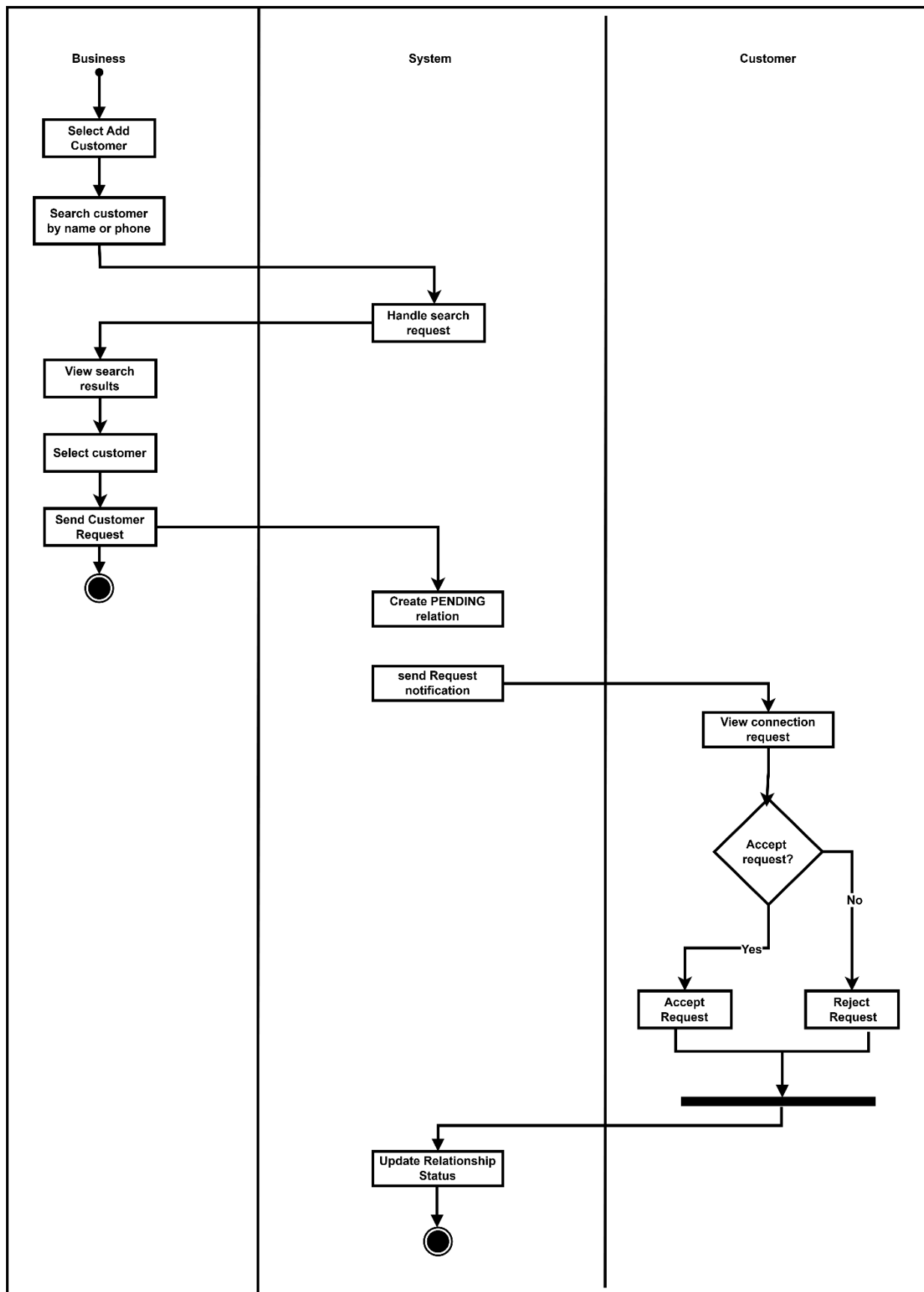


Figure 31: Activity Diagram – Customer Management

3.4.7.3 Activity Diagram – Dashboard and Analytics

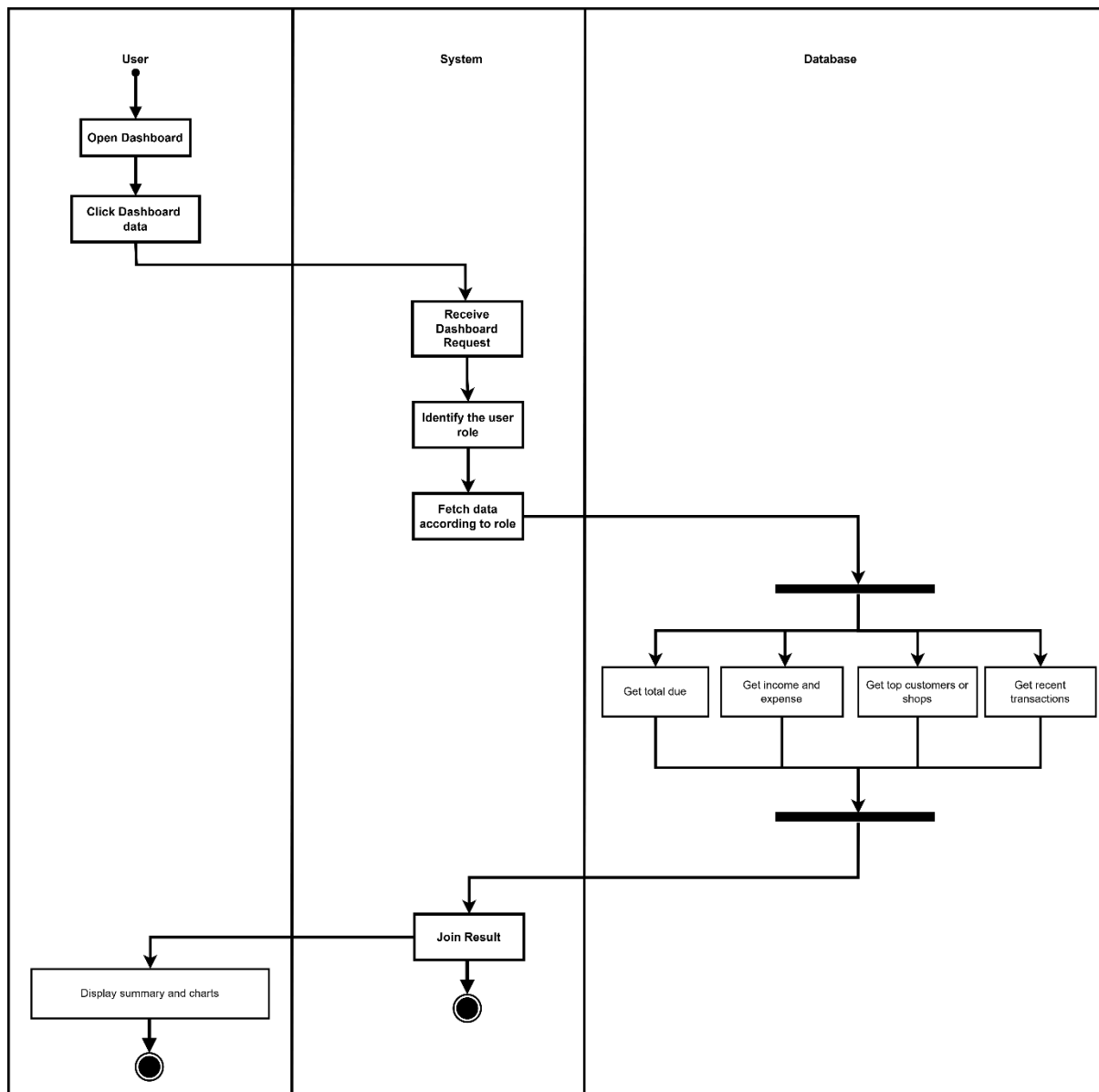


Figure 32: Activity Diagram – Dashboard and Analytics

3.4.7.4 Activity Diagram – AI Features

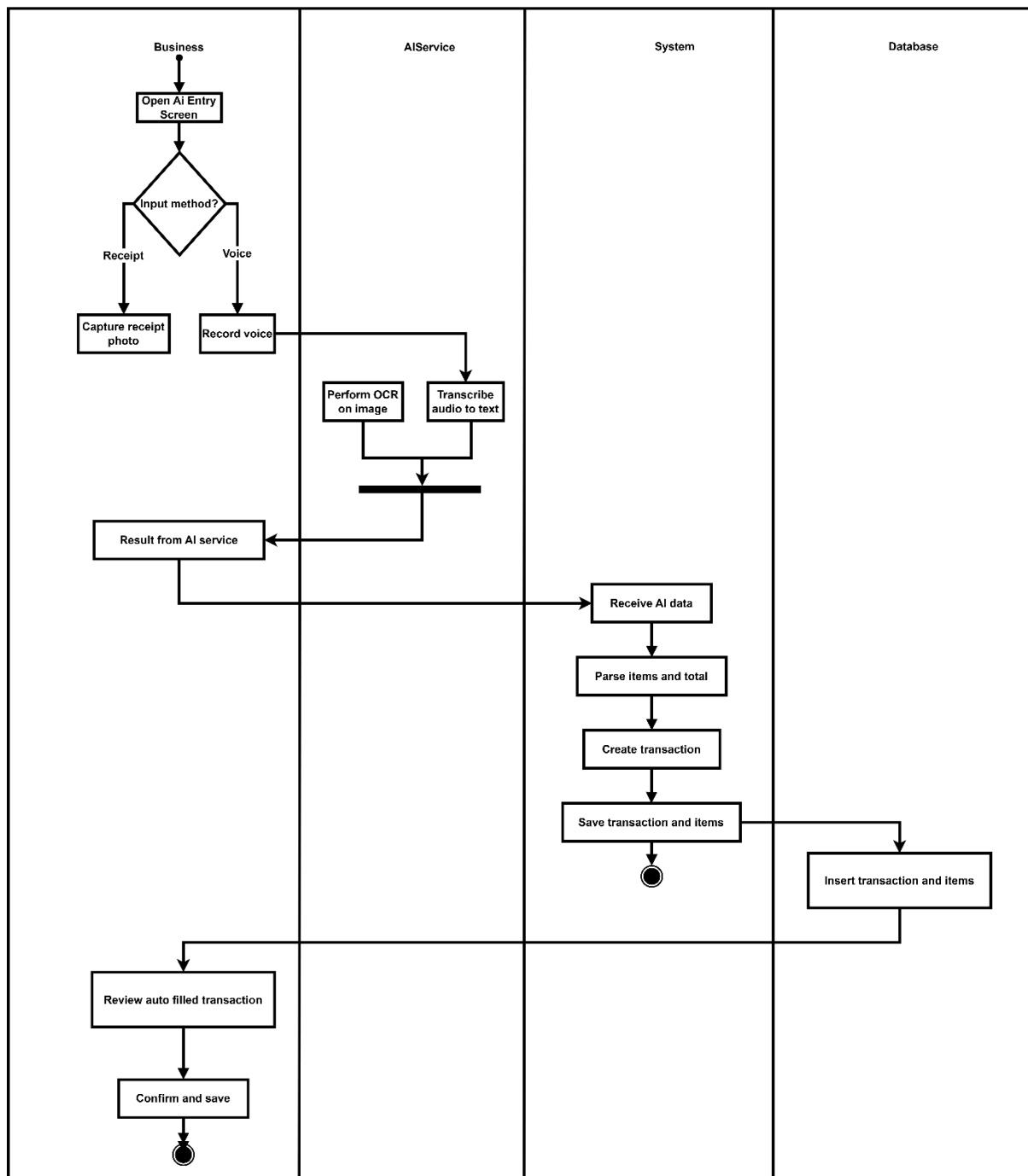


Figure 33: Activity Diagram – AI Features

3.4.7.5 Activity Diagram – Notification and Reminder

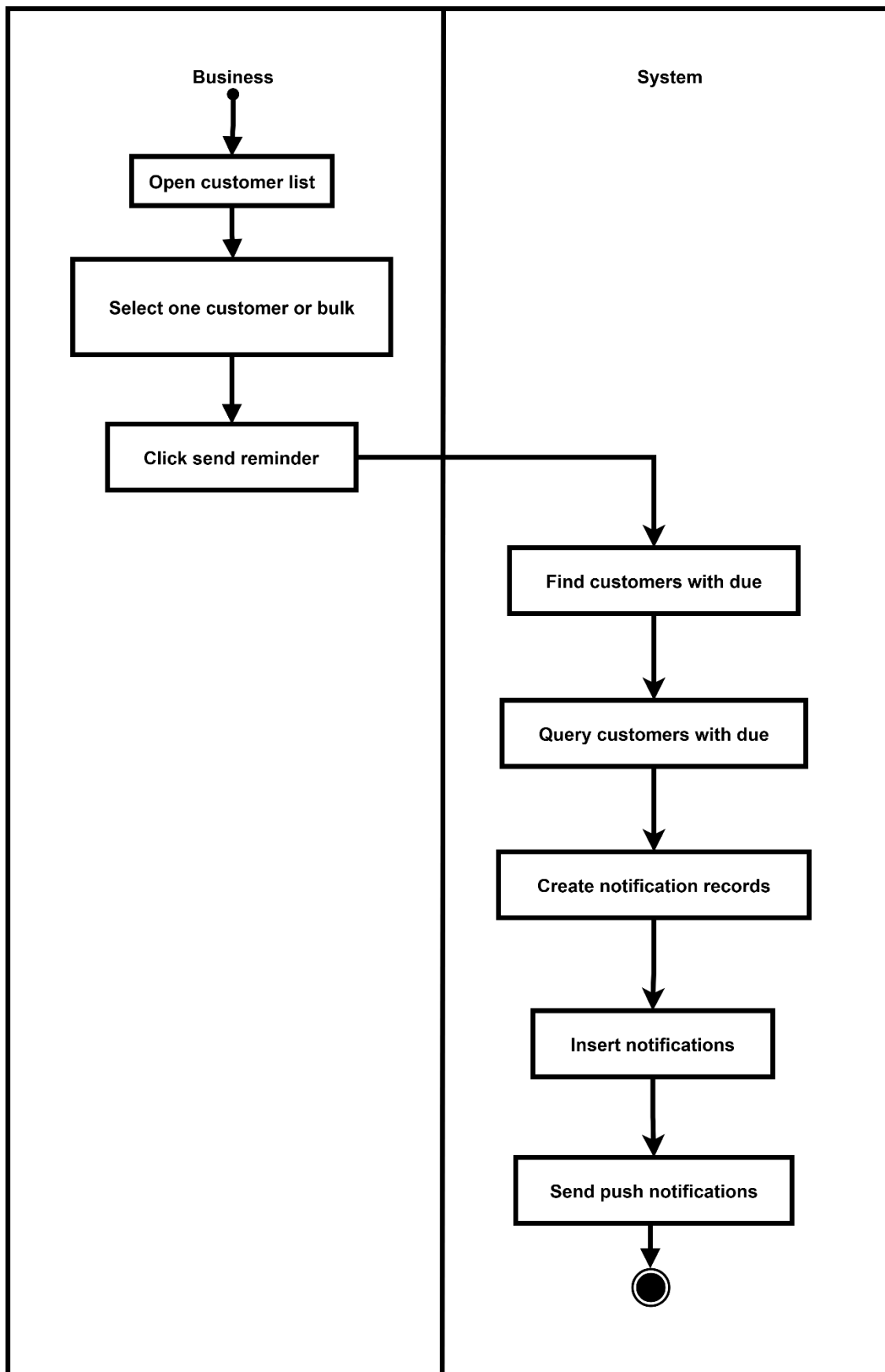


Figure 34: Activity Diagram – Notification and Reminder

3.4.8 Data Flow Diagram (DFD)

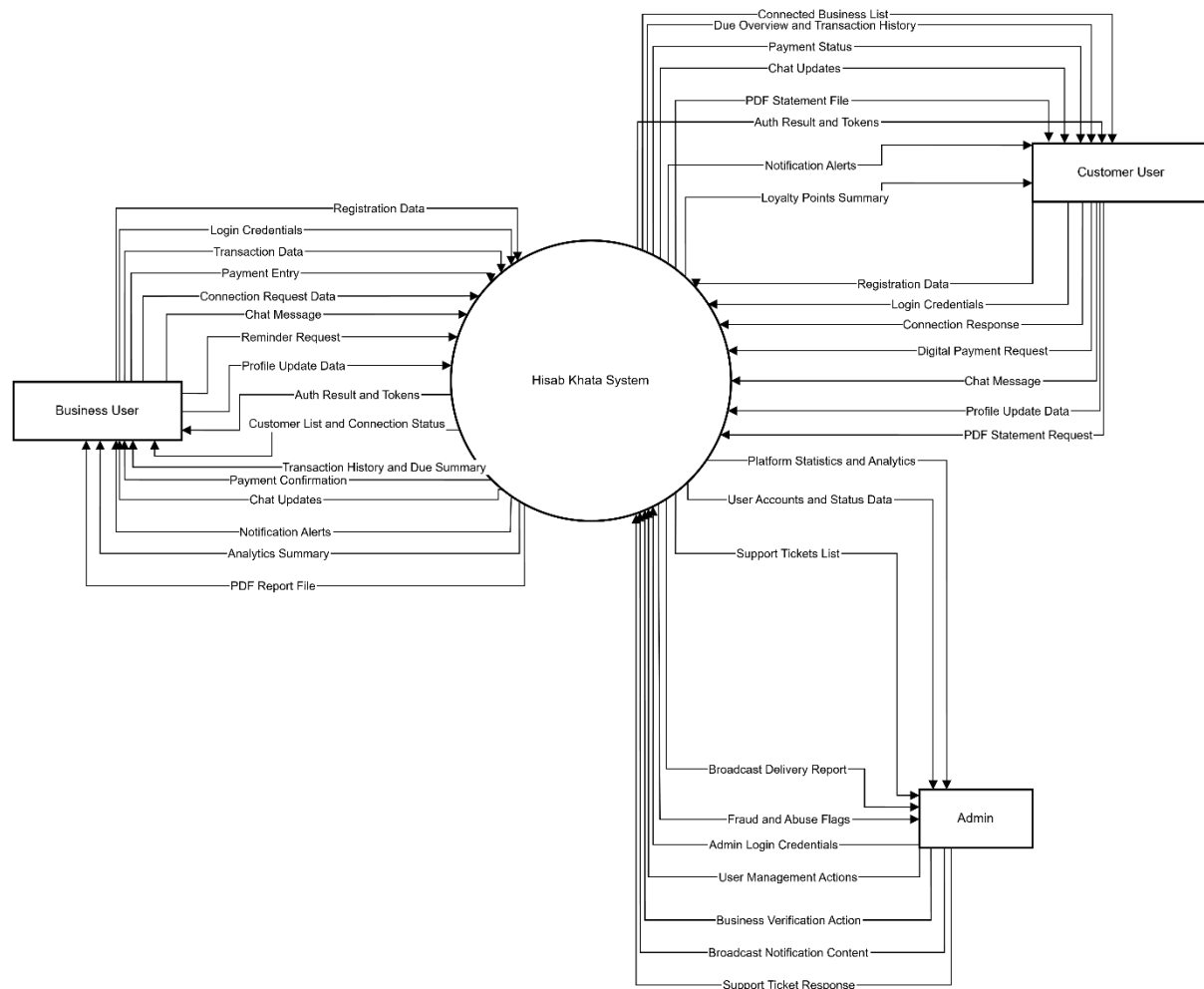


Figure 35: Dataflow Diagram

3.5 Style guide



Figure 36: Color Palette



Figure 37: Hisab Khata Logo

3.6 UI Figma Design

3.6.1 Welcome Screen



Figure 38: Welcome Scree

3.6.2 Authentication Screens

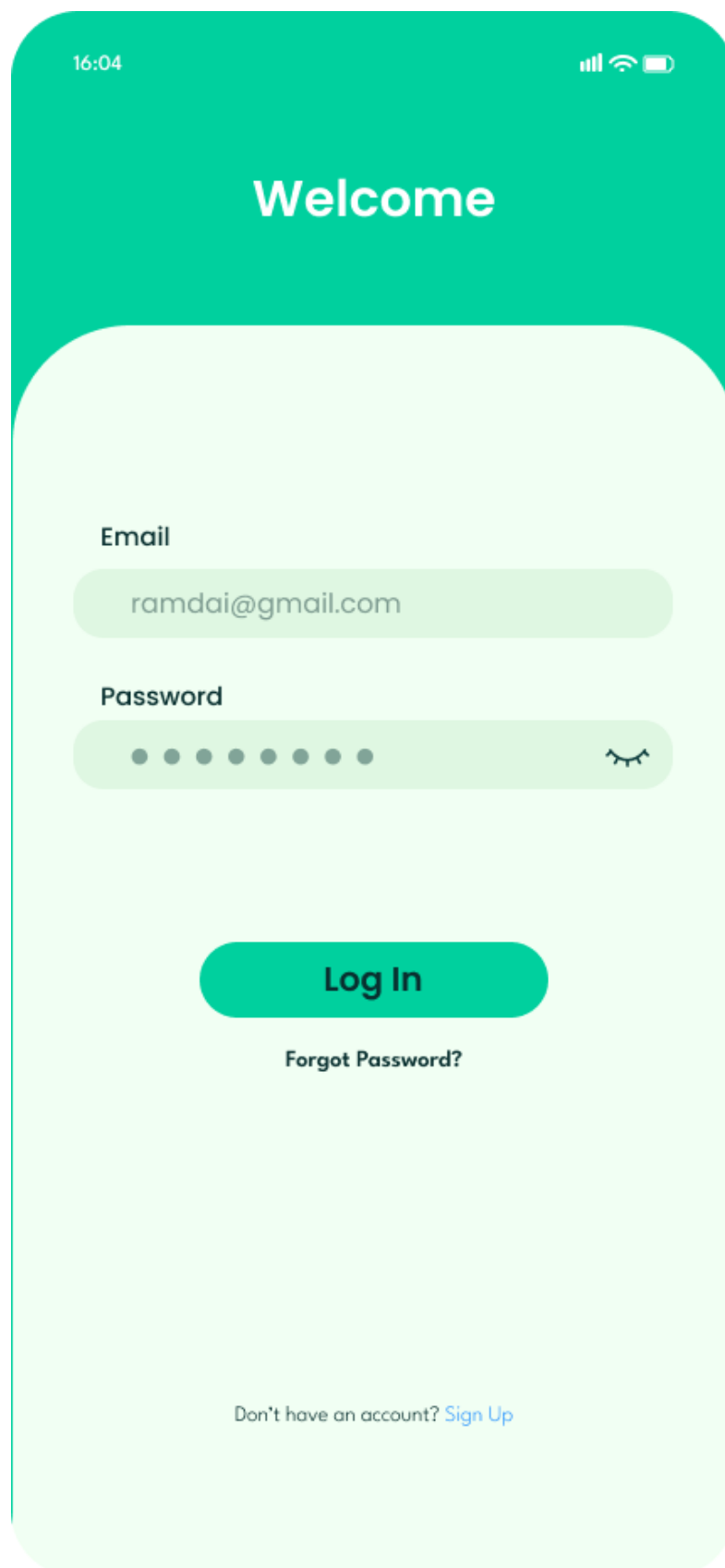


Figure 39: Login Screen

Create Account

Full Name Or Business Name

example@example.com

☒ As Business ☐ As Customer

Email

example@example.com

Mobile Number

+ 123 456 789

Password

• • • • • • • • 

Confirm Password

• • • • • • • • 

By continuing, you agree to
Terms of Use and Privacy Policy.

Sign Up

Already have an account? [Log In](#)

Figure 40: Signup Screen for business

Create Account

Full Name Or Business Name

example@example.com

As Business

As Customer

Email

example@example.com

Mobile Number

+ 123 456 789

Password

● ● ● ● ● ● ● ●

Confirm Password

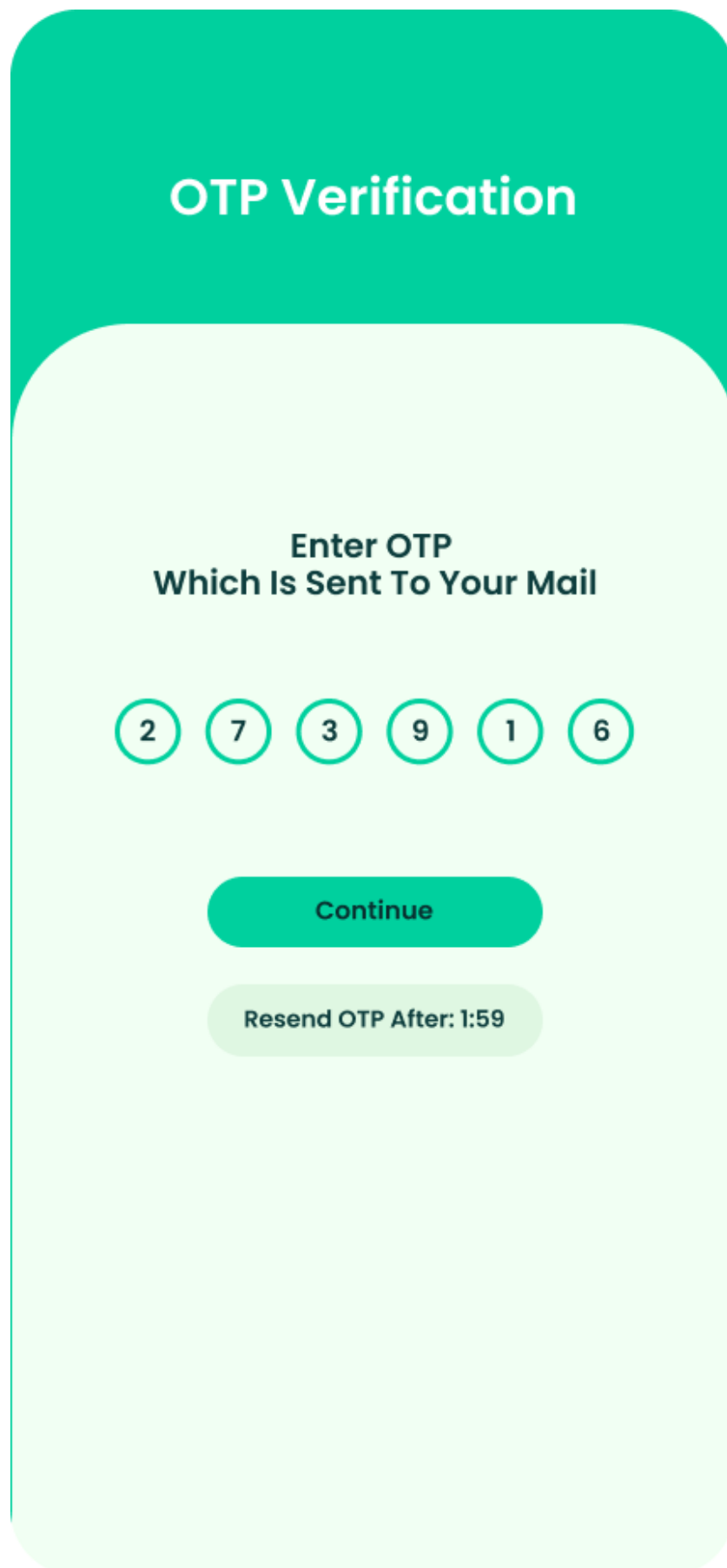
● ● ● ● ● ● ● ●

By continuing, you agree to
Terms of Use and Privacy Policy.

Sign Up

Already have an account? [Log In](#)

Figure 41: Signup Screen as Customer



The image shows a mobile application screen for OTP verification. It features a teal header with the title 'OTP Verification'. Below the header, a light green rounded rectangle contains the text 'Enter OTP Which Is Sent To Your Mail'. Underneath this text, there are six circular input fields, each containing a digit of the OTP: 2, 7, 3, 9, 1, and 6. Below the input fields, there is a teal 'Continue' button and a light green 'Resend OTP After: 1:59' button.

OTP Verification

Enter OTP
Which Is Sent To Your Mail

2 7 3 9 1 6

Continue

Resend OTP After: 1:59

Figure 42: OTP Verification Screen

3.6.3 Customer Dashboard Screens

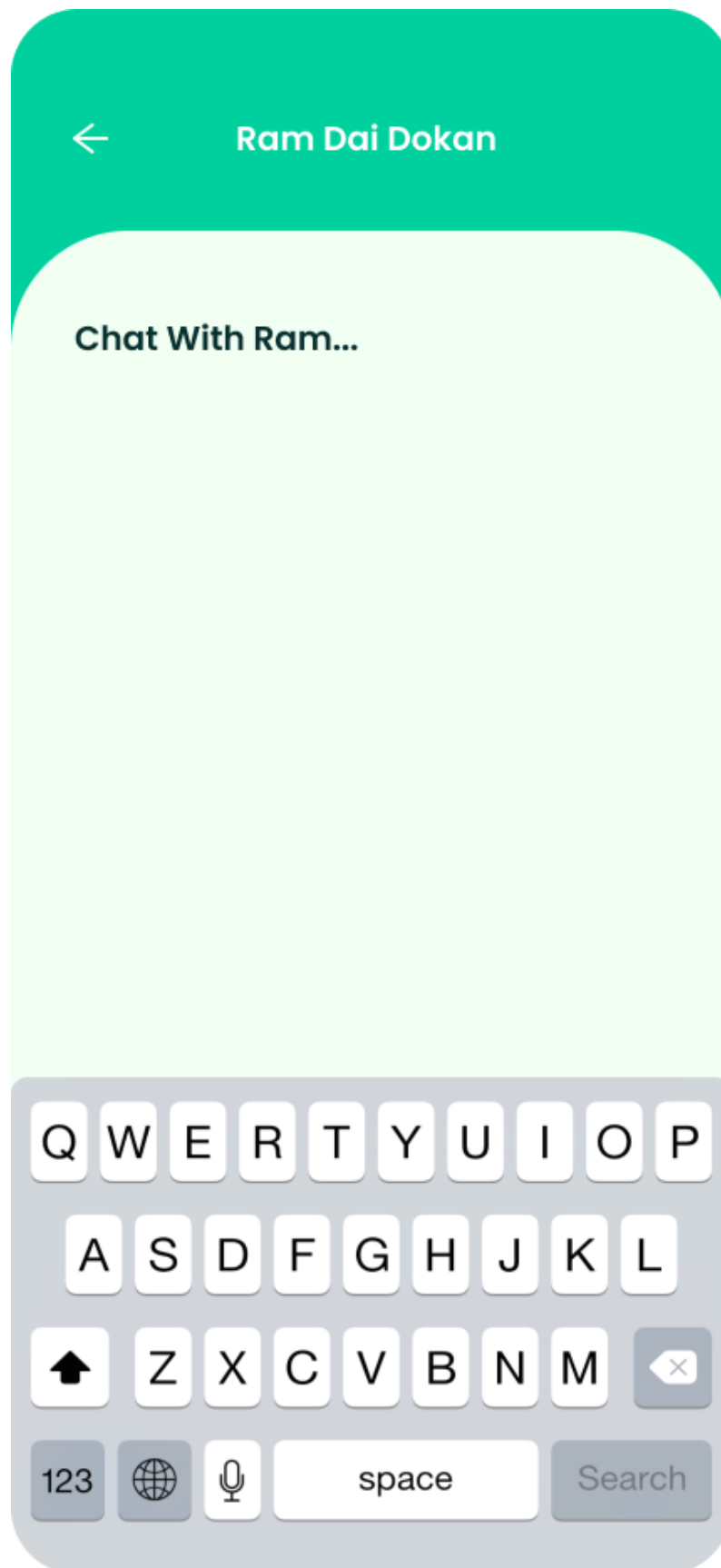


Figure 43: Chat Screen

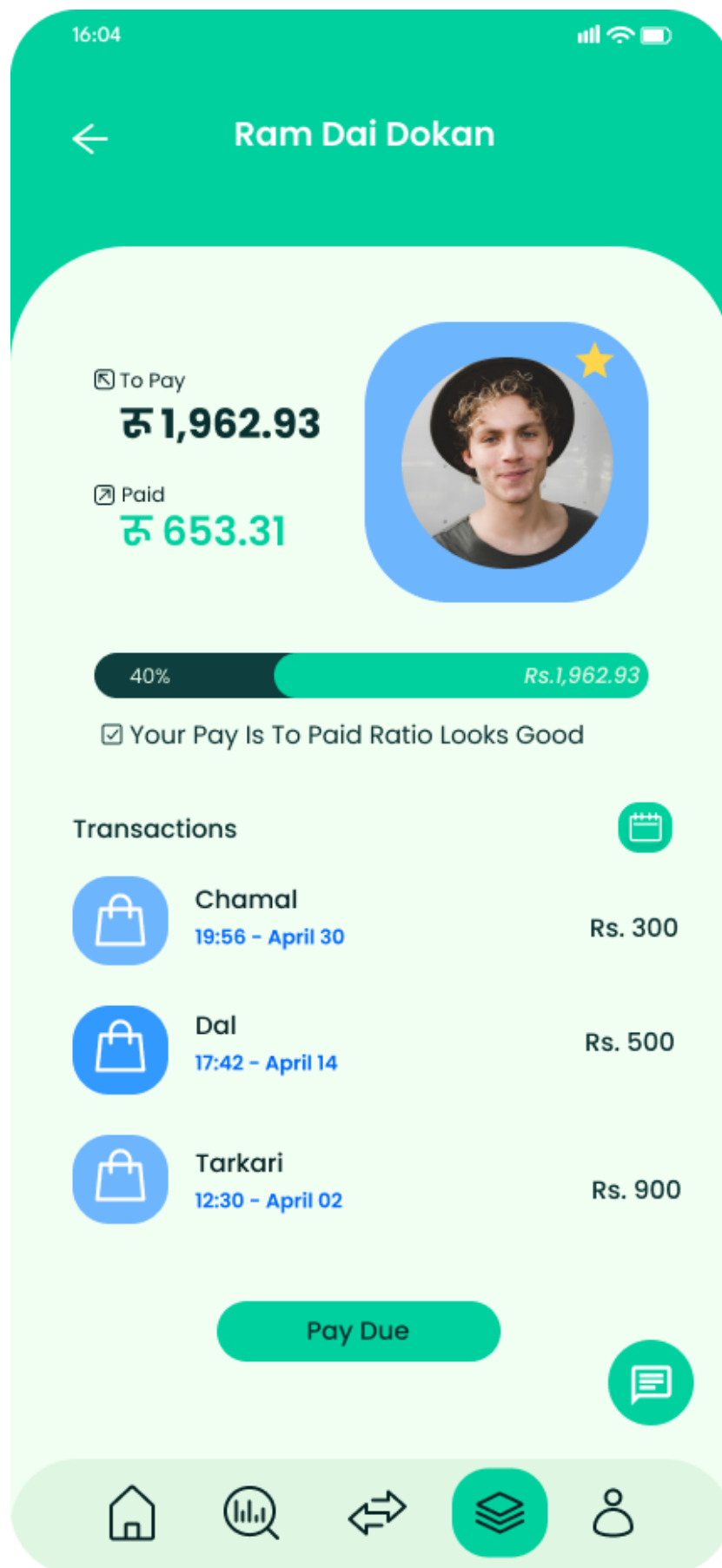


Figure 44: Business Favorite Active Screen

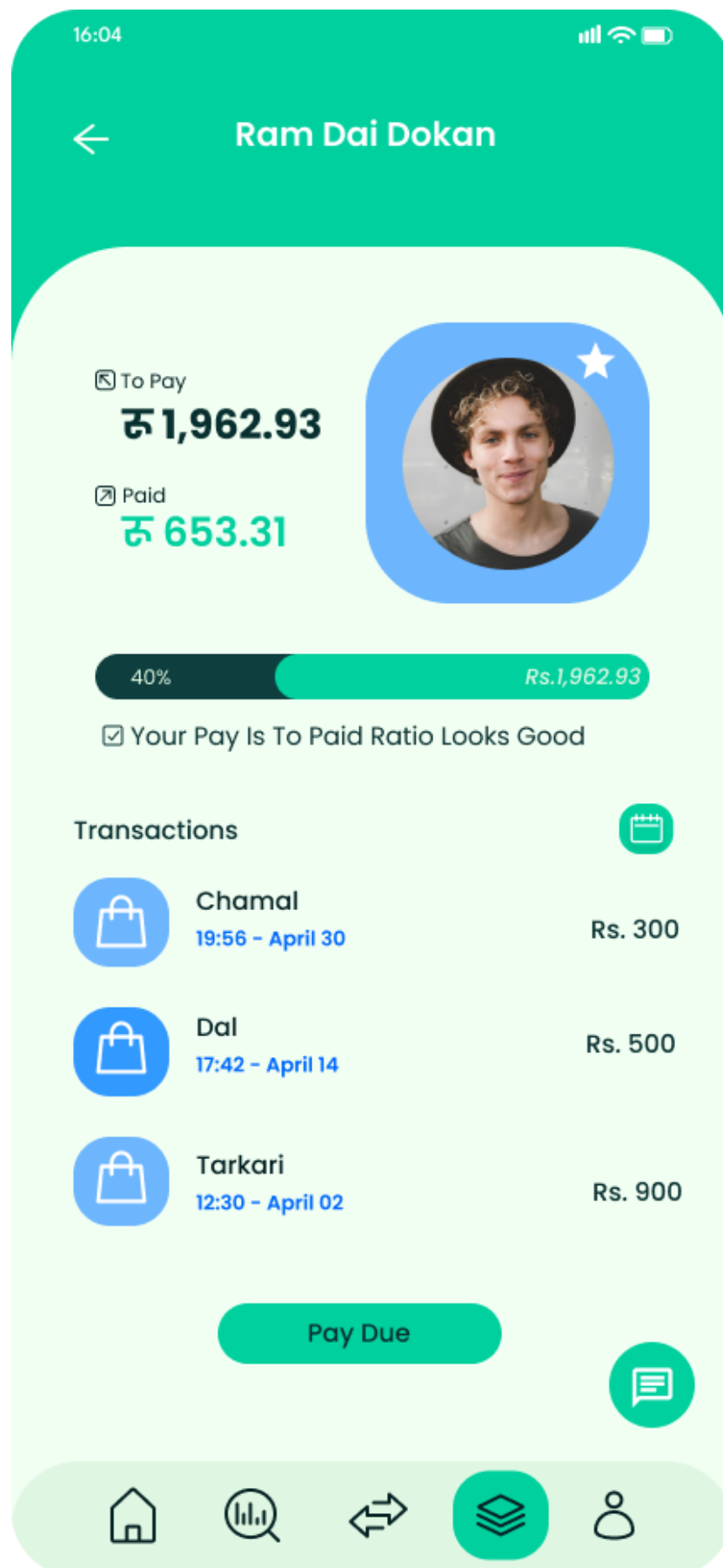


Figure 45: Business Favorite Passive Screen

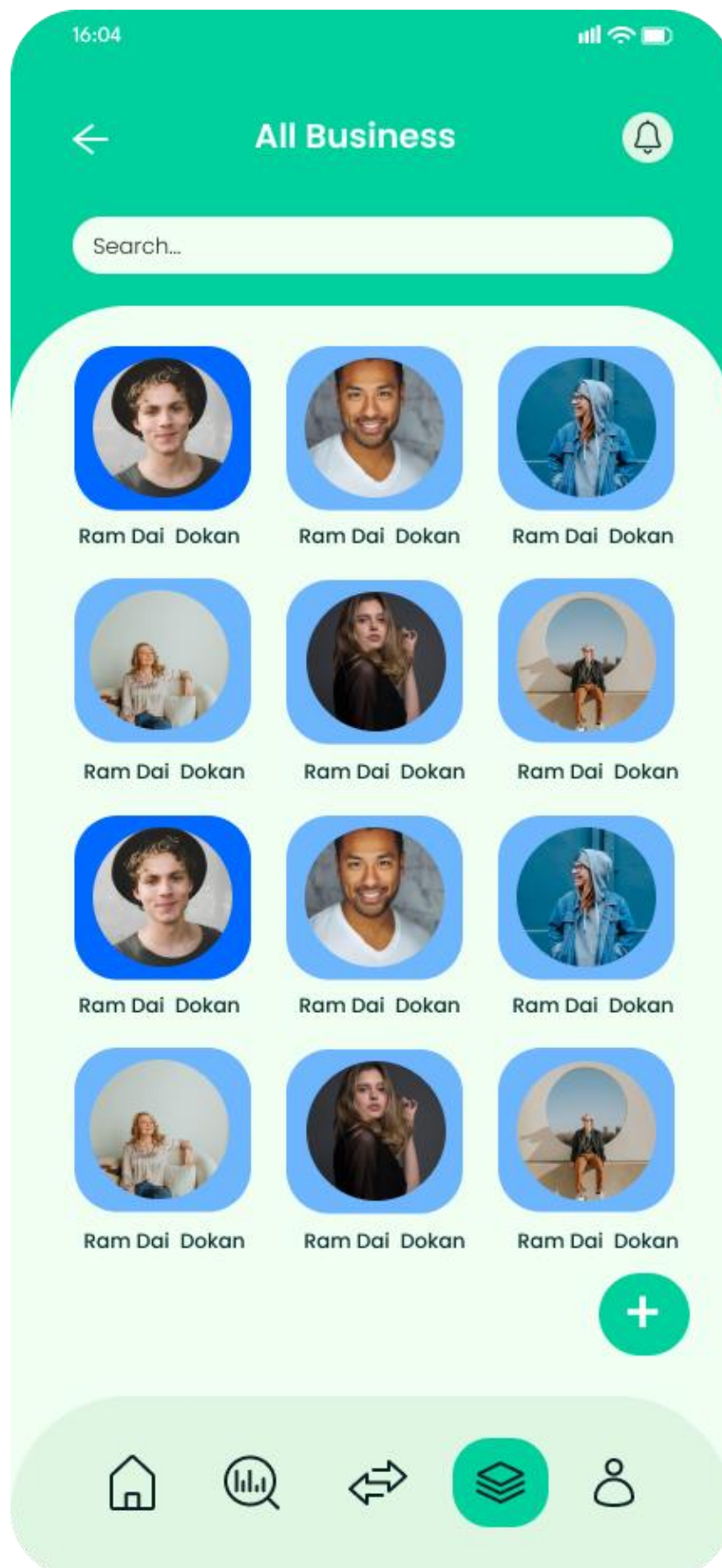
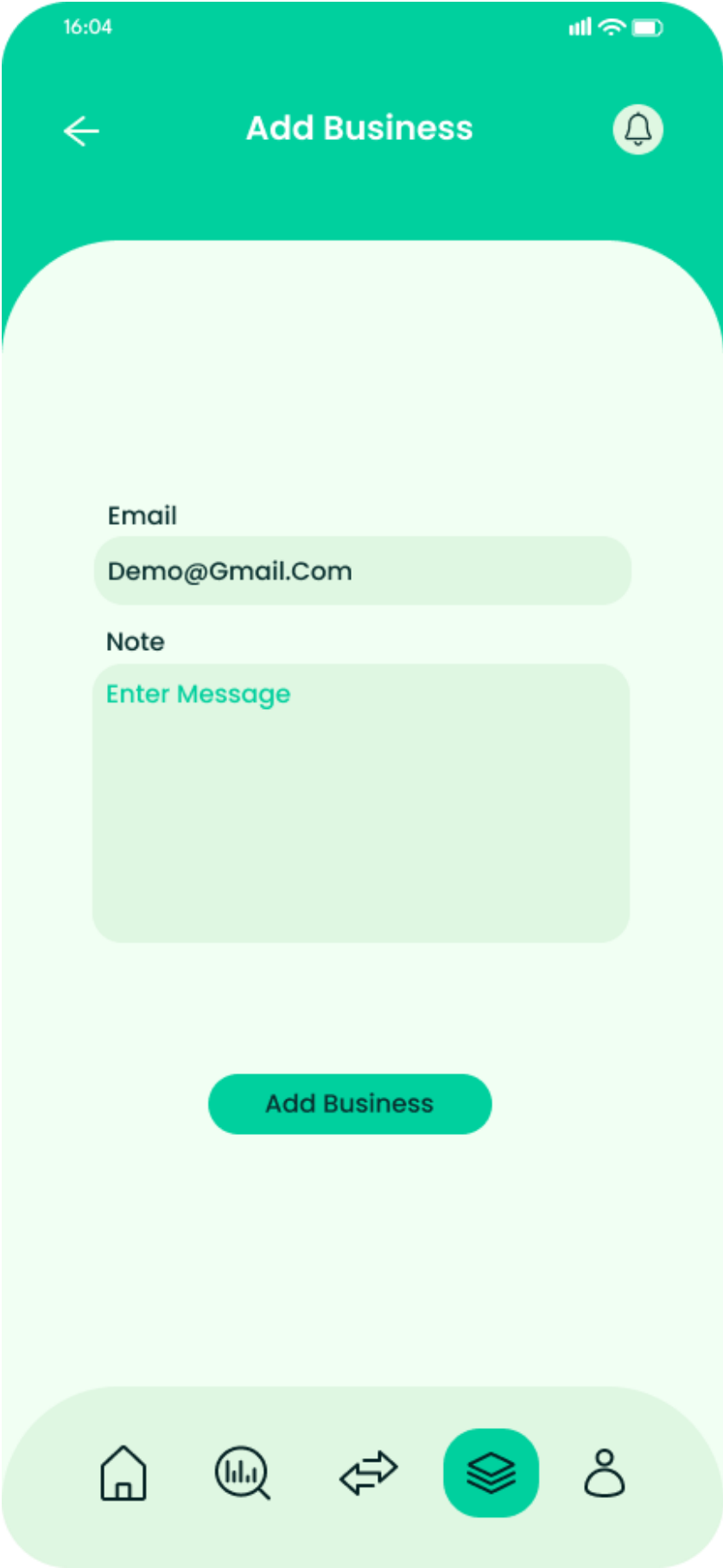


Figure 46: All Added Business List Screen



The image shows a mobile application screen titled "Add Business". At the top, there is a teal header bar with a back arrow on the left, the title "Add Business" in the center, and a bell icon on the right. Below the header, the screen has a light green background. There are two input fields: an "Email" field with the text "Demo@Gmail.Com" and a "Note" field with the placeholder text "Enter Message". Below these fields is a teal button labeled "Add Business". At the bottom of the screen is a navigation bar with five icons: a house, a bar chart, a double-headed arrow, a stack of coins (which is highlighted with a teal circle), and a person icon.

16:04

← Add Business

Email

Demo@Gmail.Com

Note

Enter Message

Add Business

Home Bar Chart Double Arrow Coins (Selected) Profile

Figure 47: Add Business Screen

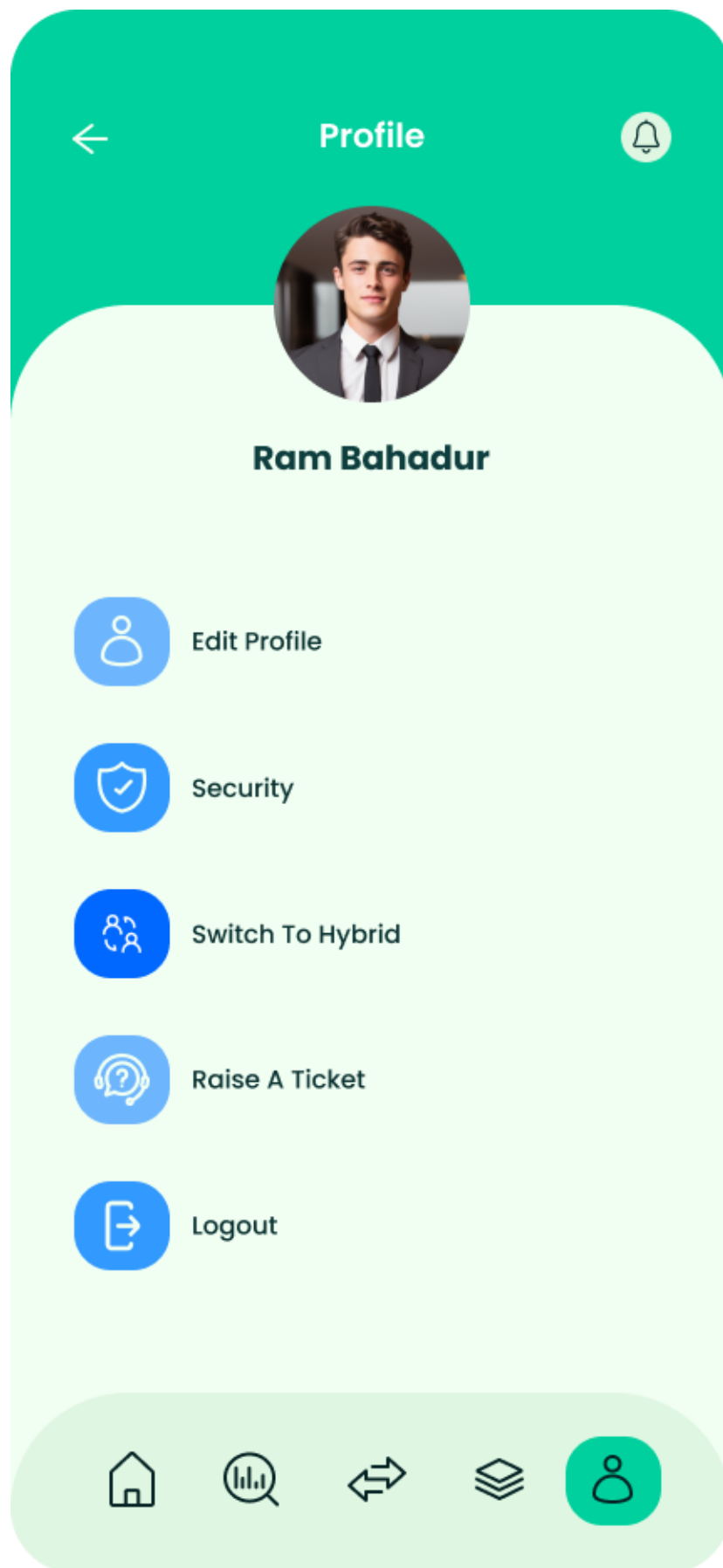


Figure 48: Customer Profile Screen

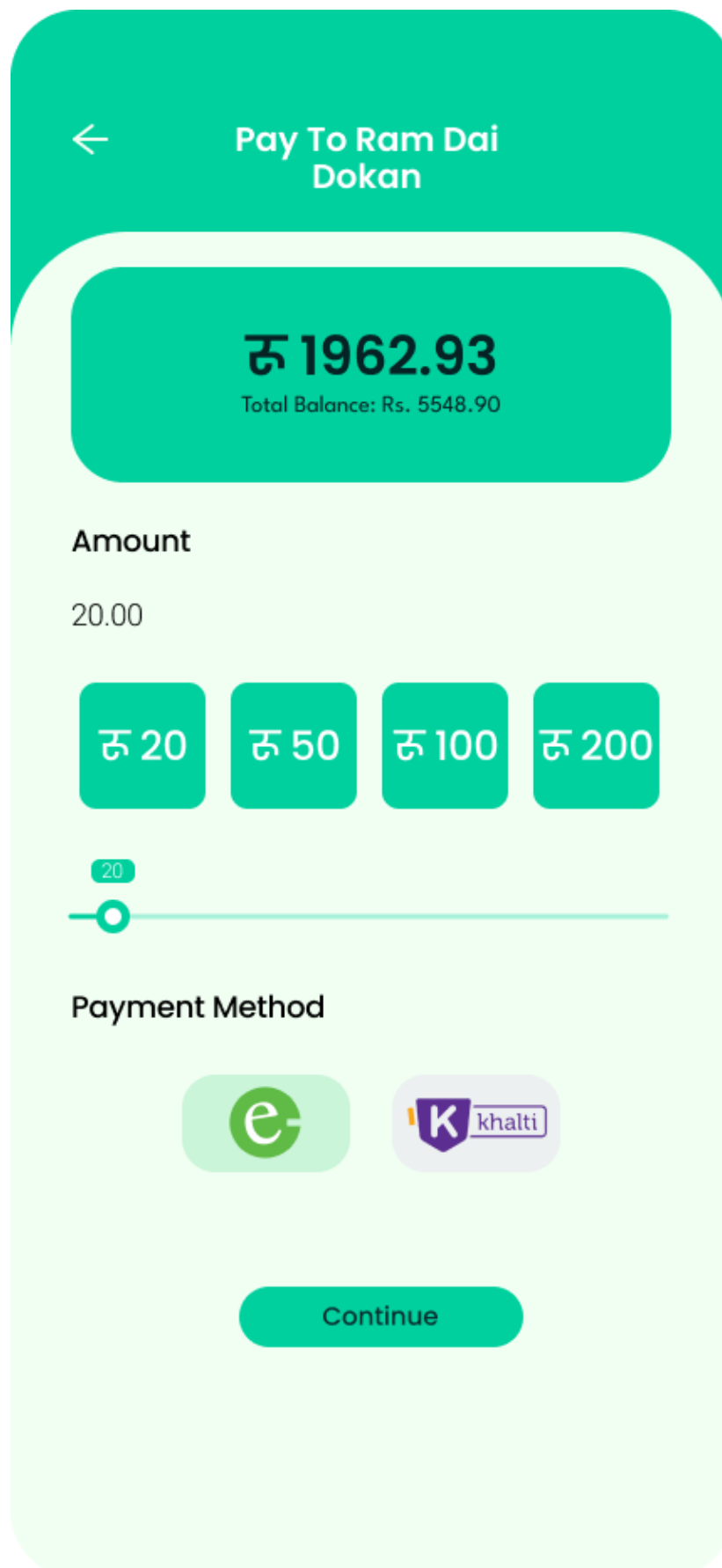


Figure 49: Payment Gateway Screen

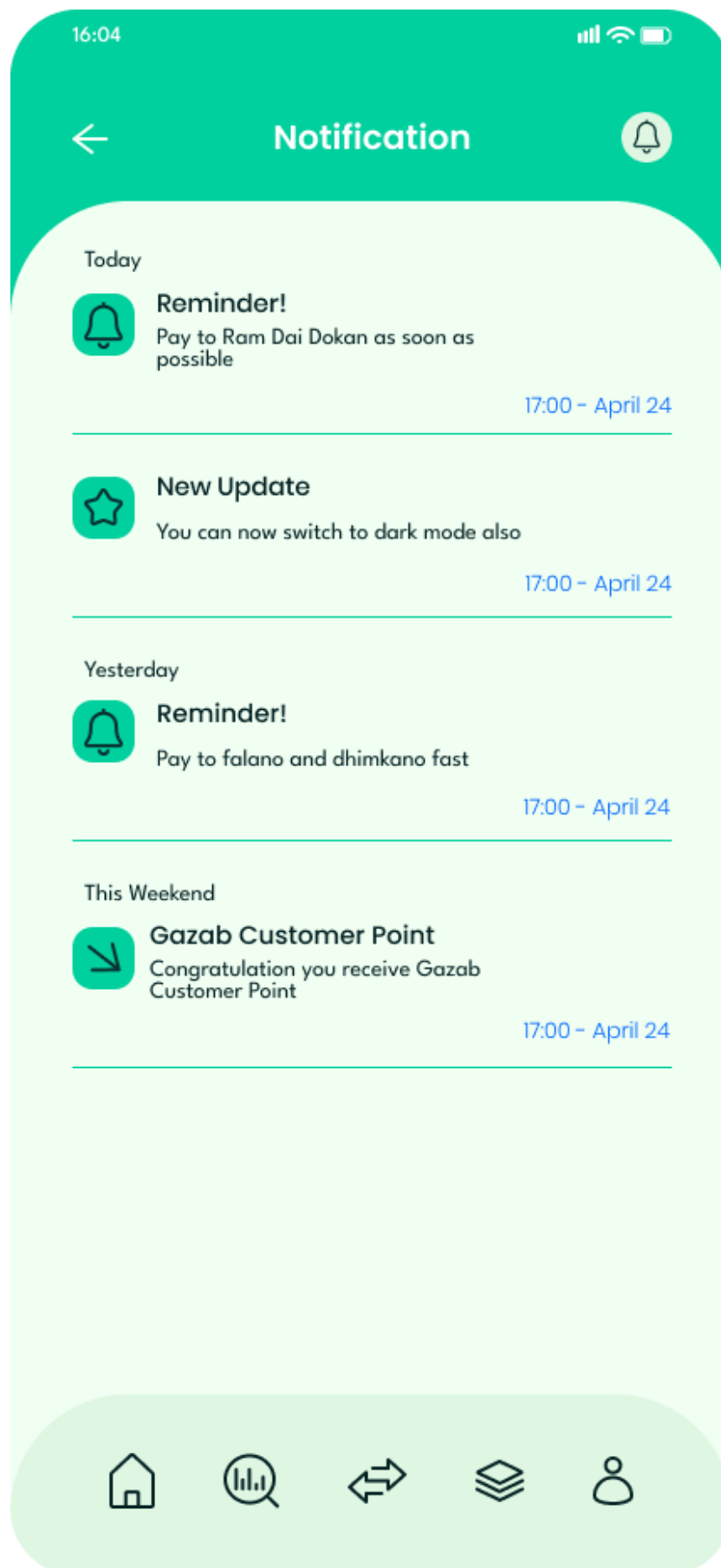


Figure 50: Notification Screen

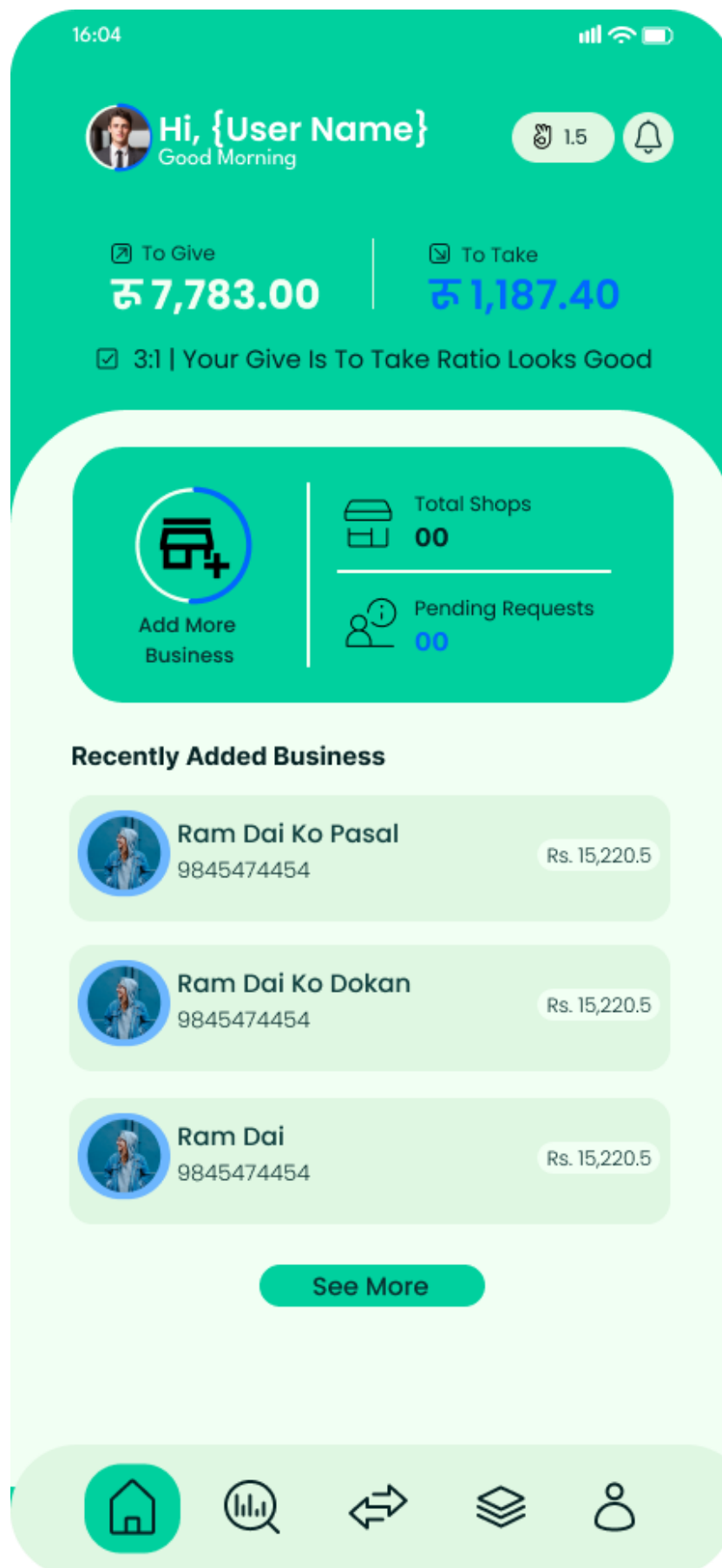
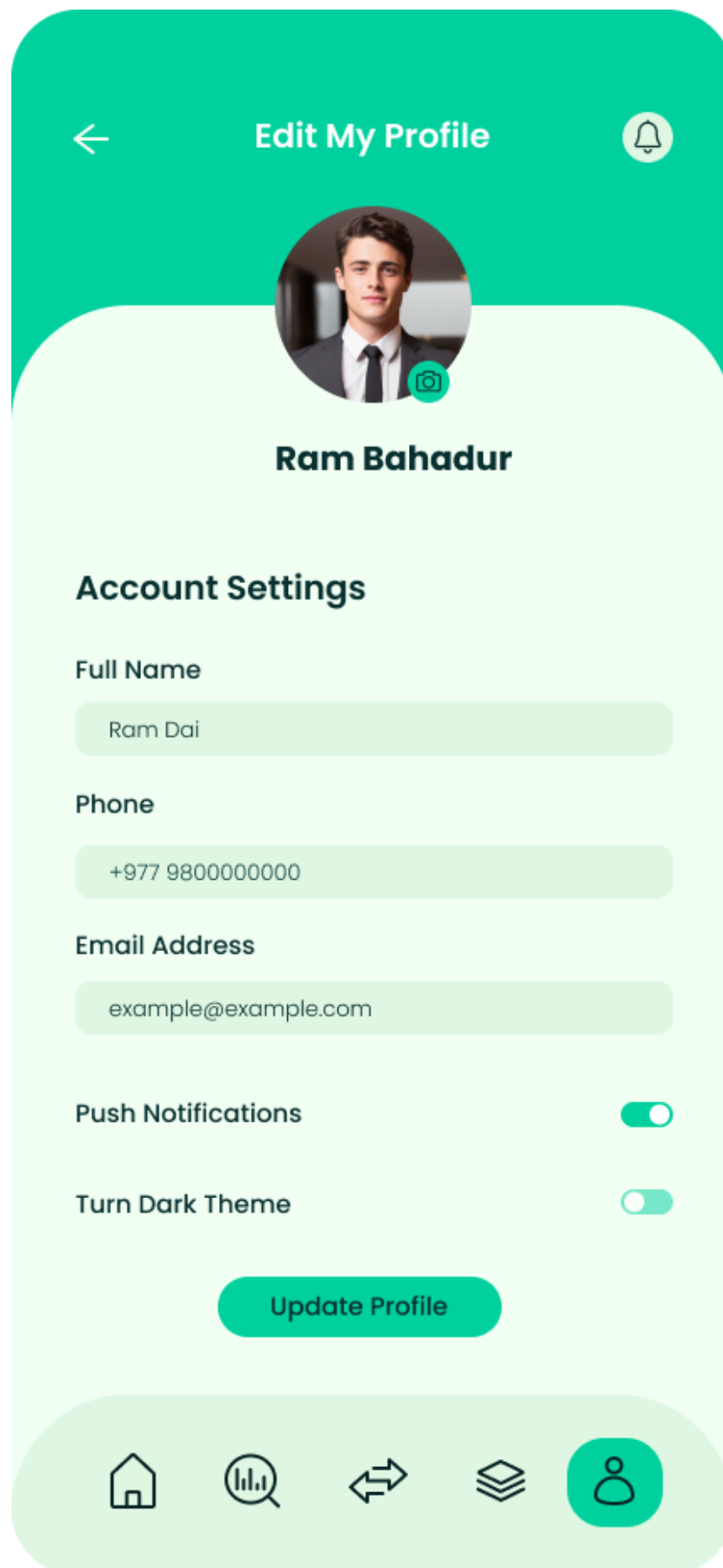


Figure 51: Customer Dashboard Screen



The image shows a mobile application screen titled "Edit My Profile". At the top, there is a teal header bar with a back arrow on the left, the title "Edit My Profile" in the center, and a bell icon on the right. Below the header, there is a circular profile picture of a man in a suit and tie. To the right of the profile picture is a small teal camera icon. Below the profile picture, the name "Ram Bahadur" is displayed in bold. Underneath the name, the section "Account Settings" is shown. This section contains four rows of settings: "Full Name" with a text input field containing "Ram Dai", "Phone" with a text input field containing "+977 9800000000", "Email Address" with a text input field containing "example@example.com", and two toggle switches: "Push Notifications" (which is turned on) and "Turn Dark Theme" (which is turned off). At the bottom of the settings section is a teal button labeled "Update Profile". At the very bottom of the screen is a navigation bar with five icons: a house, a bar chart, a double-headed arrow, a stack of papers, and a person icon (which is highlighted with a teal background).

← Edit My Profile 🔔

Ram Bahadur

Account Settings

Full Name

Ram Dai

Phone

+977 9800000000

Email Address

example@example.com

Push Notifications ☒

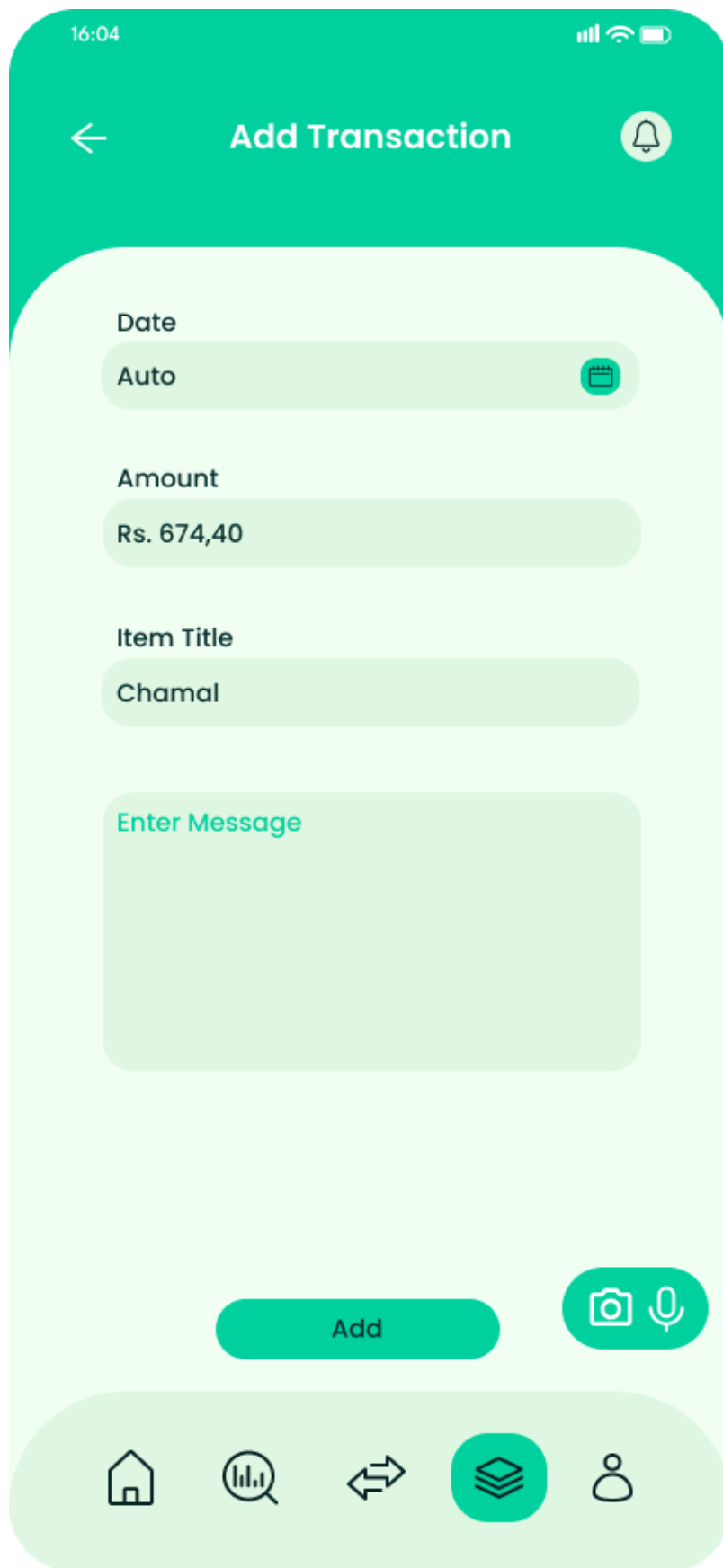
Turn Dark Theme ☐

Update Profile

Figure 52: Customer Profile Edit Screen

3.6.4 Business Dashboard Screen



The image shows a mobile application screen titled "Add Transaction". The screen has a teal header bar with a back arrow on the left, the title "Add Transaction" in the center, and a bell icon on the right. The status bar at the top shows the time 16:04 and signal, Wi-Fi, and battery icons. The main content area is a light green rounded rectangle containing four input fields: "Date" with "Auto" and a calendar icon, "Amount" with "Rs. 674,40", "Item Title" with "Chamal", and a large "Enter Message" text area. At the bottom of the form is a teal "Add" button. To the right of the "Add" button is a rounded rectangle containing a camera icon and a microphone icon. At the very bottom is a navigation bar with five icons: a house, a bar chart, a double-headed arrow, a stack of coins (which is highlighted with a teal background), and a person icon.

16:04

← Add Transaction 🔔

Date
Auto 📅

Amount
Rs. 674,40

Item Title
Chamal

Enter Message

Add 📷 🎤

🏠 📊 ↔️ 🏠 👤

Figure 53: Add Transaction Screen

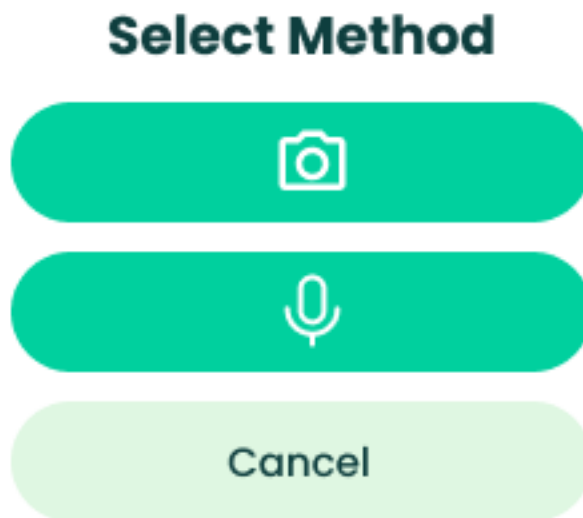
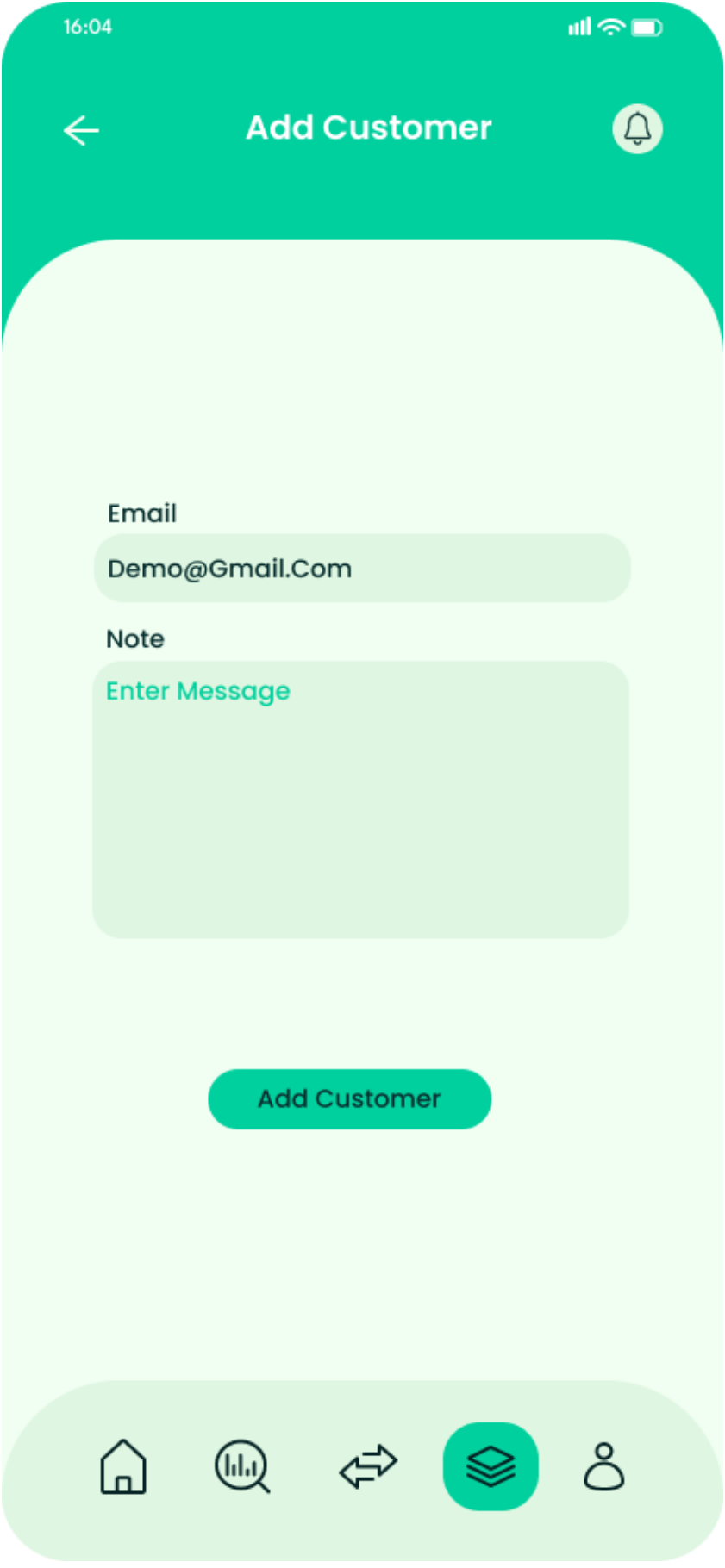


Figure 54: Add Transaction Options Screen



The image shows a mobile application screen titled "Add Customer". At the top, there is a teal header bar with a back arrow on the left, the title "Add Customer" in the center, and a bell icon on the right. The status bar at the very top shows the time "16:04" and signal/battery icons. Below the header, the screen has a light green background. There are two input fields: "Email" with the text "Demo@Gmail.Com" and "Note" with the placeholder text "Enter Message". Below these fields is a teal button labeled "Add Customer". At the bottom, there is a navigation bar with five icons: a house, a bar chart, a double-headed arrow, a stack of coins (which is highlighted with a teal circle), and a person icon.

16:04

← Add Customer

Email

Demo@Gmail.Com

Note

Enter Message

Add Customer

Navigation icons: Home, Analytics, Sync, Add Customer (Active), Profile

Figure 55: Add Customer Screen

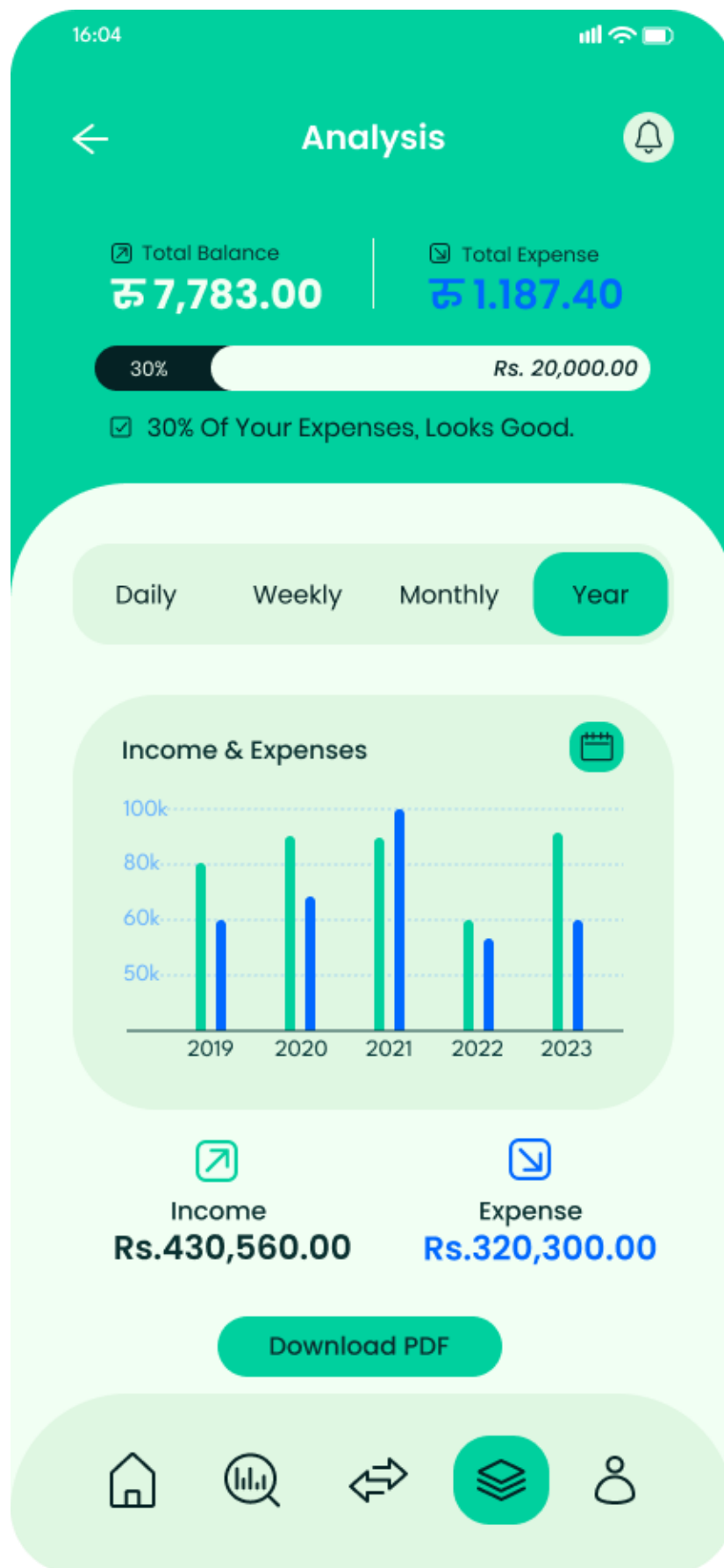


Figure 56: Analytics Screen

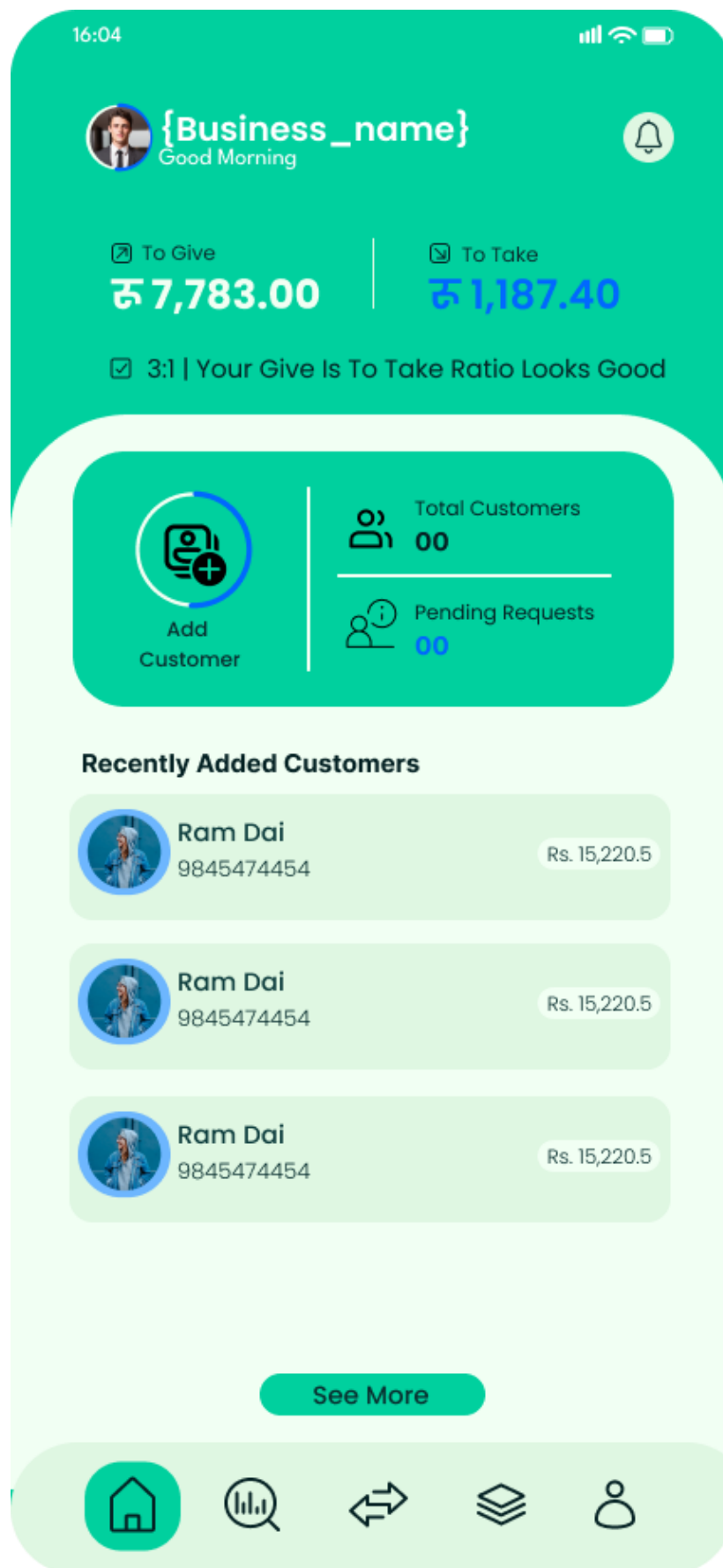


Figure 57: Business Dashboard Screen

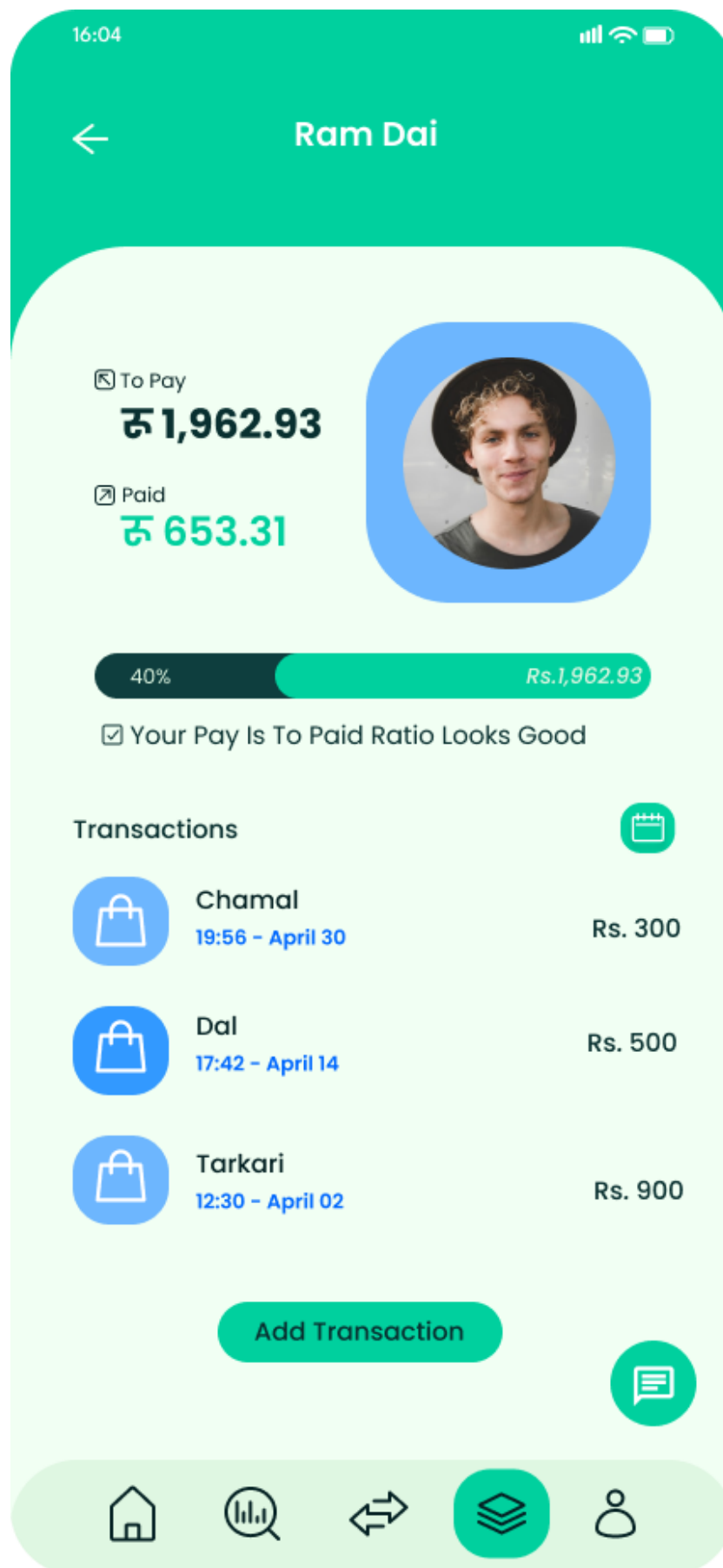


Figure 58: Customer Profile View

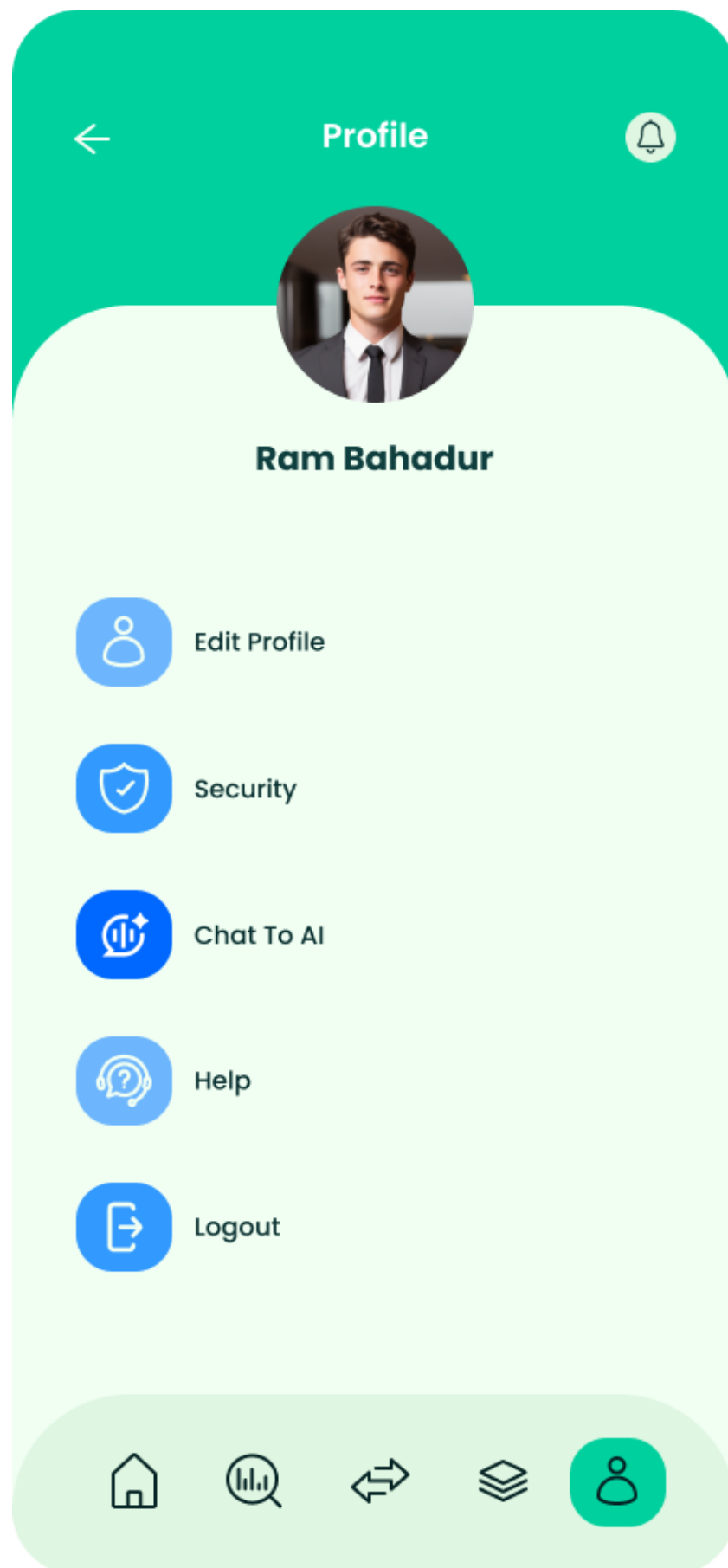


Figure 59: Business Profile Screen

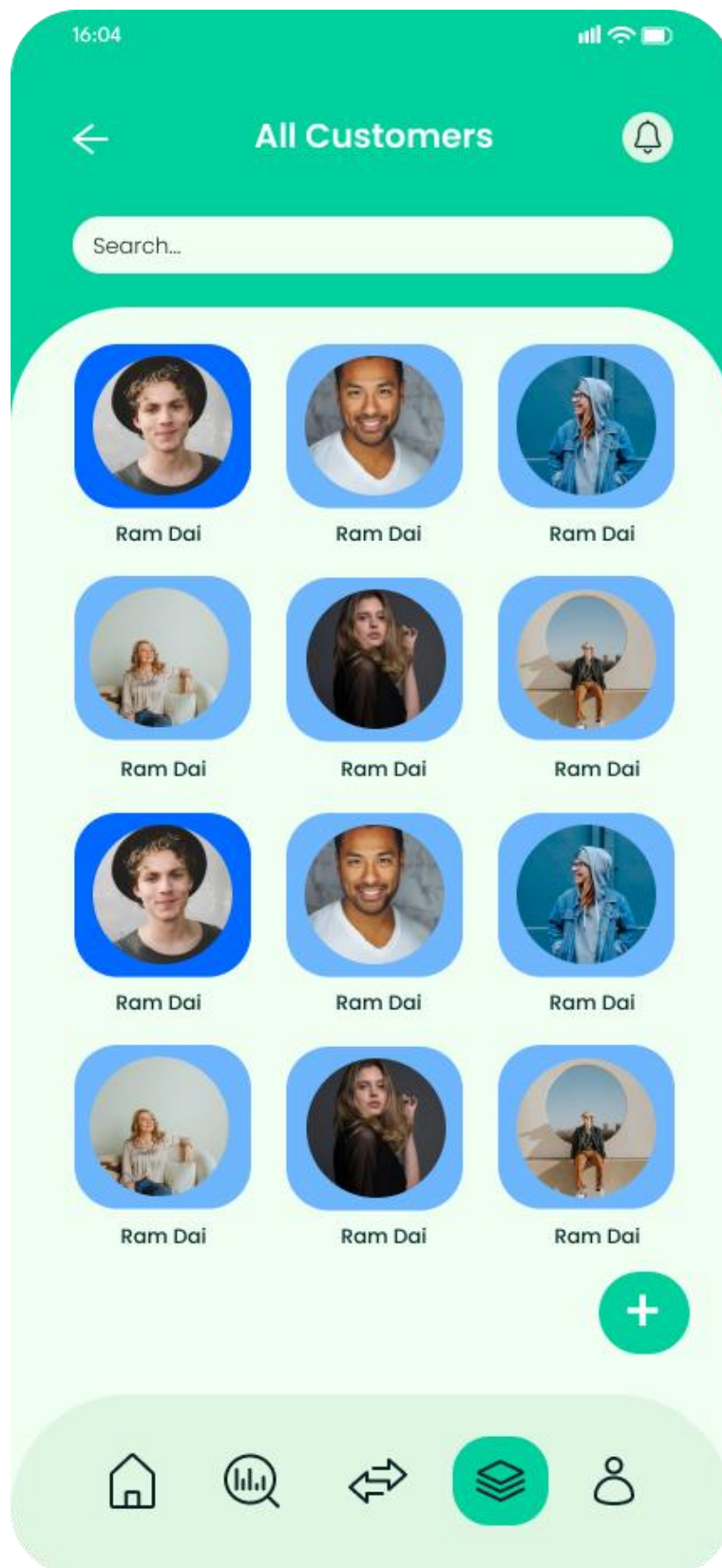


Figure 60: All Connected Customers

3.7 Velocity Chart (Completed Story Points)

Inception is completed; Elaboration is mostly completed (auth, profiles, dashboards base, request system, notifications). Construction and Transition modules (transactions, payments, chat, analytics, PDF, admin, QA) are still pending. Since RUP is followed in this project, the charts are not 100% accurate because velocity and burndown charts are mainly for agile/scrum.

Iterations used: Inception, Elaboration, Construction, Transition.

Velocity Chart (Completed Work per Phase)

Shows how much work was completed in each RUP phase.

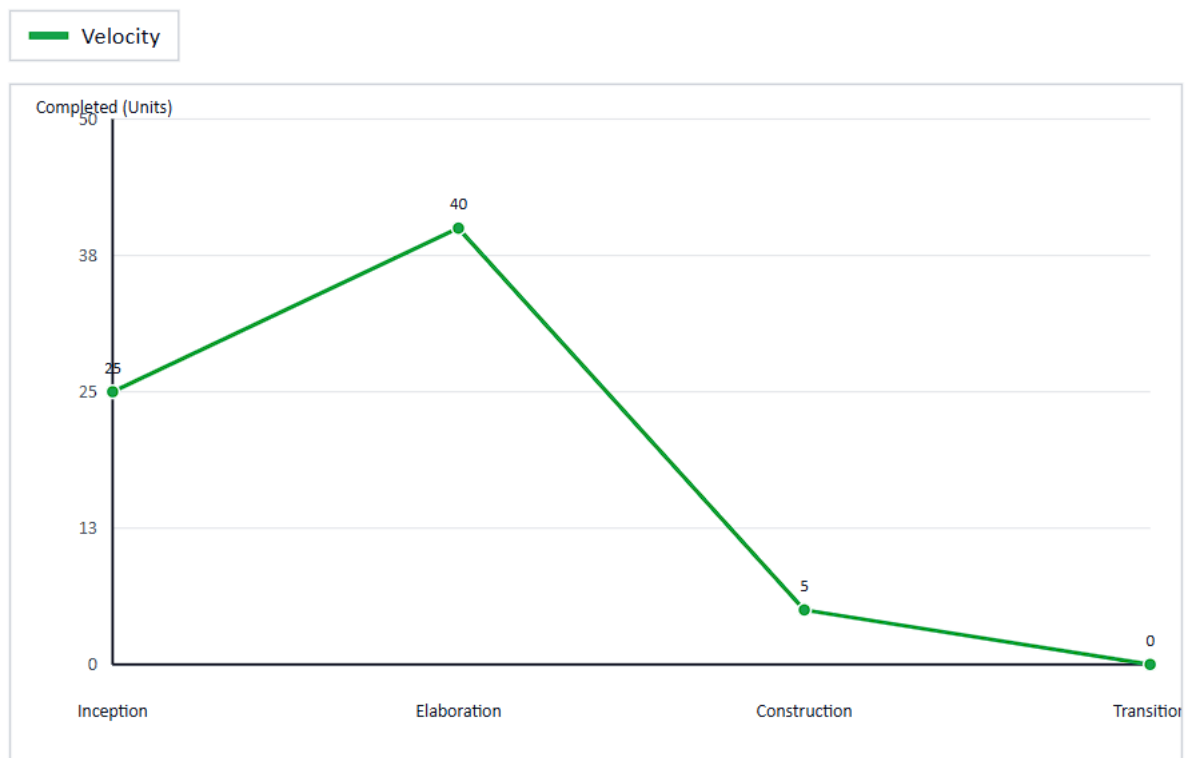


Figure 61: Velocity Chart

3.8 Burndown Chart

Burndown Chart (Remaining Work)

Compares the ideal reduction of remaining work vs the actual remaining work.

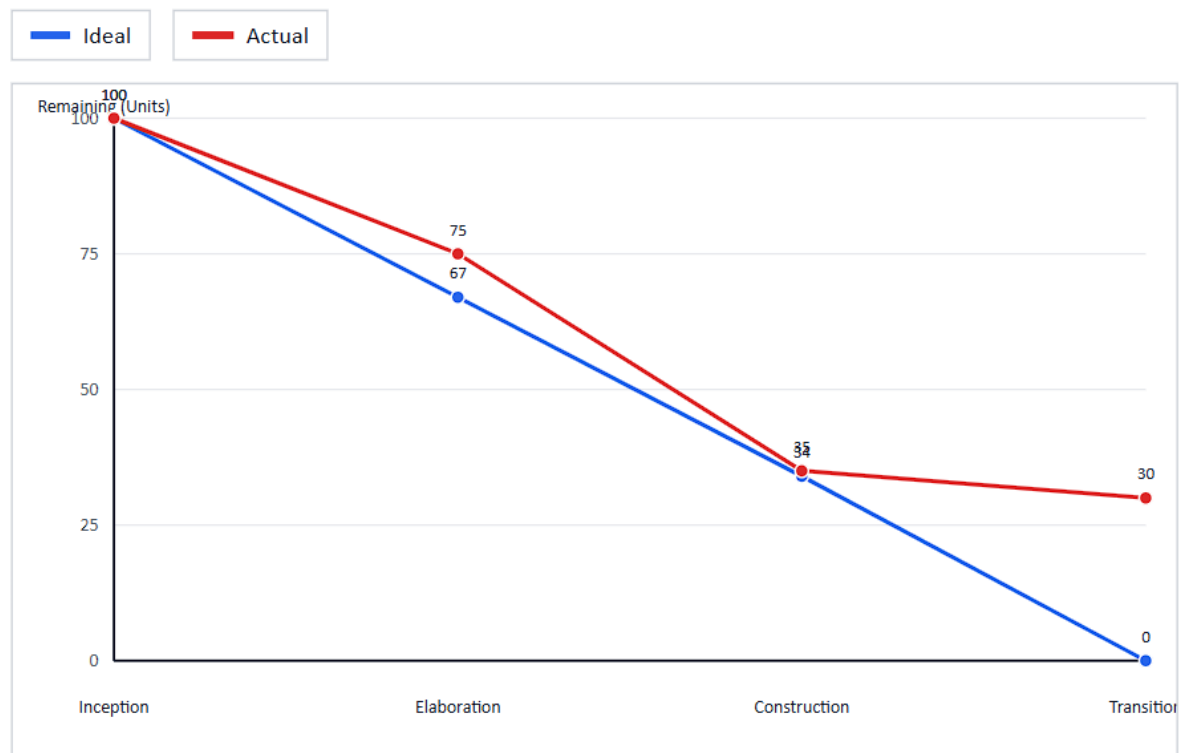


Figure 62: Burndown Chart

4 Analysis of Progress

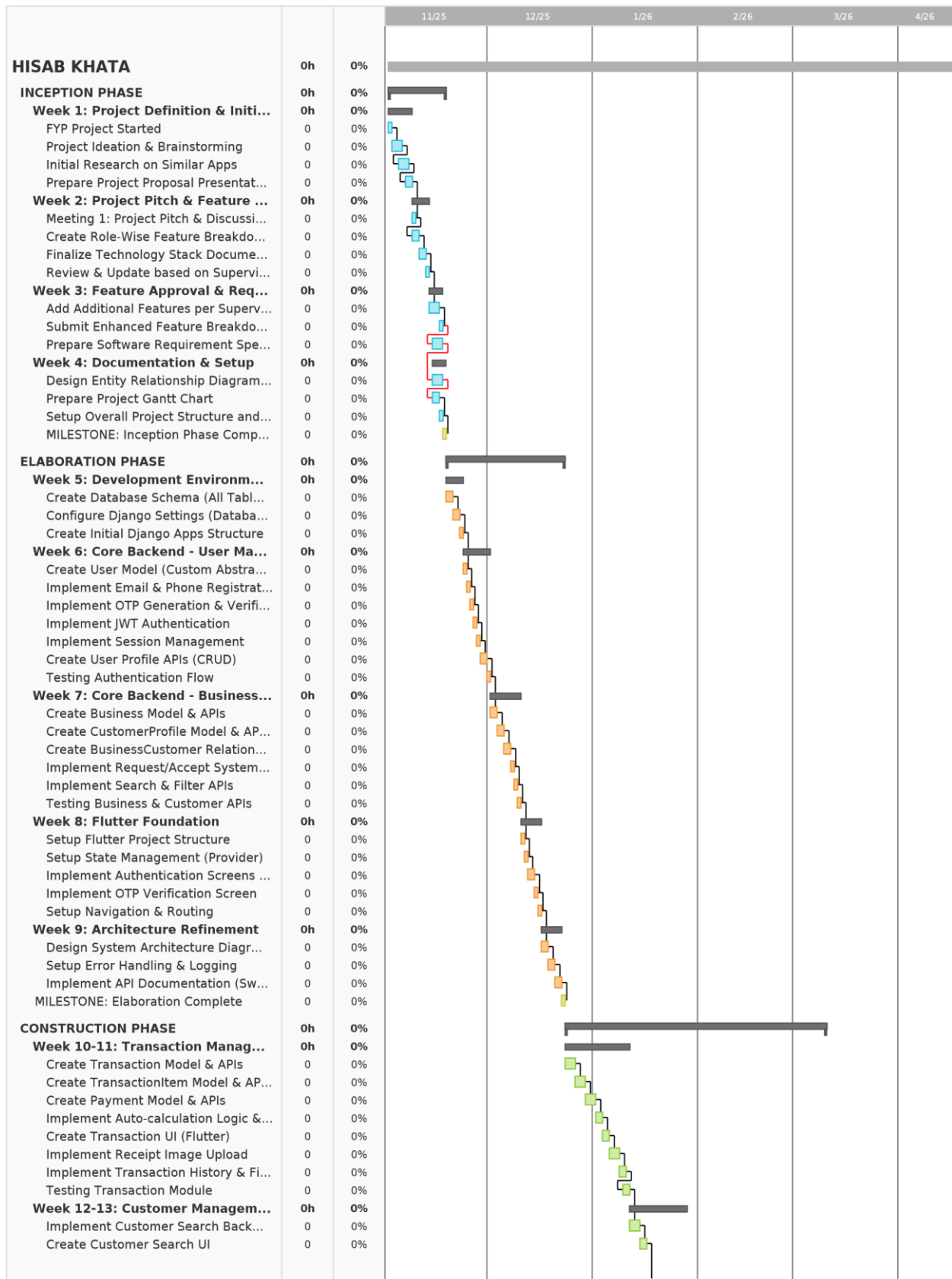
Once the project was given the green light by the supervisor, I developed the project WBS and Gantt chart to coordinate the project. Up to now, I have done the basic authentication module such as signup, login, OTP verification, OTP resend and better email templates. I also used clean architecture and BLoC on Flutter and designed customer and business dashboard with profile view/edit functionality. On the back end, I introduced connection request functionality with complete APIs, and notification models and a notification application. All the necessary UML diagrams are also done.

4.1 Progress Table

Table 2: Progress Table

SN	Task / Module	Status	Complete
1	Project setup (Flutter + Django)	Completed	100%
2	Clean Architecture setup (Flutter)	Completed	100%
3	Authentication (Signup/Login/JWT)	Completed	100%
4	OTP Verification + Resend	Completed	100%
5	User Roles (Business/Customer/Admin/Hybrid base)	In Progress	70%
6	Customer Dashboard (UI + BLoC + API integration)	In Progress	70%
7	Customer Profile (view/edit + image handling)	In Progress	75%
8	Business Dashboard (UI + BLoC + API integration)	In Progress	70%
9	Business Profile (view/edit + image handling)	In Progress	75%
10	Business Verification (Admin capability)	In Progress	85%
11	Connection Requests (send/receive/accept/reject + APIs)	In Progress	90%
12	List Connected Users API	In Progress	85%
13	Notifications (backend app + model)	In Progress	80%
14	Transaction Management (normal/credit, history, payment entry)	Not Started	0%
15	Digital Payments (Khalti/eSewa integration)	Not Started	0%
16	Real-time Chat	Not Started	0%
17	Analytics charts and insights	Not Started	0%
18	PDF Export (monthly stats, statements)	Not Started	0%
19	Support Tickets (user + admin response)	Not Started	0%
20	Admin Broadcast Notifications	Not Started	0%
21	AI Features	Not Started	0%
22	UML Diagrams (ERD, Use case, Class, 7 sequence, 7 activity, 7 collaboration)	Completed	100%

4.2 Gantt Chart





5 Future Works

To accomplish the development of the Hisab Khata system, several features are to be developed in the future to enhance the usability. According to the progress made at present, the second work will be aimed at completing the rest of the core modules and finishing off the features which have been partially completed.

Upcoming Features:

- **User Roles Completion:** Complete business customer admin hybrid role-based access control.
- **Dashboard Refinement:** Bugs, Customer and Business dashboards are incomplete, and have stable UI with correct data.
- **Transaction Management:** The normal and credit transactions, transaction history, search/filtering and recording of customer payment functions are implemented.
- **Digital Payments Integration:** Bring together Khalti and eSewa payment flows and verification and status management.
- **Real-Time Chat:** Add business-customer chat: add support with message delivery and notification.
- **Analytics and Reports:** Introduce business and customer dashboards with the help of charts, insights, and monthly statistics.
- **PDF Export:** Export statement and monthly statement in PDF format which can be downloaded.
- **Support Ticket System:** Additional support tickets creation and managerial response tickets in the administrative panel.
- **Admin Broadcast Notifications:** Should give the option of admins to broadcast notifications to all users.
- **AI Features:** Introduce the scanning of receipts and voice- Based entry of transactions that are manually verified.

6 Conclusion

Hisab Khata, is designed to revolutionize credit and transaction management of small businesses in Nepal by providing a digital platform. The system will address the long-standing problems of manual record-keeping, inefficiency, and lack of visibility through digitization of credit records, transparency of transactions, digital payments, AI-supported entry, and analytics dashboard. In this way, it will enable the shopkeepers to effectively monitor debts, payments, and business performance, and customers to keep track of their dues and trust.

Overall, Hisab Khata can become a very important enhancement of traditional credit-based transactions in the small-business sector of Nepal, which will lead to the financial inclusion, the efficiency of business operations, and more open business-customer relations. It may become a valuable resource to both shopkeepers and customers in Nepal with proper implementation, testing, and user friendly, modernizing the olden-day credit methods without losing flexibility and trust. Additionally, the project takes into account ground realities by creating the system with localization in language, payment methods etc.

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8 Appendix

8.1 SRS

Link to SRS: [link](#)

[Back to top](#)

8.2 Pre Survey

What type of user are you?
32 responses

Note: The questions will be asked according to the user you choose.

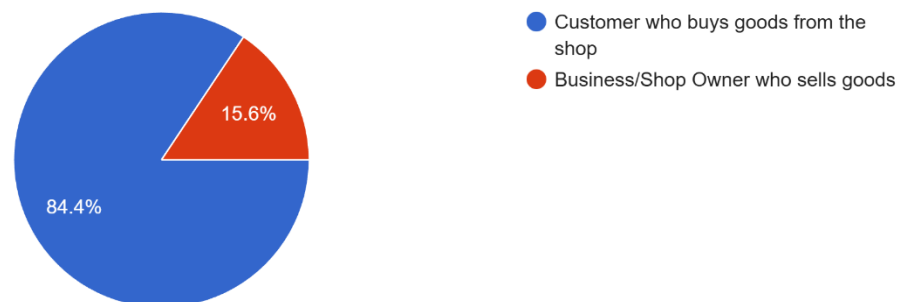


Figure 63: Pre Survey Question 1

8.2.1 Report on Customer

Do you own a smartphone?
27 responses

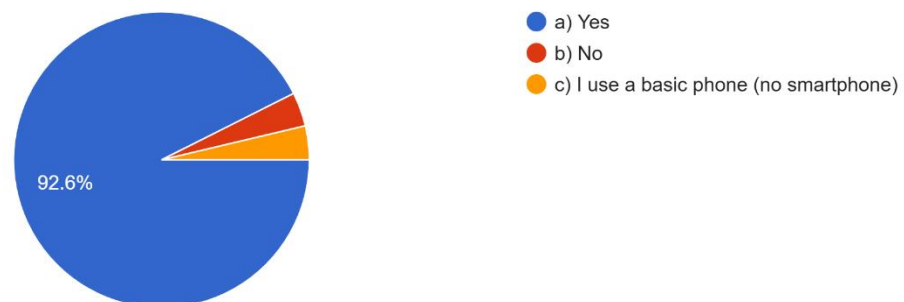


Figure 64: Pre Survey Question 2

How often do you buy goods on credit (baki/udhaaro) from shops?

27 responses

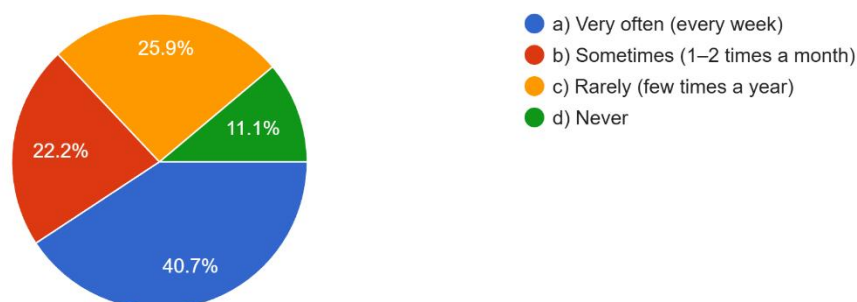


Figure 65: Pre Survey Question 3

In how many shops do you currently have credit/dues?

27 responses

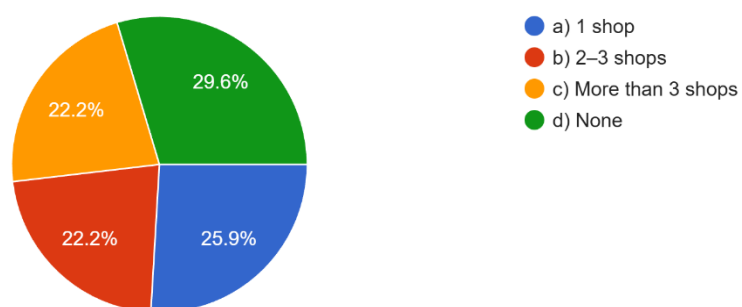


Figure 66: Pre Survey Question 4

How do you usually know how much you owe (your due amount) to a shop?

27 responses

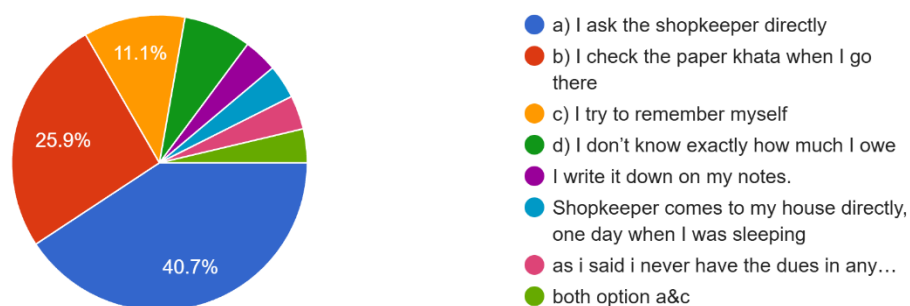


Figure 67: Pre Survey Question 5

Have you ever had confusion or disagreement with a shop about your due amount or payment history?

27 responses

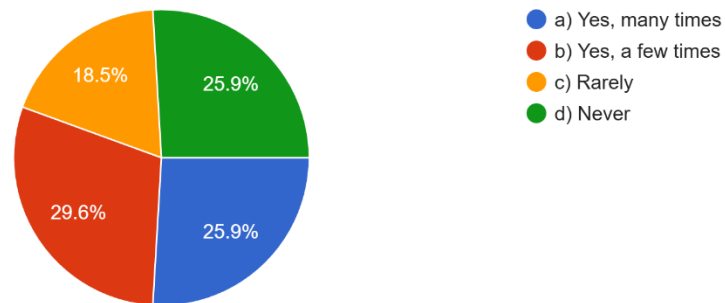


Figure 68: Pre Survey Question 6

Have you ever had confusion or disagreement with a shop about your due amount or payment history?

27 responses

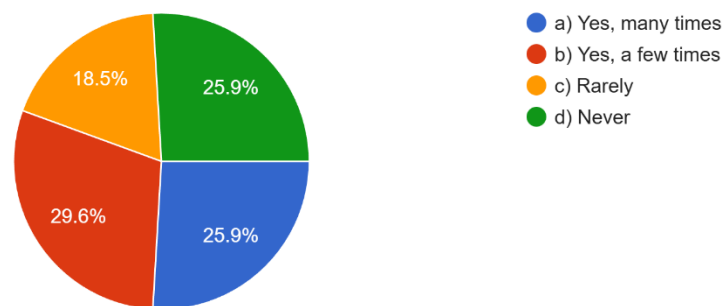


Figure 69: Pre Survey Question 7

How much do you trust that the shop's paper khata is always correct?

27 responses

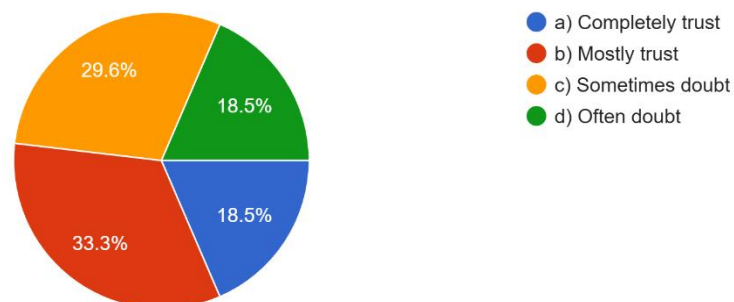


Figure 70: Pre Survey Question 8

How comfortable are you with mobile payment apps (eSewa, Khalti, bank apps)?

27 responses

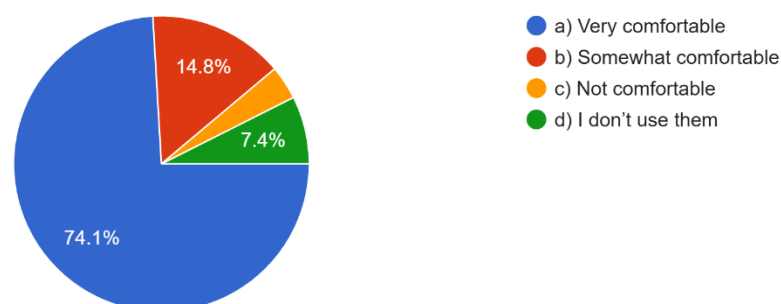


Figure 71: Pre Survey Question 9

Would you like to see your all dues with different shops in one mobile app?

27 responses

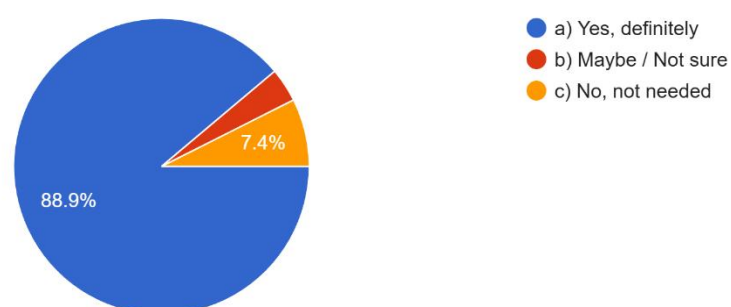


Figure 72: Pre Survey Question 10

How useful would it be for you to see full transaction history (all previous purchases and payments) with each shop?

27 responses

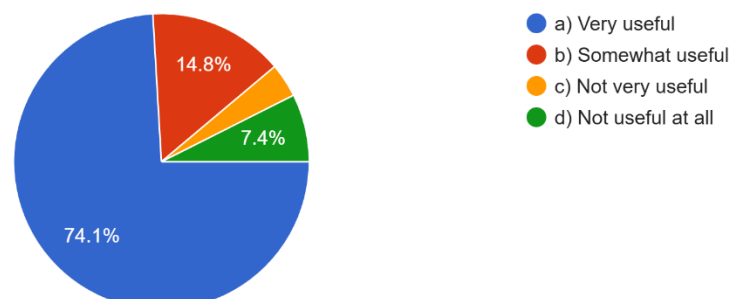


Figure 73: Pre Survey Question 11

Would you like to receive reminder notifications before your payment due date?

27 responses

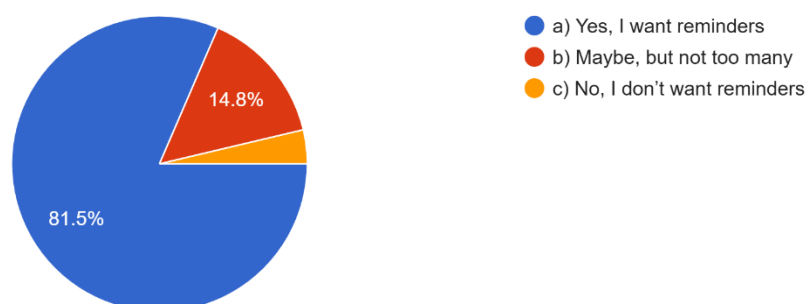


Figure 74: Pre Survey Question 12

How interested are you in paying your dues directly from the app using Khalti or eSewa?

27 responses

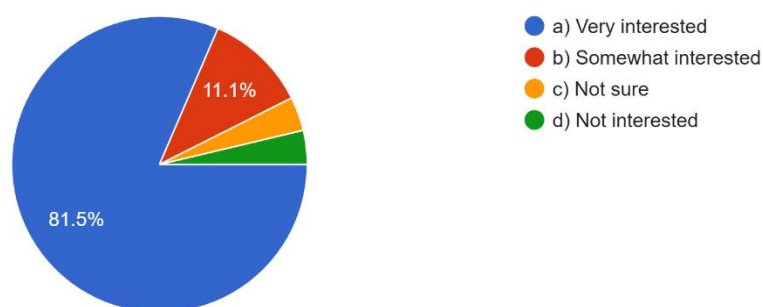


Figure 75: Pre Survey Question 13

Hisab Khata will give you "Gazab Customer Points" for paying on time, which shops can see as your trust score. How do you feel about this?

27 responses

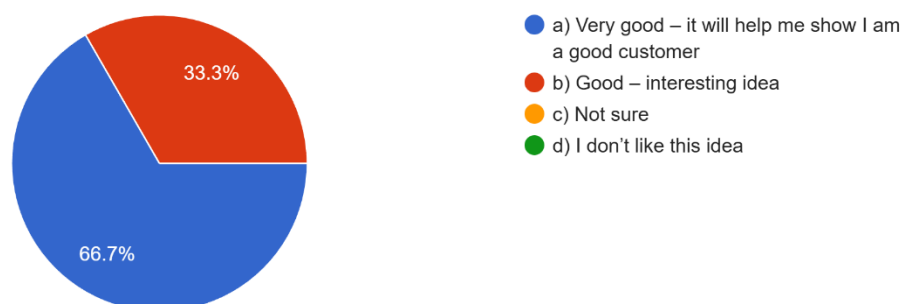


Figure 76: Pre Survey Question 14

Which language would you prefer in the app?

27 responses

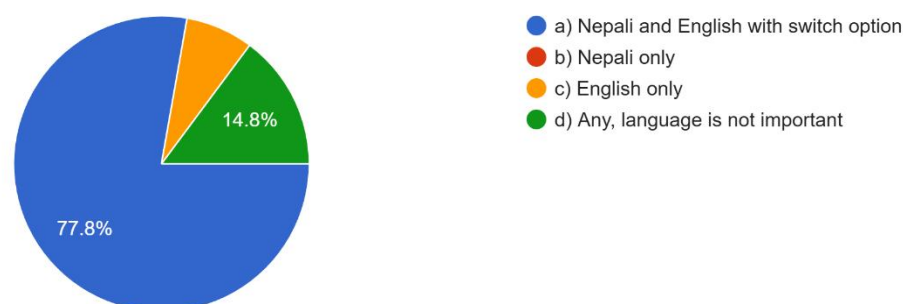


Figure 77: Pre Survey Question 15

Would you use such App in which you can connect multiple shops and see the your overall financial activity of that shop?

27 responses

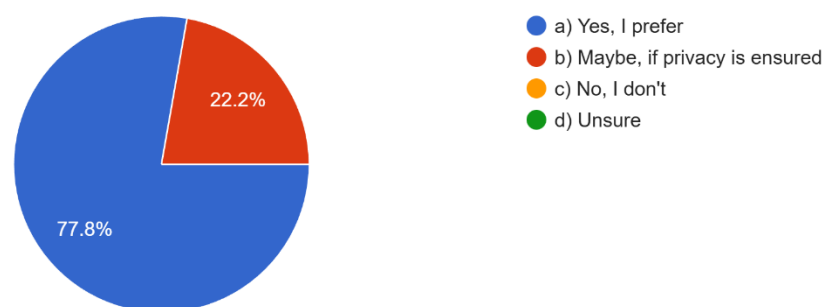


Figure 78: Pre Survey Question 16

Overall, how interested are you in using an app like Hisab Khata if your regular shop(s) started using it?

27 responses

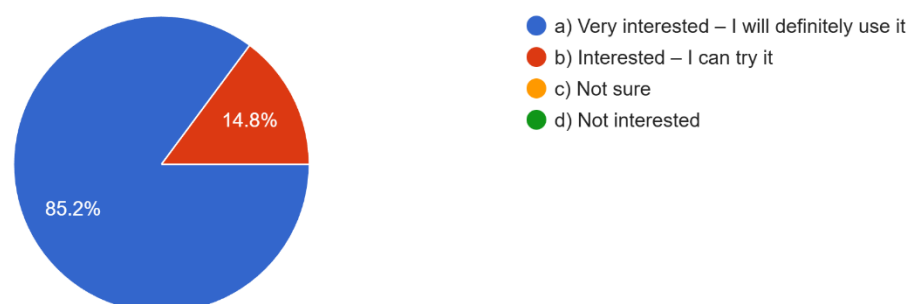


Figure 79: Pre Survey Question 17

8.2.2 Report on Business

What type of business do you run?

5 responses

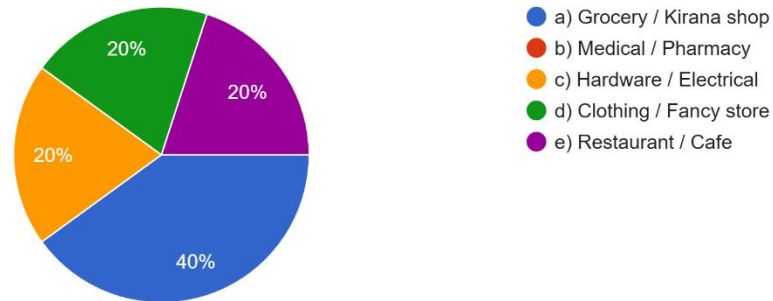


Figure 80: Pre Survey Question 18

How comfortable are you with using a smartphone for daily work?

5 responses

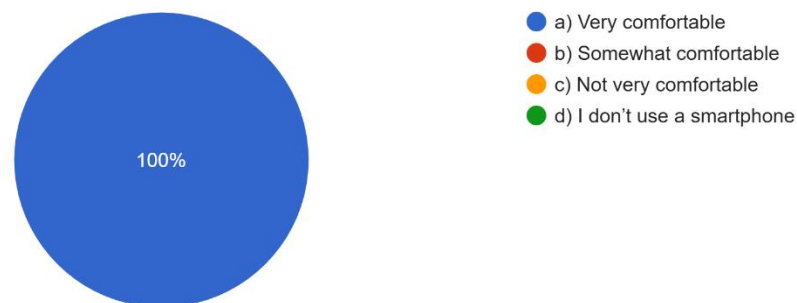


Figure 81: Pre Survey Question 19

How do you currently record customer credit (baki/udhaaro)?

5 responses

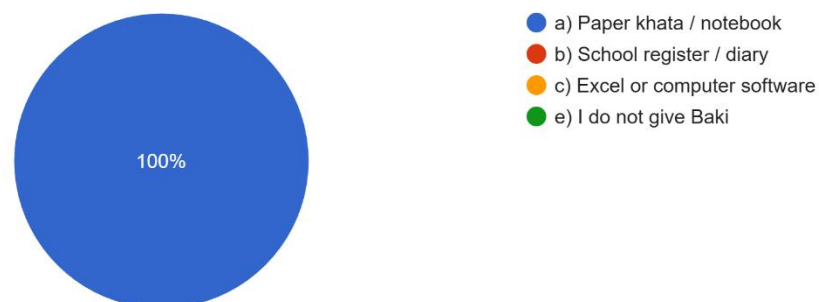


Figure 82: Pre Survey Question 20

How difficult is it to track customer dues(baki/udharo)?

5 responses



Figure 83: Pre Survey Question 21

Would you prefer customers to see their dues through an app?

5 responses

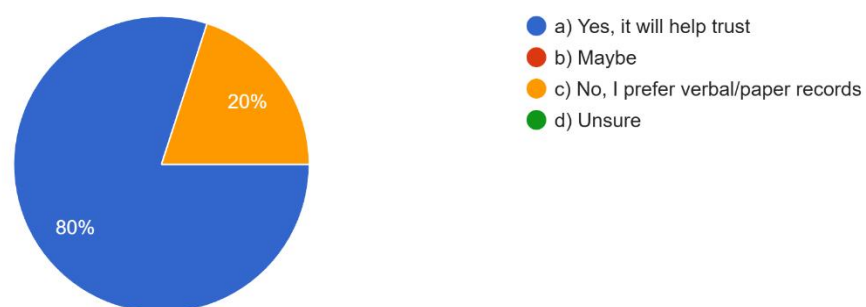


Figure 84: Pre Survey Question 22

How often do you face mistakes or confusion in your credit records?

5 responses

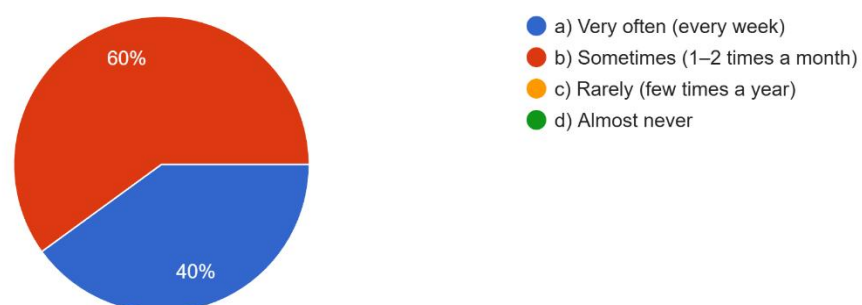


Figure 85: Pre Survey Question 23

Have you ever had disputes with customers about dues or payment history?

5 responses

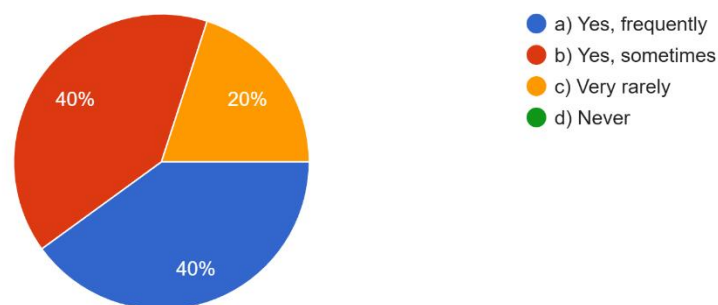


Figure 86: Pre Survey Question 24

How do you usually remind customers about their pending dues?

5 responses

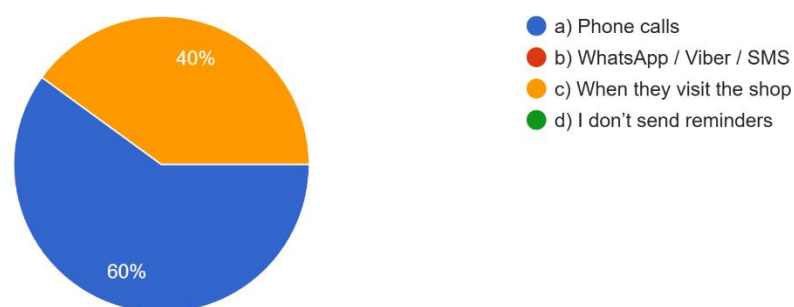


Figure 87: Pre Survey Question 25

Do you currently accept digital payments (Khalti, eSewa, etc.)?

5 responses

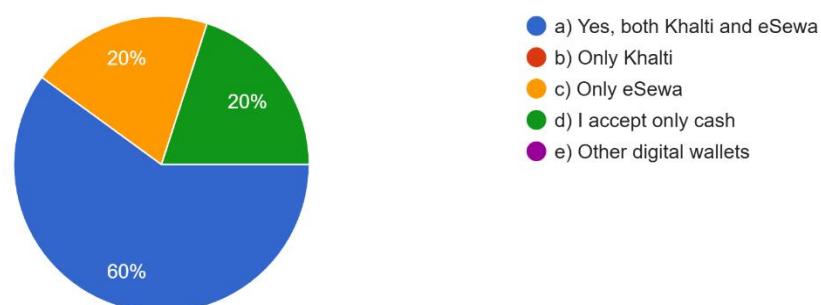


Figure 88: Pre Survey Question 26

How interested are you in using one app to manage credit records and digital payments together?

5 responses

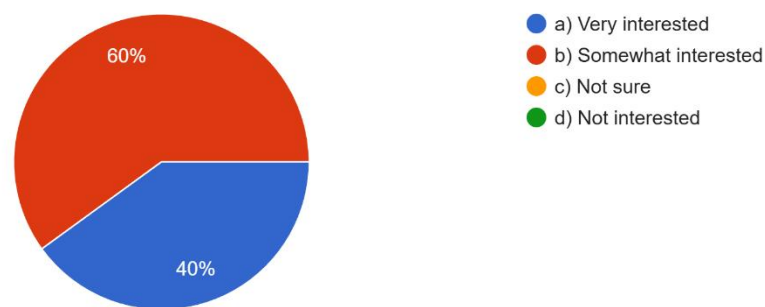


Figure 89: Pre Survey Question 26

If an app like Hisab Khata could automatically calculate total dues and show which customers owe the most, how useful would that be for you?

5 responses



Figure 90: Pre Survey Question 27

Would you like your customers to see their dues, transaction history and pay directly on their own mobile app?

5 responses

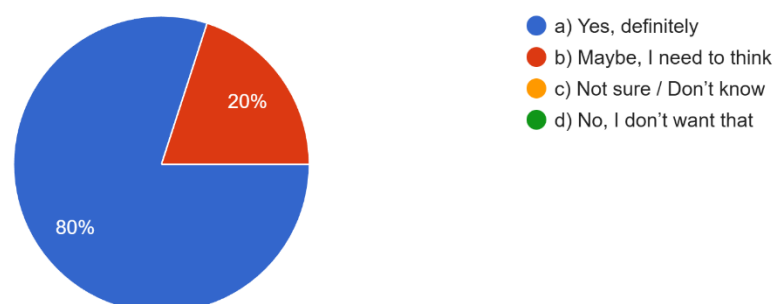


Figure 91: Pre Survey Question 28

How likely are you to use features like: automatic payment reminders (SMS/notification) sent to customers?

5 responses

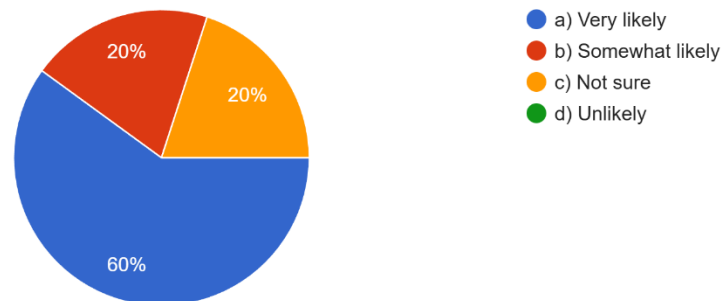


Figure 92: Pre Survey Question 29

How comfortable are you with using AI features like: 1. scanning a receipt photo to auto-fill transactions 2. speaking the transaction details (voice entry)?

5 responses

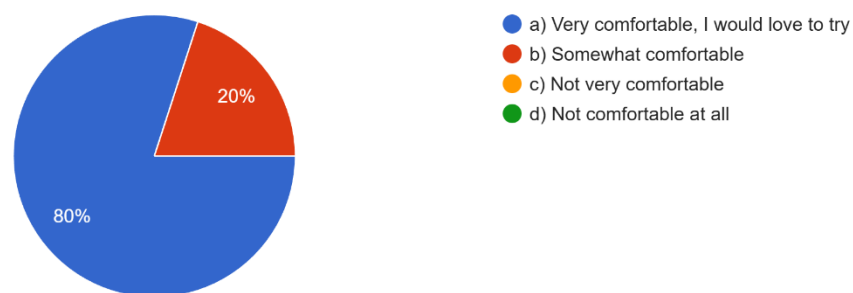


Figure 93: Pre Survey Question 30

Which features are most important for you in a credit management app?

5 responses

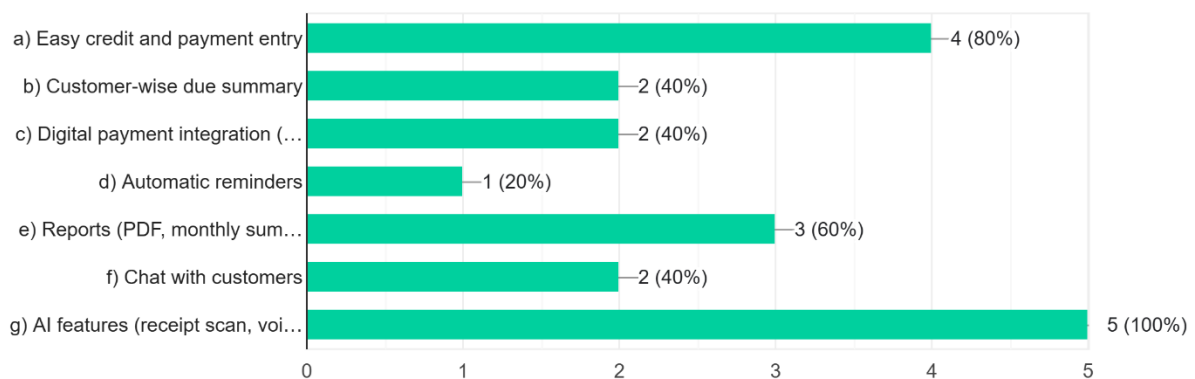


Figure 94: Pre Survey Question 31

If Hisab Khata app is: - Free to use, - In Nepali and language too - and easy to use, how likely are you to use it in your shop?

5 responses



Figure 95: Pre Survey Question 32

[Back to top](#)

8.3 Physical Pre Survey

8.3.1 Customers

8.3.1.1 Survey1

FYP Pre-survey Form: Hisab Khata
London MET ID: 23056498
Student Name: Abhishek Sharma
Project Title: Hisab Khata

Description:
 This survey is for collecting public opinions about **Hisab Khata**, a digital credit and transaction management system for small businesses and customers in Nepal. The aim is to understand how people currently manage dues, how comfortable they are with digital payments, and whether they would use a mobile app designed to make transactions clearer, faster, and more transparent.

For Customers: This short survey asks customers about your experience with shop credit and how likely you would be to use a mobile app (Hisab Khata). Choose one answer per question

- Do you own a smartphone?**
 - ☒ a. Yes
 - ☐ b. No
 - ☐ c. I use a basic phone (no smartphone)
- How often do you buy goods on credit (baki/udhaaro) from shops?**
 - ☐ a. Very often (every week)
 - ☒ b. Sometimes (1–2 times a month)
 - ☐ c. Rarely (few times a year)
 - ☐ d. Never
- In how many shops do you currently have credit/dues?**
 - ☐ a. 1 shop
 - ☒ b. 2–3 shops
 - ☐ c. More than 3 shops
 - ☐ d. None
- How do you usually know how much you owe (your due amount) to a shop?**
 - ☒ a. I ask the shopkeeper directly
 - ☐ b. I check the paper khata
 - ☐ c. I try to remember myself
 - ☐ d. I don't know exactly how much
 - ☐ e. Other: _____
- Have you ever had confusion or disagreement with a shop about your due amount or payment history?**
 - ☐ a. Yes, many times
 - ☒ b. Yes, a few times
 - ☐ c. Rarely
 - ☐ d. Never
- How much do you trust that the shop's paper khata is always correct?**
 - ☐ a. Completely trust
 - ☐ b. Mostly trust
 - ☒ c. Sometimes doubt
 - ☐ d. Often doubt
- How comfortable are you with mobile payment apps (eSewa, Khalti, bank apps)?**
 - ☒ a. Very comfortable
 - ☐ b. Somewhat comfortable
 - ☐ c. Not comfortable
 - ☐ d. I don't use them
- Would you like to see your all dues with different shops in one mobile app?**
 - ☒ a. Yes, definitely
 - ☐ b. Maybe / Not sure
 - ☐ c. No, not needed

9. How useful would it be for you to see full transaction history (all previous purchases and payments) with each shop?
- ☒ a. Very useful
- b. Somewhat useful
- c. Not very useful
- d. Not useful at all
10. Would you like to receive reminder notifications before your payment due date?
- ☒ a. Yes, I want reminders
- b. Maybe, but not too many
- c. No, I don't want reminders
11. How interested are you in paying your dues directly from the app using Khalti or eSewa?
- ☒ a. Very interested
- b. Somewhat interested
- c. Not sure
- d. Not interested
12. Hisab Khata will give you "Gazab Customer Points" for paying on time, which shops can see as your trust score. How do you feel about this?
- ☒ a. Very good
- b. Good – interesting idea
- c. Not sure
- d. I don't like this idea
13. Which language would you prefer in the app?
- ☒ a) Nepali and English with switch
- b. b) Nepali only
- c. c) English only
- d. d) Any, language is not important
14. Would you use such App in which you can connect multiple shops and see the your overall financial activity of that shop?
- ☒ a. Yes, I prefer
- b. Maybe, if privacy is ensured
- c. No, I do not
- d. Unsure
15. Overall, how interested are you in using an app like Hisab Khata if your regular shop(s) started using it?
- ☒ a. Very interested
- b. Interested – I can try it
- c. Not sure
- d. Not interested

Participant Details & Signature

Name: Diksha Ghimire
Contact Number: 9802030890
Address: Itahari - 4; Sunsari
Date 30/12/2025
Signature: *Diksha*

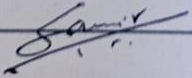
8.3.1.2 Survey2

FYP Pre-survey Form: Hisab Khata
London MET ID: 23056498
Student Name: Abhishek Sharma
Project Title: Hisab Khata
Description:
This survey is for collecting public opinions about Hisab Khata, a digital credit and transaction management system for small businesses and customers in Nepal. The aim is to understand how people currently manage dues, how comfortable they are with digital payments, and whether they would use a mobile app designed to make transactions clearer, faster, and more transparent.

For Customers: This short survey asks customers about your experience with shop credit and how likely you would be to use a mobile app (Hisab Khata). Choose one answer per question

1. Do you own a smartphone?
a. ☒ Yes b. ☐ No c. ☐ I use a basic phone (no smartphone)
2. How often do you buy goods on credit (baki/udhaaro) from shops?
a. ☐ Very often (every week) c. ☒ Rarely (few times a year)
b. ☐ Sometimes (1–2 times a month) d. ☐ Never
3. In how many shops do you currently have credit/dues?
a. ☒ 1 shop c. ☐ More than 3 shops
b. ☐ 2–3 shops d. ☐ None
4. How do you usually know how much you owe (your due amount) to a shop?
a. ☐ I ask the shopkeeper directly d. ☒ I don't know exactly how much
b. ☐ I check the paper khata e. ☐ Other: _____
c. ☐ I try to remember myself
5. Have you ever had confusion or disagreement with a shop about your due amount or payment history?
a. ☒ Yes, many times c. ☐ Rarely
b. ☐ Yes, a few times d. ☐ Never
6. How much do you trust that the shop's paper khata is always correct?
a. ☐ Completely trust c. ☐ Sometimes doubt
b. ☒ Mostly trust d. ☐ Often doubt
7. How comfortable are you with mobile payment apps (eSewa, Khalti, bank apps)?
a. ☐ Very comfortable c. ☐ Not comfortable
b. ☐ Somewhat comfortable d. ☒ I don't use them
8. Would you like to see your all dues with different shops in one mobile app?
a. ☐ Yes, definitely c. ☐ No, not needed
b. ☒ Maybe / Not sure

9. How useful would it be for you to see full transaction history (all previous purchases and payments) with each shop?
- a. Very useful
☒ b. Somewhat useful
c. Not very useful
d. Not useful at all
10. Would you like to receive reminder notifications before your payment due date?
- a. Yes, I want reminders
b. Maybe, but not too many
☒ c. No, I don't want reminders
11. How interested are you in paying your dues directly from the app using Khalti or eSewa?
- ☒ a. Very interested
b. Somewhat interested
c. Not sure
d. Not interested
12. Hisab Khata will give you "Gazab Customer Points" for paying on time, which shops can see as your trust score. How do you feel about this?
- ☒ a. Very good
b. Good – interesting idea
c. Not sure
d. I don't like this idea
13. Which language would you prefer in the app?
- a. a) Nepali and English with switch
b. b) Nepali only
☒ c. c) English only
d. d) Any, language is not important
14. Would you use such App in which you can connect multiple shops and see the your overall financial activity of that shop?
- ☒ a. Yes, I prefer
b. Maybe, if privacy is ensured
c. No, I do not
d. Unsure
15. Overall, how interested are you in using an app like Hisab Khata if your regular shop(s) started using it?
- ☒ a. Very interested
b. Interested – I can try it
c. Not sure
d. Not interested

Participant Details & SignatureName: Samir NepalContact Number: 9842153267Address: Sundarharanicha - 5Date 19 / 12 / 2025Signature: 

8.3.1.3 Survey3

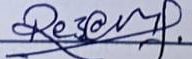
FYP Pre-survey Form: Hisab Khata
 London MET ID: 23056498
 Student Name: Abhishek Sharma
 Project Title: Hisab Khata

Description:
 This survey is for collecting public opinions about Hisab Khata, a digital credit and transaction management system for small businesses and customers in Nepal. The aim is to understand how people currently manage dues, how comfortable they are with digital payments, and whether they would use a mobile app designed to make transactions clearer, faster, and more transparent.

For Customers: This short survey asks customers about your experience with shop credit and how likely you would be to use a mobile app (Hisab Khata). Choose one answer per question

- Do you own a smartphone?
 a. ☒ Yes b. ☐ No c. ☐ I use a basic phone (no smartphone)
- How often do you buy goods on credit (baki/udhaaro) from shops?
 a. ☐ Very often (every week) c. ☒ Rarely (few times a year)
 b. ☐ Sometimes (1–2 times a month) d. ☐ Never
- In how many shops do you currently have credit/dues?
 a. ☒ 1 shop c. ☐ More than 3 shops
 b. ☐ 2–3 shops d. ☐ None
- How do you usually know how much you owe (your due amount) to a shop?
 a. ☒ I ask the shopkeeper directly d. ☐ I don't know exactly how much
 b. ☐ I check the paper khata e. ☐ Other: _____
 c. ☐ I try to remember myself
- Have you ever had confusion or disagreement with a shop about your due amount or payment history?
 a. ☒ Yes, many times c. ☐ Rarely
 b. ☐ Yes, a few times d. ☐ Never
- How much do you trust that the shop's paper khata is always correct?
 a. ☒ Completely trust c. ☐ Sometimes doubt
 b. ☐ Mostly trust d. ☐ Often doubt
- How comfortable are you with mobile payment apps (eSewa, Khalti, bank apps)?
 a. ☐ Very comfortable c. ☒ Not comfortable
 b. ☐ Somewhat comfortable d. ☐ I don't use them
- Would you like to see your all dues with different shops in one mobile app?
 a. ☐ Yes, definitely c. ☒ No, not needed
 b. ☐ Maybe / Not sure

9. How useful would it be for you to see full transaction history (all previous purchases and payments) with each shop?
- a. Very useful ☒ b. Somewhat useful ☐ c. Not very useful ☒ d. Not useful at all ☐
10. Would you like to receive reminder notifications before your payment due date?
- a. Yes, I want reminders ☒ b. Maybe, but not too many ☐ c. No, I don't want reminders ☐
11. How interested are you in paying your dues directly from the app using Khalti or eSewa?
- a. Very interested ☒ b. Somewhat interested ☐ c. Not sure ☒ d. Not interested ☐
12. Hisab Khata will give you "Gazab Customer Points" for paying on time, which shops can see as your trust score. How do you feel about this?
- a. Very good ☒ b. Good – interesting idea ☐ c. Not sure ☐ d. I don't like this idea ☐
13. Which language would you prefer in the app?
- a. a) Nepali and English with switch ☐ b. b) Nepali only ☒ c. c) English only ☐ d. d) Any, language is not important ☐
14. Would you use such App in which you can connect multiple shops and see the your overall financial activity of that shop?
- a. Yes, I prefer ☒ b. Maybe, if privacy is ensured ☐ c. No, I do not ☐ d. Unsure ☐
15. Overall, how interested are you in using an app like Hisab Khata if your regular shop(s) started using it?
- a. Very interested ☒ b. Interested – I can try it ☐ c. Not sure ☐ d. Not interested ☐

Participant Details & SignatureName: Nirajan DevjaContact Number: 9702004153Address: ItahariDate 12 / 30 / 2025Signature: 

8.3.2 Business

8.3.2.1 Survey1

FYP Pre-survey Form: Hisab Khata
London MET ID: 23056498
Student Name: Abhishek Sharma
Project Title: Hisab Khata
Description:
 This survey is for collecting public opinions about **Hisab Khata**, a digital credit and transaction management system for small businesses and customers in Nepal. The aim is to understand how people currently manage dues, how comfortable they are with digital payments, and whether they would use a mobile app designed to make transactions clearer, faster, and more transparent.

For Business: This short survey asks shop owners about credit, record-keeping, and whether they would adopt a digital system like Hisab Khata. Choose one answer per question.

- What type of business do you run?**

a. Grocery / Kirana shop	d. Clothing / Fancy store
b. Medical / Pharmacy	e. Restaurant / Café
☒ c. Hardware / Electrical	f. Other: _____
- How comfortable are you with using a smartphone for daily work?**

a. Very comfortable	☒ c. Not very comfortable
b. Somewhat comfortable	d. I don't use a smartphone
- How do you currently record customer credit (baki/udhaaro)?**

a. Paper khata / notebook	d. I do not give Baki
☒ b. Register / diary	e. Other: _____
c. Excel or computer software	
- How difficult is it to track customer dues(baki/udharo)?**

a. A. Very difficult	c. C. Easy
☒ b. B. Somewhat difficult	d. D. Not applicable
- Would you prefer customers to see their dues through an app?**

a. Yes, it will help trust	c. No, I prefer verbal/paper records
☒ b. Maybe	d. Unsure
- How often do you face mistakes or confusion in your credit records?**

a. Very often (every week)	☒ c. Rarely (few times a year)
b. 1-2 times a month	d. Almost never
- Have you ever had disputes with customers about dues or payment history?**

a. Yes, frequently	☒ c. Very rarely
b. Yes, sometimes	d. Never
- How do you usually remind customers about their pending dues?**

a. Phone calls	c. When they visit the shop
☒ b. WhatsApp / Viber / SMS	d. Other: _____

9. Do you currently accept digital payments (Khalti, eSewa, etc.)?

a. Yes, both Khalti and eSewa
b. Only Khalti
c. Only eSewa
d. I accept only cash
e. Other digital wallets

10. How interested are you in using one app to manage credit records and digital payments together?

a. Very interested
b. Somewhat interested
c. Not sure
d. Not interested

11. If an app like Hisab Khata could automatically calculate total dues and show which customers owe the most, how useful would that be for you?

a. Very useful
b. Somewhat useful
c. Not very useful
d. Not useful at all

12. Would you like your customers to see their dues, transaction history and pay directly on their own mobile app?

a. Yes, definitely
b. Maybe, I need to think
c. Not sure / Don't know
d. No, I don't want that

13. How likely are you to use features like: automatic payment reminders (SMS/notification) sent to customers?

a. Very likely
b. Somewhat likely
c. Not sure
d. Unlikely

14. How comfortable are you with using AI features like: scanning a receipt to auto-fill transactions and speaking the transaction details (voice entry)?

a. Very comfortable
b. Somewhat comfortable
c. Not very comfortable
d. Not comfortable at all

15. Which features are most important for you in a credit management app? (Choose Multiple)

a. ☒ Easy credit and payment entry
b. ☒ Customer-wise due summary
c. ☒ Digital payment integration
d. ☒ Automatic reminders
e. ☒ Reports (Monthly PDFs)
f. ☒ Chat and AI Features

Participant Details & Signature

Name: Mukunda Jamar Katel

Contact Number: 9824921311

Address: Rachi

Date: 22/28/2026

Signature: [Signature]

8.3.2.2 Survey2

FYP Pre-survey Form: Hisab Khata
London MET ID: 23056498
Student Name: Abhishek Sharma
Project Title: Hisab Khata

Description:
 This survey is for collecting public opinions about **Hisab Khata**, a digital credit and transaction management system for small businesses and customers in Nepal. The aim is to understand how people currently manage dues, how comfortable they are with digital payments, and whether they would use a mobile app designed to make transactions clearer, faster, and more transparent.

For Business: This short survey asks shop owners about credit, record-keeping, and whether they would adopt a digital system like Hisab Khata. Choose one answer per question.

- What type of business do you run?**

a. ☒ Grocery / Kirana shop	d. ☐ Clothing / Fancy store
b. ☐ Medical / Pharmacy	e. ☐ Restaurant / Café
c. ☐ Hardware / Electrical	f. ☐ Other: _____
- How comfortable are you with using a smartphone for daily work?**

a. ☒ Very comfortable	c. ☐ Not very comfortable
b. ☐ Somewhat comfortable	d. ☐ I don't use a smartphone
- How do you currently record customer credit (baki/udhaaro)?**

a. ☒ Paper khata / notebook	d. ☐ I do not give Baki
b. ☐ Register / diary	e. ☐ Other: _____
c. ☐ Excel or computer software	
- How difficult is it to track customer dues(baki/udharo)?**

a. ☒ A. Very difficult	c. ☐ C. Easy
b. ☐ B. Somewhat difficult	d. ☐ D. Not applicable
- Would you prefer customers to see their dues through an app?**

a. ☒ Yes, it will help trust	c. ☐ No, I prefer verbal/paper records
b. ☐ Maybe	d. ☐ Unsure
- How often do you face mistakes or confusion in your credit records?**

a. ☒ Very often (every week)	c. ☐ Rarely (few times a year)
b. ☐ 1–2 times a month	d. ☐ Almost never
- Have you ever had disputes with customers about dues or payment history?**

a. ☒ Yes, frequently	c. ☐ Very rarely
b. ☐ Yes, sometimes	d. ☐ Never
- How do you usually remind customers about their pending dues?**

a. ☒ Phone calls	c. ☐ When they visit the shop
b. ☐ WhatsApp / Viber / SMS	d. ☐ Other: _____

9. Do you currently accept digital payments (Khalti, eSewa, etc.)?

a. ☒ Yes, both Khalti and eSewa
b. Only Khalti
c. Only eSewa
d. I accept only cash
e. Other digital wallets

10. How interested are you in using one app to manage credit records and digital payments together?

a. ☒ Very interested
b. Somewhat interested
c. Not sure
d. Not interested

11. If an app like Hisab Khata could automatically calculate total dues and show which customers owe the most, how useful would that be for you?

a. ☒ Very useful
b. Somewhat useful
c. Not very useful
d. Not useful at all

12. Would you like your customers to see their dues, transaction history and pay directly on their own mobile app?

a. ☒ Yes, definitely
b. Maybe, I need to think
c. Not sure / Don't know
d. No, I don't want that

13. How likely are you to use features like: automatic payment reminders (SMS/notification) sent to customers?

a. ☒ Very likely
b. Somewhat likely
c. Not sure
d. Unlikely

14. How comfortable are you with using AI features like: scanning a receipt to auto-fill transactions and speaking the transaction details (voice entry)?

a. ☒ Very comfortable
b. Somewhat comfortable
c. Not very comfortable
d. Not comfortable at all

15. Which features are most important for you in a credit management app? (Choose Multiple)

a. ☒ Easy credit and payment entry
b. ☐ Customer-wise due summary
c. ☒ Digital payment integration
d. ☐ Automatic reminders
e. ☒ Reports (Monthly PDFs)
f. ☐ Chat and AI Features

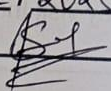
Participant Details & Signature

Name: Sandip rhadka

Contact Number: 9815321215

Address: Itahari-09 sunsari

Date 28/12/2025

Signature: 

8.3.2.3 Survey3

FYP Pre-survey Form: Hisab Khata
London MET ID: 23056498
Student Name: Abhishek Sharma
Project Title: Hisab Khata

Description:
 This survey is for collecting public opinions about **Hisab Khata**, a digital credit and transaction management system for small businesses and customers in Nepal. The aim is to understand how people currently manage dues, how comfortable they are with digital payments, and whether they would use a mobile app designed to make transactions clearer, faster, and more transparent.

For Business: This short survey asks shop owners about credit, record-keeping, and whether they would adopt a digital system like Hisab Khata. Choose one answer per question.

- What type of business do you run?**
 - a. Grocery / Kirana shop
 - b. Medical / Pharmacy
 - c. Hardware / Electrical
 - d. ☒ Clothing / Fancy store
 - e. Restaurant / Café
 - f. Other: _____
- How comfortable are you with using a smartphone for daily work?**
 - a. Very comfortable
 - b. Somewhat comfortable
 - c. ☒ Not very comfortable
 - d. I don't use a smartphone
- How do you currently record customer credit (baki/udhaaro)?**
 - a. Paper khata / notebook
 - b. Register / diary
 - c. Excel or computer software
 - d. ☒ I do not give Baki
 - e. Other: _____
- How difficult is it to track customer dues(baki/udharo)?**
 - a. ☒ A. Very difficult
 - b. B. Somewhat difficult
 - c. C. Easy
 - d. D. Not applicable
- Would you prefer customers to see their dues through an app?**
 - a. Yes, it will help trust
 - b. Maybe
 - c. ☒ No, I prefer verbal/paper records
 - d. Unsure
- How often do you face mistakes or confusion in your credit records?**
 - a. ☒ Very often (every week)
 - b. 1–2 times a month
 - c. Rarely (few times a year)
 - d. Almost never
- Have you ever had disputes with customers about dues or payment history?**
 - a. Yes, frequently
 - b. Yes, sometimes
 - c. ☒ Very rarely
 - d. Never
- How do you usually remind customers about their pending dues?**
 - a. ☒ Phone calls
 - b. WhatsApp / Viber / SMS
 - c. When they visit the shop
 - d. Other: _____

9. Do you currently accept digital payments (Khalti, eSewa, etc.)?

a. ☒ Yes, both Khalti and eSewa d. ☐ I accept only cash

b. ☐ Only Khalti e. ☐ Other digital wallets

c. ☐ Only eSewa

10. How interested are you in using one app to manage credit records and digital payments together?

a. ☒ Very interested c. ☐ Not sure

b. ☐ Somewhat interested d. ☐ Not interested

11. If an app like Hisab Khata could automatically calculate total dues and show which customers owe the most, how useful would that be for you?

a. ☐ Very useful e. ☒ Not very useful

b. ☐ Somewhat useful d. ☐ Not useful at all

12. Would you like your customers to see their dues, transaction history and pay directly on their own mobile app?

a. ☒ Yes, definitely c. ☐ Not sure / Don't know

b. ☐ Maybe, I need to think d. ☐ No, I don't want that

13. How likely are you to use features like: automatic payment reminders (SMS/notification) sent to customers?

a. ☐ Very likely e. ☒ Not sure

b. ☐ Somewhat likely d. ☐ Unlikely

14. How comfortable are you with using AI features like: scanning a receipt to auto-fill transactions and speaking the transaction details (voice entry)?

a. ☒ Very comfortable c. ☐ Not very comfortable

b. ☐ Somewhat comfortable d. ☐ Not comfortable at all

15. Which features are most important for you in a credit management app? (Choose Multiple)

a. ☒ Easy credit and payment entry d. ☐ Automatic reminders

b. ☐ Customer-wise due summary e. ☐ Reports (Monthly PDFs)

c. ☐ Digital payment integration f. ☐ Chat and AI Features

Participant Details & Signature

Name: Sujan Sharma

Contact Number: 9849321722

Address: Ilharni

Date 29 / 12 / 2025

Signature: Sujan

8.4 Product Backlogs

8.4.1 Sprint 1 (Foundation: Auth, Profiles, Basic Dashboards)

Table 3: Sprint 1 (Foundation: Auth, Profiles, Basic Dashboards)

ID	User Story	Priority	Sprint	Story Point
US001	As a user, I want to register using email and phone so that I can create an account.	High	1	5
US002	As a user, I want to verify account using OTP so that the account becomes active.	High	1	5
US003	As a user, I want to resend OTP so that I can verify if OTP expires.	Medium	1	3
US004	As a user, I want to log in using email and password so that I can access the system.	High	1	5
US005	As a user, I want the app to validate email, phone, and password so that incorrect data is prevented.	Medium	1	3
US006	As a customer, I want to view my dashboard so that I can see my due overview.	High	1	5
US007	As a business, I want to view my dashboard so that I can see total due summary.	High	1	5
US008	As a customer, I want to view and edit my profile so that my details stay updated.	High	1	5
US009	As a business, I want to view and edit my profile so that customers can identify my shop.	High	1	5
US010	As a user, I want to upload profile photo using camera or gallery so that profile looks professional.	Medium	1	3

8.4.2 Sprint 2 (Customer Management: Connections + Lists)

Table 4: Sprint 2 (Customer Management: Connections + Lists)

ID	User Story	Priority	Sprint	Story Point
US011	As a business, I want to search users by email or phone so that I can find customers quickly.	High	2	5
US012	As a business, I want to send a connection request using email so that I can connect with a customer.	High	2	5
US013	As a customer, I want to see received connection requests so that I can respond to them.	High	2	5
US014	As a customer, I want to accept or reject a connection request so that I control my connections.	High	2	5
US015	As a business, I want to see sent requests so that I can track pending connections.	Medium	2	3
US016	As a business, I want to see received requests so that I can accept new customers.	Medium	2	3
US017	As a business, I want to list all connected customers so that I can manage my network.	High	2	5
US018	As a customer, I want to list all connected businesses so that I can track all dues.	High	2	5

8.4.3 Sprint 3 (Transaction Management: Normal + Credit + History)

Table 5: Sprint 3 (Transaction Management: Normal + Credit + History)

ID	User Story	Priority	Sprint	Story Point
US019	As a business, I want to create a normal transaction so that sales are recorded.	High	3	8
US020	As a business, I want to create a credit transaction so that due sales are recorded.	High	3	8
US021	As a business, I want to add multiple items in a transaction so that bills are accurate.	High	3	8
US022	As a business, I want to attach receipt image so that transaction proof is available.	Medium	3	3
US023	As a business, I want to view transaction history so that I can check past records.	High	3	5
US024	As a business, I want to filter transactions by date and type so that I can find records quickly.	Medium	3	3
US025	As a business, I want to search transactions by customer name so that I can locate bills easily.	Medium	3	3
US026	As a customer, I want to view transaction history per business so that I have transparency.	High	3	5
US027	As a business, I want the system to auto-calculate due so that manual errors are reduced.	High	3	5

8.4.4 Sprint 4 (Payments + Notifications + Reminders)

Table 6: Sprint 4 (Payments + Notifications + Reminders)

ID	User Story	Priority	Sprint	Story Point
US028	As a customer, I want to make digital payment so that I can pay without cash.	High	4	13
US029	As a business, I want to record customer payment so that due decreases correctly.	High	4	8
US030	As a system, I want to integrate Khalti so that online payment is supported.	High	4	13
US031	As a system, I want to integrate eSewa so that online payment is supported.	High	4	13
US032	As a customer, I want due reminder notifications so that I do not miss payments.	High	4	5
US033	As a business, I want to send payment reminder to a customer so that collection becomes easier.	Medium	4	3
US034	As a business, I want to send bulk reminders so that many customers are notified at once.	Medium	4	5
US035	As a system, I want to store notification history so that users can review alerts.	Medium	4	3

8.4.5 Sprint 5 (Chat + Analytics + PDF Reports + Loyalty)

Table 7: Sprint 5 (Chat + Analytics + PDF Reports + Loyalty)

ID	User Story	Priority	Sprint	Story Point
US036	As a business, I want to chat with a customer so that I can discuss due and payments.	Medium	5	8
US037	As a customer, I want to chat with a business so that I can clarify transactions.	Medium	5	8
US038	As a business, I want analytics charts so that I can see paid vs pending amounts.	Medium	5	8
US039	As a customer, I want analytics summary so that I can track spending and dues.	Low	5	5
US040	As a business, I want to export monthly PDF report so that I can keep records offline.	Medium	5	8
US041	As a customer, I want to export statement PDF so that I can maintain proof of payments.	Medium	5	5
US042	As a business, I want loyalty points for customers so that trusted customers are rewarded.	Low	5	5
US043	As a customer, I want to view loyalty points so that I can track my trust score.	Low	5	3
US044	As a customer, I want to mark favorite businesses so that I can access them quickly.	Low	5	3

8.4.6 Sprint 6 (Admin Panel + Support Tickets + AI Enhancements)

Table 8: Sprint 6 (Admin Panel + Support Tickets + AI Enhancements)

ID	User Story	Priority	Sprint	Story Point
US045	As an admin, I want to view platform statistics so that I can monitor system usage.	High	6	8
US046	As an admin, I want to verify businesses so that only trusted shops are verified.	High	6	5
US047	As an admin, I want to activate or deactivate users so that abuse can be controlled.	High	6	5
US048	As an admin, I want to manage support tickets so that user problems are resolved.	High	6	8
US049	As a user, I want to create support tickets so that I can report issues.	Medium	6	5
US050	As an admin, I want to broadcast notifications so that all users get important updates.	Medium	6	5
US051	As a business, I want receipt scanning OCR so that transaction entry becomes faster.	Medium	6	13
US052	As a business, I want voice based quick transaction entry so that I can add transaction hands free.	Medium	6	13
US053	As a system, I want to log admin actions so that audit tracking is maintained.	Medium	6	3

Contingency Plan

In case there is any form of delay, core functions like Transaction Management, Dashboard Stability, User Roles and Admin Controls will be given priority first. Such advanced features as AI features, live chat, and complete analytics can be reallocated to subsequent Phase to make the main system functional and deliverable on time.

8.5 Acronyms and Abbreviations

Table 9: Acronyms and Abbreviations

Acronym	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
APK	Android Package Kit
ERD	Entity-Relationship Diagram
FCM	Firebase Cloud Messaging
JWT	JSON Web Token
OCR	Optical Character Recognition
OTP	One Time Password
REST	Representational State Transfer
RUP	Rational Unified Process
SDK	Software Development Kit
SMB	Small and Medium-sized Business
SRS	Software Requirements Specification
UI/UX	User Interface / User Experience
WBS	Work Breakdown Structure

8.6 Market context of digital

The digital transformation of Nepal is increasing day by day:

- Smartphone coverage: 73.2% (2023) of the population.
- Users of the internet: 65.3 of the total population.
- Adoption of digital payments: Digital wallet usage increases up to 200 percent (2022-2024).

This presents a great opportunity to a mobile-first solution that would meet the needs of Nepali shopkeepers specifically (My Republica, 2023).

[Back to top...](#)

8.7 Milestone Diagram

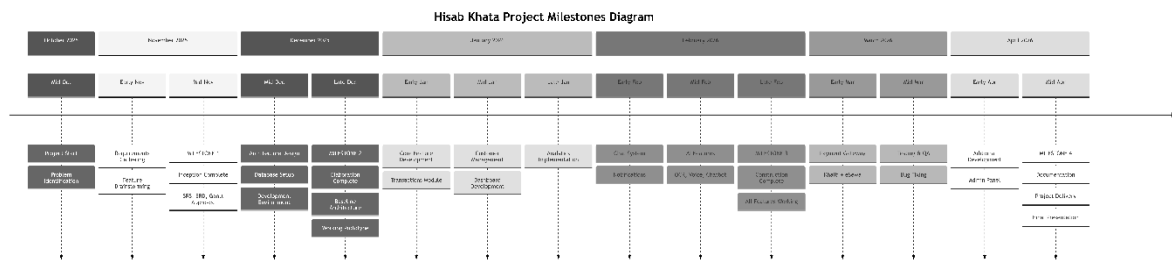


Figure 96: Milestone Diagram

8.8 WBS

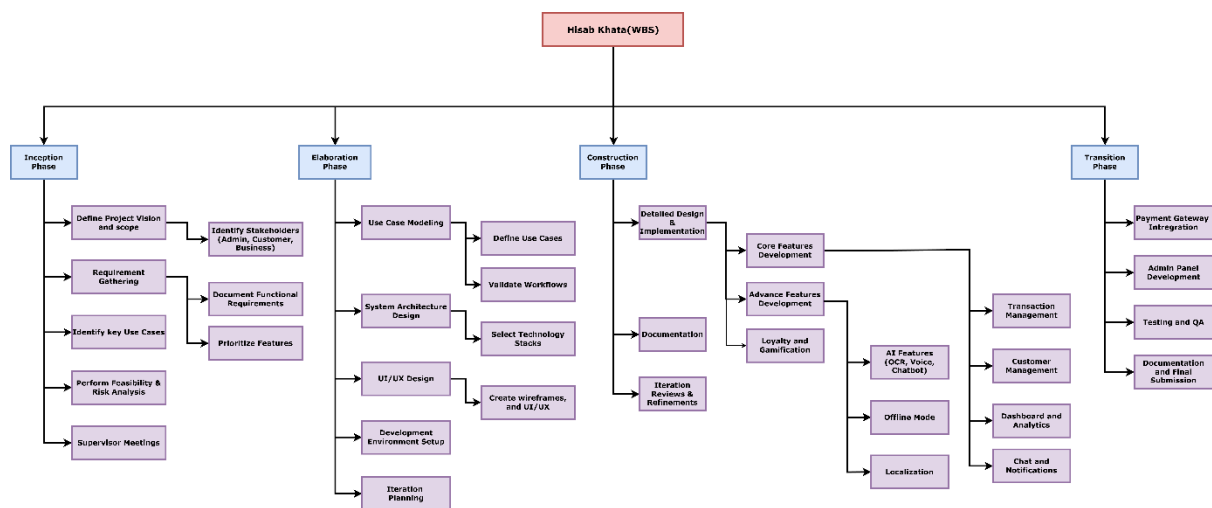


Figure 97: WBS Diagram

8.9 Reason for not selecting waterfall model

- It lacks flexibility and adaptability. Once the phase is done there is not going back option or going back may be difficult and costly.
- Customers cannot give feedback because testing will be later and there is little chance of updating based on client's feedback.
- There is no proper working software until near the end. For a project like Hisab Khata, early working prototype with minimum core functionalities would be useful for testing.

[Go back...](#)

8.10 Reasons for not selecting Agile/Scrum methodology

- It demands a great degree of discipline, communication and self-organizing team.
- Reduce the focus on documentation where minimal documentation may be a disadvantage to a system such as Hisab Khata where reliability, auditability, future maintainability, and perhaps more formal support is required.

[Go back...](#)

8.11 Reasons for not selecting Prototype based methodology

- Prototyping provides excellent early feedback and user-centric design validation, both user-facing features and usability are desirable but the complexity of Hisab Khata (backend, payment, AI, multi-role support) demands a good architectural foundation and design. Simple prototype-based approach is dangerous in the sense that it can result in a low-quality end system or technical debt given time limitations.

[Go back...](#)

8.12 Reason for selecting RUP

RUP offers the right balance. It is structured but flexible, architectural and adaptable, documented and flexible. Its focus on early architecture, cyclical development, and capacity to customize workflows is in line with the requirements of the project. Also, it focusses on documentation and traceability and provides a sense of clarity and maintainability, which is advantageous not only in evaluation but also in the event of the project being continued in the future.

[Go back...](#)